

Who Benefits from Benefits?

Municipal Consequences of Federal Cash Transfers

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Abstract

Cash transfer programs are centrally designed to limit local discretion. Yet in Brazil's *Bolsa Família*, eligibility rules and benefit values are managed federally while beneficiary registration is operated municipally through local welfare offices (*CRAS*). A 2009 federal reform revised the cross-municipal allocation formula, generating exogenous variation in authorized enrollment that we use to instrument realized expansion across the 2,105 municipalities (approximately 38 percent of the country) where the sitting mayor sought reelection in 2012. A one-standard-deviation increase in instrumented enrollment raises incumbent vote share by 2.94 percentage points, large enough to neutralize the typical inter-election incumbency disadvantage. The mechanism is gated at two binding margins: federal authorization translates weakly into realized expansion without a pre-existing *CRAS*, and approximately 93 percent of the within-*CRAS* identifying mass concentrates where the incoming mayor replaced the front-line manager between 2008 and 2009. Three quarters of those replacements bring in individuals new to the municipal welfare bureaucracy rather than internal reshuffles. The substitutes are not pre-selected on prior partisan affiliation but run for elected office at higher rates than stayers and align with the mayor's coalition, consistent with program expansion raising the visibility of the managerial post and with those individuals leveraging it to launch local electoral careers. Centralized design does not shield delivery from local politics.

Keywords: cash transfers; electoral returns; state capacity; *Bolsa Família*.

JEL Codes: D72, H53, H75, I38, O17.

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1 Introduction

Governments increasingly rely on cash transfer programs to alleviate poverty while limiting local political discretion. By standardizing eligibility rules, fixing benefit values by federal decree, and processing payments through centralized banking systems, these programs are designed to reduce the scope for patronage and discretionary allocation (Fiszbein and Schady, 2009). Yet access often depends on local bureaucracies controlled by elected officials. Whether control over program registration rather than program design is sufficient to generate electoral returns is the question we address.

We study this question in the context of Brazil's *Bolsa Família*, one of the largest conditional cash transfer programs in the world.¹ Although eligibility rules, benefit size, and payments are determined by the federal government, municipalities remain responsible for identifying and registering eligible households, mostly through local welfare centers, the CRAS (*Centro de Referência de Assistência Social*). This institutional arrangement creates a narrow but important margin of local discretion: while mayors cannot alter transfer amounts, redirect funds, or withhold payments, they control the administrative pipeline through which households may gain access to the program.

This margin operated through a registration process that required active municipal effort. Identifying potentially eligible families meant locating them, conducting household interviews, and collecting documentation; municipal authorities controlled the local registration apparatus, which gave them influence over the pace, intensity, and reach of enrollment efforts.² In practice, two municipalities facing similar levels of underlying need could generate very different realized coverage depending on how actively the local administration pursued registration.

Our empirical strategy exploits a 2009 reform that revised the formula used to compute municipal *Bolsa Família* quotas. Throughout this period, each municipality faced a federally determined quota, that is, a ceiling on the number of families that could be served under the program.³ The 2009 reform updated the formula used to calcu-

¹By June 2006, *Bolsa Família* covered 11.1 million families, approximately 46 million people, or about 25% of the Brazilian population. By June 2015, the program supported 13.8 million families, with an average monthly benefit of R\$167.15 and a budget equivalent to 0.5% of GDP.

²Hellmann (2015) describes enrollment as a municipal responsibility carried out in three stages: identification and localization of eligible families, household interviews, and digital entry of the information into the federal system. He reports that as of September 2014, 77% of registrations were made without a home visit. Municipalities were responsible for preparing the local structure for registration, training personnel, promoting awareness campaigns, and designing *active search* strategies to reach eligible families outside the existing registry.

³Lindert et al. (2007) explains that *Bolsa Família* maintained municipal program quotas based on local poverty estimates. Original municipal allocations were constructed by combining the 2001 PNAD with

late those municipal ceilings using census and survey information, generating large and heterogeneous changes in the number of families each municipality was authorized to serve.⁴ Crucially, the reform expanded the ceiling, not the registry: although already-registered eligible families were absorbed automatically, the magnitude of the expansion left a large residual margin of unfilled slots whose realization depended on local governments' effort to identify, locate, and register eligible families not yet in the federal database.

We combine administrative data on quotas, realized beneficiary counts, and local social-assistance infrastructure with municipal electoral data from the subsequent election.⁵ We use the federally induced quota shock as an instrument for realized beneficiary expansion and estimate its effect on incumbent electoral performance. The first stage is strong: municipalities receiving larger quota shocks expanded coverage substantially more than those receiving smaller shocks. This is the empirical counterpart to the institutional fact that federal authorization required local action to materialize.

Our findings trace the reform along four stages: the first-stage relationship between federal authorization and realized enrollment; the intention-to-treat effect of the authorization shock on electoral outcomes; the average causal response of electoral outcomes to realized enrollment that combines the two; and the institutional conditions under which these effects emerge.

The first stage confirms that federal authorization traveled to realized enrollment through the channel the institutional design contemplates. A one-standard-deviation increase in the quota shock, roughly 1,447 additional authorized slots, expanded realized beneficiary coverage by 0.21 standard deviations, approximately 1,015 additional families. Having established that the authorization shock moves realized enrollment, we ask what the reform itself delivers electorally, that is, the effect of expanding the federal ceiling, before separating out how much of that ceiling municipalities chose to fill. This intention-to-treat effect is 0.60 percentage points on the incumbent's vote share: the electoral consequence of authorizing more beneficiaries, integrating over the local take-up response documented in the first stage.

Scaling the intention-to-treat effect by the first stage gives the average causal re-

the 2000 Census, generating an initial national target of 11.2 million families when the program was created in 2003.

⁴Hellmann (2015) notes that *Bolsa Família* administered municipal quotas using estimates of poor families developed in partnership with IBGE and reports that, in 2009, the federal government refined the measurement of the target population by incorporating methodologies to capture income volatility among vulnerable households. These studies formed a major expansion of coverage, incorporating more than 1.6 million additional families.

⁵The analysis focuses on the 2008 and 2012 municipal elections and covers 2,105 municipalities.

sponse of electoral outcomes to realized beneficiary expansion. The 0.60 percentage points reported above correspond to the 1,015 additional families that each 1,447-slot authorization shock translated into on average; equivalently, an additional 1,000 realized beneficiaries raise the incumbent's vote share by approximately 0.59 percentage points. In standardized terms, a one-standard-deviation increase in instrumented enrollment, about 4,949 additional families, raises vote share by 2.94 percentage points. The estimate is robust to alternative instrument thresholds, different measurement windows for the endogenous variable, other outcome definitions, distinct weighting schemes, and inverse-probability weighting for selection into the rerunning sample.

These returns travel through the institutional layer that converts authorization into delivery. In the 690 municipalities without a pre-existing CRAS in 2008, the quota shock generates a first stage roughly a factor of three weaker than in municipalities with a CRAS, and the electoral effect cannot be identified at conventional precision for the no-CRAS subgroup. The Frisch-Waugh-Lovell decomposition concentrates approximately nine-tenths of the headline identifying mass on the 1,415 CRAS municipalities, where the implementation chain operates.

Within CRAS municipalities, a second binding margin operates through manager turnover. The within-CRAS first stage attenuates sharply where the *de facto* front-line manager who runs the CRAS unit day-to-day did not change between the 2008 and 2009 waves of the survey. Identification weights concentrate approximately 93 percent of the within-CRAS identifying mass on the manager-turnover subsample, and we report cell-by-cell Anderson-Rubin confidence sets confirming that within the no-turnover CRAS cell, the IV coefficient is not identified at conventional precision. The Wald test of equality between turnover and no-turnover first-stage coefficients rejects, and three quarters of the manager changes captured by our definition bring in individuals new to the municipal welfare bureaucracy rather than internal reshuffles.

The federal authorization causes substitution at incumbent CRAS units without causing new unit openings; the substitutes the shock attracts treat the post differently from inherited managers. Substitutes are not pre-selected on prior partisan affiliation, but they run for elected office at higher rates than stayers, mostly for city council, and they file new affiliations with parties in the mayor's electoral coalition rather than joining the mayor's party.

These findings place the paper in conversation with four related literatures. We speak first to the political economy of cash transfer programs. A large body of work shows that incumbents benefit electorally from the expansion of social transfers ([Manacorda et al., 2011](#); [De La O, 2013](#)). Much of this literature studies settings in which

local discretion is deliberately minimized, relying on centrally determined thresholds, randomized assignment, or other exogenous variation to identify electoral effects. We complement this work by studying a setting in which the program architecture is centralized but the last mile of delivery is not. Our findings suggest that centralizing transfer rules and payment flows is not sufficient to depoliticize redistribution when access to the program depends on a locally controlled registration bureaucracy. Even when mayors cannot alter eligibility criteria or benefit levels, control over the administrative pipeline through which households gain program access can still generate measurable electoral returns.

Our paper is especially closely related to [Frey \(2019\)](#), but the two designs identify electoral returns under sharply different degrees of mayoral involvement in the implementation chain. Frey exploits a discontinuity in federal funding for the *Family Health Program*, a primary-care program in which teams (a doctor, a nurse, an assistant nurse, and community health agents) follow a federally specified protocol of routine home visits to poor households. His mechanism runs through an information channel embedded in those visits: as part of the protocol, the teams disseminate *Bolsa Família* eligibility information, and new enrollment follows mechanically, without any discretionary action by the municipal executive. We exploit the complementary margin. In our setting, converting expanded federal authorization into realized enrollment requires active effort from the municipal welfare staff that the mayor hires, deploys, and directs to identify, locate, and register eligible families; the channel is the local administration itself. The two designs also identify on largely disjoint subsamples: Frey’s variation concentrates in small, low-development municipalities crossing the funding threshold; ours identifies on the complementary set with room to expand under the 2009 quota reform. Within his sample, coverage growth driven by an outreach channel that does not require mayoral effort reduces incumbency advantage and shifts the composition of challengers toward less clientelistic parties. Within ours, coverage growth driven by a channel that requires mayoral effort produces positive returns for the incumbent.

Second, we contribute to the literature on political accountability under shared governance. A central question in this literature is whether voters can correctly attribute responsibility when multiple levels of government jointly produce policy outcomes ([Przeworski et al., 1999](#); [Besley, 2006](#)). [Ferraz and Finan \(2008\)](#) show that voters punish corrupt mayors when randomly assigned audits reveal misuse of federal funds, especially where local media facilitate information transmission. [Avis et al. \(2018\)](#) show that audits also reduce corruption through a legal disciplining channel. Our setting differs because the relevant information concerns positive delivery rather than malfea-

sance. Voters observe the expansion of a valued program and update their beliefs about the incumbent's competence and effort. The mechanism we document is therefore closer to credit claiming than to sanctioning, but it operates through the same logic: voters use observable policy outcomes to infer politician type. In that sense, our results speak to a broader literature on attribution in multi-level settings by showing that voters can reward local incumbents even when the policy is federally funded and formally centralized, so long as local effort remains salient in implementation.

Third, we contribute to the literature on state capacity and the bureaucratic foundations of policy delivery. [Besley and Persson \(2009, 2011\)](#) establish state capacity as a precondition for credible policy delivery, and [Finan et al. \(2017\)](#) survey the personnel and bureaucratic infrastructure through which public programs are implemented. Our findings speak to a basic margin within this framework: the *existence* of local administrative infrastructure. Municipalities without a pre-existing CRAS converted essentially none of their quota shock into additional beneficiaries; municipalities with a CRAS translated the shock into expansion at a material rate. The electoral consequences inherit the presence threshold: the political returns to federal policy accrue only where local capacity exists to channel them.

Fourth, conditional on the presence of a delivery apparatus, our within-CRAS findings on bureaucratic turnover speak to the literature on the selection and politicization of frontline bureaucrats. The 2009 federal authorization shock causes turnover of the *de facto* CRAS manager, and the substitutes it attracts are not pre-selected on prior partisan affiliation. Two complementary mechanisms rationalize the pattern. First, program expansion raises the *de facto* attractiveness of the managerial post and pulls a higher-ability pool of candidates into it, consistent with [Dal Bó et al. \(2013\)](#). Second, once in the post, those individuals run for elected office at higher rates than stayers and align with the mayor's electoral coalition rather than switching to the mayor's exact party; this echoes [Colonnelli et al. \(2020\)](#), in which public-sector positions serve as channels into electoral careers, though our finding is sign-reversed since substitutes are not pre-selected on partisan ties. The pattern thus departs from the loyalist-allocation equilibrium in [Robinson and Verdier \(2013\)](#) and from the hiring-discretion turnover in [Akhtari et al. \(2022\)](#): here a federal program shock drives the turnover, and politicization happens downstream rather than at hiring. [Frey and Santarrosa \(2025\)](#) corroborate this reading: *Bolsa Família* interviewers who remain in the post align politically with the mayor while serving, with the effect concentrated where program exposure is highest. The position itself becomes politicized as the program expands; individuals are drawn into the post by the visibility the program creates, and once there they end

up in politics.

The remainder of the paper is organized as follows. Section 2 describes the institutional setting of *Bolsa Família*, the quota system, the 2009 reform, and the data sources. Section 3 develops the conceptual framework: an augmented model of the federal-to-local implementation chain that isolates two binding margins for delivery, the presence of local welfare infrastructure and the alignment of front-line bureaucrats, and predicts that the electoral return to program expansion concentrates where both are active. Section 4 presents the empirical strategy, defines the instrument, and discusses the identifying assumptions. Section 5 reports the main results, the robustness checks, and the heterogeneity decomposition. Section 6 tests the framework’s mechanism predictions against the data and documents the chain of evidence on the substitute managers the shock attracts. Section 7 concludes.

2 Background and Data

This section describes the *Bolsa Família* program, focusing on institutional aspects essential for understanding its local implementation dynamics and political implications. The first subsection lays out the program’s design, the registration process, the quota mechanism, and the 2009 federal reform that generates our identifying variation. The second introduces the four data sources we combine, and the third reports the descriptive statistics used throughout the paper.

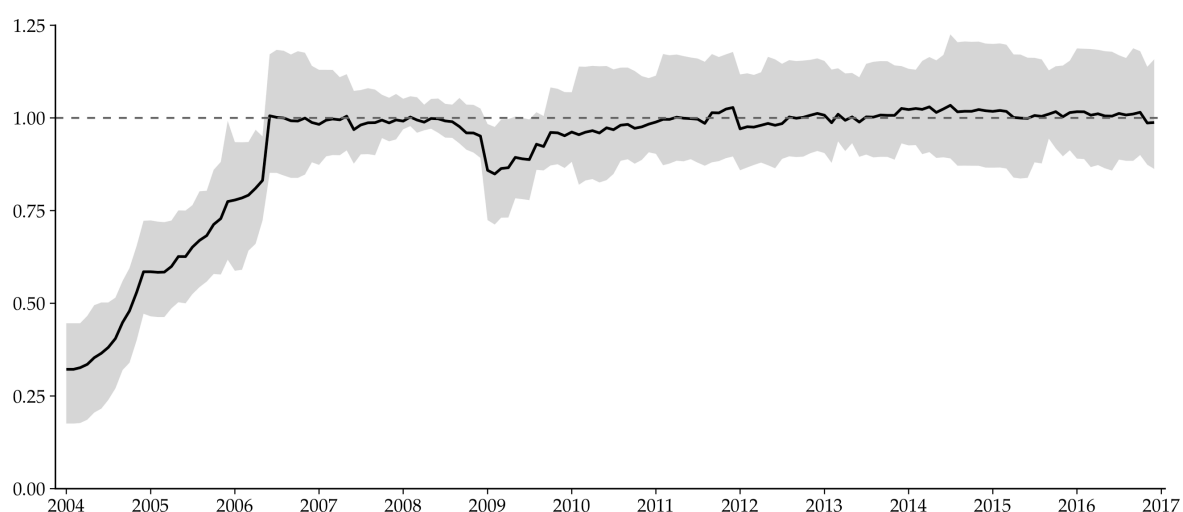
2.1 Institutional Background

Launched in 2004, *Bolsa Família* consolidated several pre-existing social assistance programs into a single conditional cash transfer system, becoming Brazil’s flagship poverty alleviation policy. The program provides monthly transfers to families below two income thresholds. Households with per capita income at or below R\$68 per month were classified as extremely poor and received an unconditional basic benefit of R\$68, plus conditional variable payments of R\$22 per child under 16 (up to three) and R\$33 per adolescent aged 16–17 (up to two), conditional on school attendance and routine health checkups. Families with per capita income between R\$68 and R\$137 qualified only for the child- and adolescent-linked variable benefits. The maximum monthly transfer per family was R\$200 and the average benefit stood at approximately R\$95. By September 2009, the program reached approximately 12.4 million families, or roughly a quarter of the Brazilian population.

Central to the program's operations is the *Cadastro Único* (*CadÚnico*), a national registry of low-income households that serves as the gateway to *Bolsa Família* and dozens of other social policies. The registry targets families with per capita income below half the minimum wage (R\$255 in 2010), a threshold substantially higher than the program's own eligibility lines, so that the pool of registered families considerably exceeds the set of actual beneficiaries.

Eligibility criteria are nationally uniform, benefit values are fixed by federal decree, and payments are processed and deposited directly into beneficiaries' accounts by the state-owned bank *Caixa Econômica Federal*. Mayors cannot alter transfer amounts, redirect funds, or withhold payments. What municipal governments control is the registration process itself.

Figure 1: Realized Enrollment as a Share of Authorized Quota



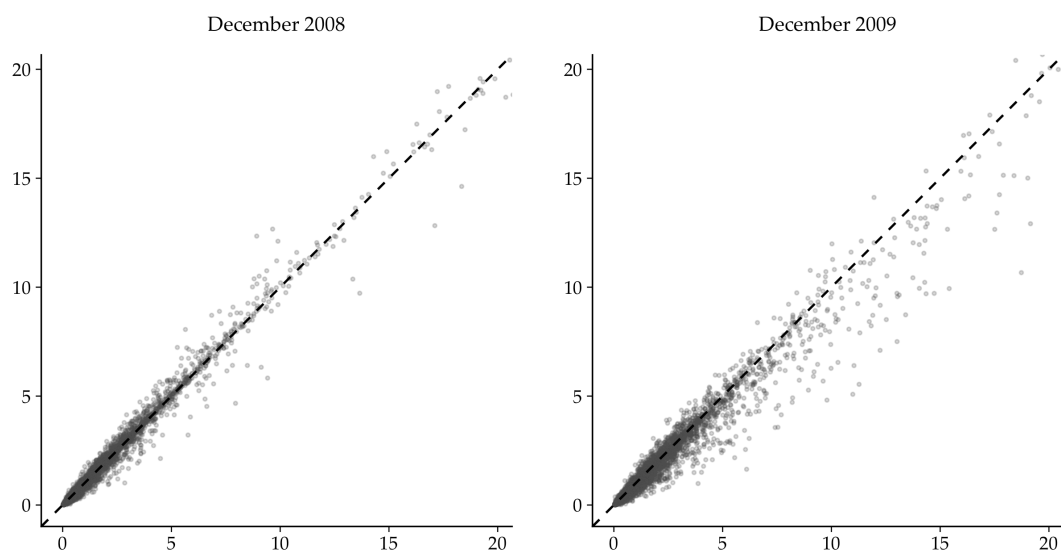
Note: Each point on the line is the cross-municipal mean of the ratio of realized *Bolsa Família* beneficiaries to the authorized quota in a given month, with each municipality weighted by its monthly authorized quota. The shaded band spans the weighted 25th to 75th percentile. The dashed horizontal line marks full take-up (ratio = 1). Before 2007, municipalities operated well below their authorized ceilings; the sharp convergence toward 1 reflects the rapid administrative build-up of the program in its first years. After 2007, the ratio fluctuates around 1, with a pronounced dip in mid-2009 as the reform's expanded quotas outpaced realized enrollment, the identifying variation we exploit.

Registration in the *Cadastro Único* requires a face-to-face interview: a social worker fills out a standardized questionnaire covering household composition, housing conditions, assets, employment, and income. In urban areas, families typically present themselves at a *Centro de Referência de Assistência Social* (CRAS). In rural and remote areas, however, eligible families cannot realistically be expected to travel to a registration point. There, coverage depends on active outreach: municipal teams must physically locate potential beneficiaries, travel to communities with limited road access, no tele-

phone coverage, and no internet, and conduct household interviews on site. Federal regulations encourage this practice by recommending that interviews take place at the family's home, both to facilitate verification and to reach populations that would otherwise remain invisible to the registry. In 2009, there were approximately 5,800 CRAS units distributed across roughly 4,000 of Brazil's 5,568 municipalities, and all data collection was done on paper.

Eligibility alone does not guarantee benefit receipt. Access to *Bolsa Família* depends on the availability of municipal quotas: caps on the number of beneficiaries per municipality, defined nationally and revised approximately every three years based on poverty estimates derived from census and survey data. These quotas function as official upper bounds on municipal enrollment, but the constraint is not always immediately binding. In practice, municipalities frequently operate near their ceiling, oscillating modestly above or below it from month to month before the federal priority mechanism reallocates unused slots.

Figure 2: Authorized Quota vs. Realized Enrollment

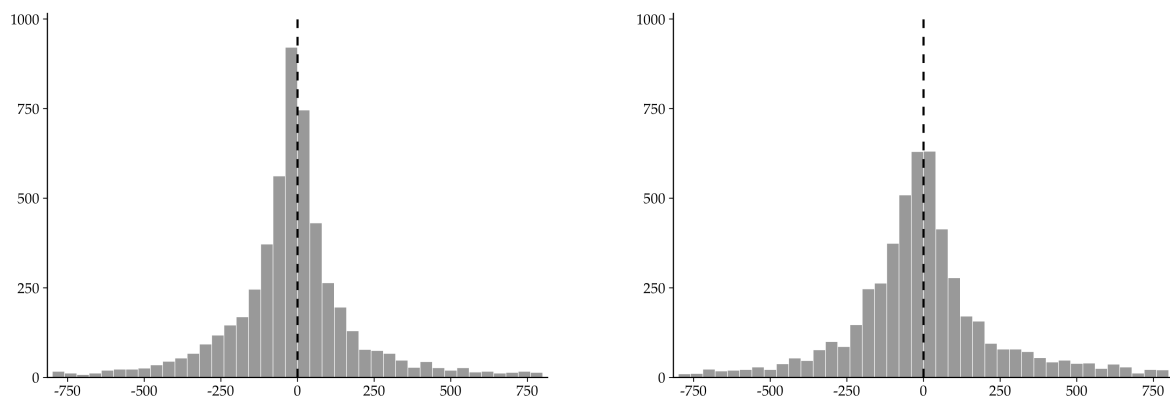


Note. Each dot is a municipality. The horizontal axis shows the federally authorized quota and the vertical axis shows the number of families actually enrolled in *Bolsa Família*, both in thousands. The dashed line marks the 45-degree locus of full take-up (realized = authorized). In December 2008, most municipalities lie close to or slightly below the 45-degree line, indicating that authorizations were approximately binding. In December 2009, following the federal quota reform, a substantial mass of municipalities shifts to the right of the 45-degree line, reflecting the expansion of authorized quotas that had not yet been translated into realized enrollment. The gap between the 45-degree line and the scatter in 2009 is the slack that our instrument exploits. Both axes are capped at the 99th percentile of the cross-municipal distribution (approximately 20,000 families); the small number of large urban municipalities above this threshold are excluded from the frame but not from any analysis.

This priority index considers each municipality's ratio of current beneficiaries to its

quota and the number of eligible-but-unserved families, creating a ratchet that, over time, tends to bring enrollment in line with the quota. The key implication is that the quota is neither a guarantee nor a purely notional figure: it constrains enrollment when binding, yet effective coverage also depends on the administrative initiative of local governments, creating a discretionary margin where political will and organizational capacity determine how many eligible families actually receive benefits.

Figure 3: Distribution of Slack Between Authorized Quota and Realized Enrollment



Note. Each panel shows the distribution of slack, defined as the authorized quota minus realized enrollment, across municipalities in December 2009, immediately following the federal reform. The left panel uses the official authorized quota; the right panel uses the counterfactual quota. Positive values indicate that a municipality received an authorization it had not yet translated into realized enrollment. The dashed vertical line marks zero slack. Both distributions are trimmed at the 5th and 95th percentiles to limit the influence of extreme observations. The mass of positive values in both panels confirms that the reform expanded quotas beyond contemporaneous enrollment levels for a substantial share of municipalities.

In 2009, the federal government revised the methodology used to compute municipal quotas, producing a heterogeneous reallocation of authorized beneficiary slots across municipalities. The previous approach, in place since 2006, distributed slots in proportion to each municipality's estimated number of poor families from the 2000 Census, adjusted for intercensal population growth. The 2009 revised methodology accounts for income volatility among families near the eligibility line, a feature of transient poverty that earlier projections systematically undercount ([Gerard et al., 2025](#)).

Alongside the methodological update, all municipal quotas were scaled by a factor of 1.18 to account for structural income volatility near the poverty threshold. Because this multiplicative factor is constant, it contributes no cross-municipal variation to the instrument; all identifying variation in the quota shock derives from the updated municipal poverty estimates and the change in income thresholds. Together, these changes added nearly 1.6 million new authorized slots in 2009. The reform expanded the ceiling, not the registry: although already-registered eligible families were absorbed auto-

matically, the magnitude of the expansion left a large residual margin of unfilled slots whose realization depended on the municipal government's effort to identify, locate, and register eligible families not yet in the federal database.

2.2 Data Sources

Our empirical analysis combines four municipal-level data sources: administrative records on *Bolsa Família*, electoral returns, census and household-survey data, and administrative records on local social-assistance infrastructure.

We begin with monthly administrative records from the *Ministério do Desenvolvimento Social* on *Bolsa Família* quotas, realized coverage, and total transfers from 2004 to 2016. Our main analysis uses the 2008 and 2012 windows to construct the treatment and instrument variables; the earlier (2004) and later (2016) years are reserved for placebo and persistence robustness exercises reported in the appendix. These data allow us to measure both authorized and realized program expansion.

We then use electoral records from the *Tribunal Superior Eleitoral* (TSE) covering municipal elections from 2000 to 2016. These data provide candidate-level vote totals, turnout, and detailed candidate characteristics, including education, prior occupation, and party affiliation.

To reconstruct counterfactual municipal quotas under the pre-reform allocation rule, we combine the 2000 Population Census from the *Instituto Brasileiro de Geografia e Estatística* (IBGE) with *Pesquisa Nacional por Amostra de Domicílios* (PNAD)-based estimates. These sources provide the inputs needed to approximate the quota formula that would have prevailed absent the reform.

Finally, we use administrative records from the *Censo SUAS* covering the universe of CRAS units in 2008 and 2009. In addition to information on the identity and tenure of directors and staff, these data include measures of local infrastructure and service provision, such as whether units conduct interviews and active searches, as well as the size of the population they serve.

2.3 Descriptive Statistics

The sample consists of 2,105 Brazilian municipalities observed in the 2008 and 2012 elections, those where the 2008 incumbent sought reelection. Table 1 reports descriptive statistics for the full population of municipalities and for this regression sample.

Three features stand out. The 2008 winner’s vote share averages 56.3 percent in the full population and 52.9 percent in the regression sample; the tighter dispersion reflects the rerunning restriction, which selects against incumbents at the extremes of the 2008 outcome distribution. *Bolsa Família* enrollment and the quota shock are both heavily right-skewed (BF mean 1,936 vs. median 947 in 2009), so both enter the regressions standardized. Census 2000 covariates have nearly identical means across panels, indicating the rerunning restriction does not select on the baselines we control for.

Table 1: Descriptive Statistics

	N	Min	Mean	Median	Max	SD
Panel A: All Municipalities						
Incumbent Won (2012)	4,730	0.00	55.35	100.00	100.00	49.72
Winner Vote Share (2008) (pp)	11,008	20.44	56.26	53.95	100.00	13.98
Realized Additional BF Expansion (2009)	11,008	-9,845.50	146.41	25.25	34,402.17	889.31
Excess Additional Beneficiaries (2009)	11,008	-5,282.54	378.03	144.65	113,606.29	1,990.32
Population (2000)	11,008	795.00	30,061.56	10,313.00	10,277,518.00	181,976.90
Gini (2000) (%)	11,008	30.00	54.73	55.00	87.00	6.82
Poor Population (2000) (%)	11,008	5.17	63.96	65.77	99.00	20.69
Income per capita (2000) (R\$)	11,008	62.65	338.02	308.11	1,759.76	192.21
Formal Workers Population (2000) (%)	11,008	1.92	36.07	34.44	86.38	18.13
Urban Population (2000) (%)	11,008	0.00	58.88	59.53	100.00	23.35
Panel B: Regression Sample						
Incumbent Won (2012)	4,210	0.00	56.34	100.00	100.00	49.60
Winner Vote Share (2008) (pp)	4,210	24.06	52.92	52.69	99.16	9.41
Realized Additional BF Expansion (2009)	4,210	-889.25	145.41	27.83	34,402.17	947.24
Excess Additional Beneficiaries (2009)	4,210	-5,282.54	349.80	145.78	50,052.59	1,446.66
Population (2000)	4,210	795.00	27,876.69	10,556.00	5,773,130.00	145,264.20
Gini (2000) (%)	4,210	30.00	54.84	55.00	80.00	6.73
Poor Population (2000) (%)	4,210	7.66	64.10	66.78	99.00	20.97
Income per capita (2000) (R\$)	4,210	69.20	337.21	300.12	1,759.76	195.03
Formal Workers Population (2000) (%)	4,210	2.89	35.78	33.99	86.38	18.28
Urban Population (2000) (%)	4,210	0.00	58.77	59.20	100.00	23.30

Notes: Panel A covers the universe of municipalities with a 2008 winner vote share and complete Census 2000 baseline controls; Panel B is the regression sample, municipalities where the 2008 incumbent sought reelection, subject to the data availability. Realized Additional BF Expansion is the muni-level inter-election change in the twelve-month moving average of beneficiary families between 2008 and 2012. Excess Additional Beneficiaries is the difference between the 2009 federal quota and the counterfactual quota, the quota that would have prevailed under the pre-reform allocation rule. The large standard deviation of this variable relative to its mean reflects substantial right skew driven by large urban municipalities; the regressions use a standardized version. The SD is smaller in Panel B than in Panel A because the regression sample drops municipalities without an incumbent rerunner. Full distributional statistics are reported in Appendix B.3.

3 Conceptual Framework

This section develops a simple framework for the political returns to welfare expansion in low-income settings like the one we study. The framework has two moving parts: a

multiplicative implementation chain in which federal authorization can only translate into realized expansion when the local administrative apparatus is in place, and a two-channel voter response in which beneficiaries reward the program materially and all voters update on the incumbent’s competence from observed implementation.

We proceed in three steps. The benchmark is a standard political-agency model in which local capacity is a single smoothly distributed index. The augmented model decomposes capacity into two institutionally grounded margins, physical infrastructure (the local CRAS apparatus) and bureaucratic alignment of the front-line manager, combined multiplicatively; the headline first stage is essentially identified where both margins are active. We close with a voter-response block that delivers the linear vote-share form and a partisan-attribution extension under which the inference channel sharpens when the mayor’s party also operates the federal program. The framework gives a parsimonious structure for interpreting the heterogeneity exercises, the partisan asymmetry, and the mechanism analysis that follows.

3.1 Agents

There are three classes of agents: a federal government, a continuum of municipalities indexed by m , each governed by an incumbent mayor, and a continuum of voters in each municipality indexed by i . Time runs over two periods, $t \in \{0, 1\}$, corresponding to the 2008 and 2012 municipal elections. The 2009 reform of *Bolsa Família*’s allocation formula occurs between the two periods.

Federal government. The federal government is non-strategic. It sets two policy parameters that are exogenous from the perspective of municipalities. First, it sets a per-family transfer value $\bar{b}$, uniform across municipalities and possibly differing between $t = 0$ and $t = 1$.⁶ Second, it sets a beneficiary quota Q_m^t for each municipality, computed from a national allocation rule based on census poverty estimates and intercensal population growth. We treat Q_m^t as exogenous and write the inter-election quota shock as

$$\Delta Q_m \equiv Q_m^1 - Q_m^{1,cf} \quad (1)$$

⁶Strictly, $\bar{b}$ summarizes a federally fixed schedule that pays a basic benefit to extremely poor families plus variable supplements per child under 16 and per adolescent aged 16–17. The schedule is uniform across municipalities and not subject to mayoral discretion, so $\bar{b}$ can be read either as a family-level transfer under a representative composition or as the population average under the local demographic distribution. Neither reading changes what follows, since the mayor cannot alter the schedule.

where $Q_m^{1,cf}$ is the counterfactual quota that would have prevailed under the pre-reform allocation rule.

Mayor. The incumbent mayor in municipality m inherits an administrative capacity K_m^0 fixed before period $t = 0$, parameterized by a legacy floor $\psi_0 \geq 0$, the passive utilization that operates through informal channels and prior administrative pathways even without a dedicated apparatus, and a ceiling $\bar{\psi} > 0$ that an idealized CRAS at full operational intensity would add on top of the floor.

Conditional on this inherited capacity, the mayor chooses discretionary effort $a_m \geq 0$ (active outreach by welfare staff, opening additional service points, mobilizing the registration apparatus, and pushing data updates through the federal pipeline) at a marginal cost that depends on his unobserved type $\theta_m \in \{0, 1\}$ representing competence, in the political-agency tradition surveyed by [Ashworth \(2012\)](#):

$$c(\theta_m) = c_0 - c_1 \cdot \theta_m, \quad c_0 > c_1 > 0. \quad (2)$$

Type is drawn by nature, observed by the mayor, and unobserved by voters; competent mayors thus face a lower marginal cost of effort and choose strictly higher a_m in equilibrium.

Voters. Voters care about two things: whether they receive the transfer, and whether the incumbent is competent.⁷ The material channel reflects the documented response of beneficiaries to direct programmatic transfers in cash-transfer settings ([Manacorda et al., 2011](#); [Zucco, 2013](#); [Avis et al., 2018](#)). The competence channel follows the retrospective-voting tradition: voters use observed government performance to update their beliefs about an unobserved politician type. Voter i 's utility differential from supporting the incumbent versus the challenger is

$$\Delta U_{im} = \alpha d_{im} + \delta \Pr(\theta_m = 1 \mid s_{im}) + \eta_{im}, \quad (3)$$

where $d_{im} \in \{0, 1\}$ indicates whether household i in municipality m is a beneficiary, s_{im} is a noisy signal of program implementation, $\alpha > 0$ is the material reward to receiving the transfer, $\delta > 0$ is the value voters place on competent incumbents, and η_{im} is an

⁷The model is agnostic about voter sophistication in separating federal from local contributions. The local component of delivery is directly observable: voters see CRAS operating, *busca ativa* happening, and registrations being processed locally. Whether voters additionally separate the federal component (benefit value, set by decree) is immaterial for the electoral effect: at most, a less sophisticated voter would also misattribute federal contributions, but the locally observable activity remains the proximate trigger.

idiosyncratic ideology shock.

Aggregate vote share. Aggregating individual choices across the continuum of voters under probabilistic voting in the tradition of [Lindbeck and Weibull \(1987\)](#), with the idiosyncratic shocks η_{im} smoothing across the ideology distribution, the muni-level vote share for the incumbent takes the linear form

$$V_m = V_0 + \lambda_1 B_m + \lambda_2 e_m, \quad (4)$$

where $\lambda_1 > 0$ aggregates the material channel, with each beneficiary reducing resistance to supporting the incumbent through α in equation (3), and $\lambda_2 \geq 0$ aggregates the inference channel, with visible implementation raising the posterior $\Pr(\theta_m = 1)$ for all voters, beneficiaries and non-beneficiaries alike.

3.2 Benchmark Model: Single Capacity Index

In the benchmark, capacity K_m^0 is summarized by a scalar $\psi_m \in [0, \infty)$, the inherited capacity that translates into effort-free quota utilization. Realized program coverage is

$$e_m = \psi_m + a_m, \quad B_m = e_m \cdot Q_m, \quad (5)$$

where e_m is the quota utilization rate and B_m is the number of realized beneficiaries.

The mayor's objective combines the electoral payoff, the convex cost of discretionary effort, and an institutional compliance pressure that penalizes deviations from full quota utilization, reflecting the federal monitoring of municipal performance against the authorized ceiling:

$$U_m = V_m - \frac{1}{2}c(\theta_m)a_m^2 - \frac{1}{2}\tau(e_m - 1)^2, \quad (6)$$

where V_m is the muni-level vote share from equation (4) and $\tau > 0$ parameterizes the enforcement pressure to remain near the quota benchmark.

Proposition 1 (Optimal Effort and Realized Expansion). *In the benchmark model, the unique optimal discretionary effort is*

$$a_m^*(\theta_m) = \frac{(\lambda_1 Q_m + \lambda_2) + \tau(1 - \psi_m)}{c(\theta_m) + \tau}. \quad (7)$$

Equilibrium realized coverage and beneficiaries are

$$e_m^*(\theta_m) = \psi_m + a_m^*(\theta_m), \quad B_m^*(\theta_m) = e_m^*(\theta_m) \cdot Q_m. \quad (8)$$

The first stage of the empirical design satisfies

$$\frac{\partial B_m^*}{\partial Q_m} = e_m^*(\theta_m) + Q_m \cdot \frac{\lambda_1}{c(\theta_m) + \tau} > 0 \quad (9)$$

for all municipalities with $\psi_m \geq 0$, $Q_m > 0$, and $a_m^* > 0$.

Proof. The argument is straightforward for the attentive reader: standard first-order conditions in a_m yield the expression above. Full details are presented in Appendix F. ■

The benchmark model predicts that the federal quota shock expands realized enrollment in every municipality, with the magnitude of expansion varying smoothly with capacity ψ_m , the size of the quota Q_m , and the mayor's competence θ_m . Heterogeneity in the first stage is therefore a matter of degree rather than of kind: the partial derivative $\partial B_m^*/\partial Q_m$ is strictly positive in every municipality with $\psi_m \geq 0$, $Q_m > 0$, and $a_m^* > 0$, so within-subgroup first stages cannot collapse to zero. The benchmark cannot accommodate a partition in which one subgroup's first stage approaches zero while another remains strongly identified, since $\partial B_m^*/\partial Q_m$ remains bounded below by the positive effort-response term $Q_m \cdot \lambda_1/[c(\theta_m) + \tau]$ even at the corner $\psi_m = a_m^* = 0$. Combined with $\lambda_1 > 0$ in equation (4), the model also predicts a positive headline 2SLS coefficient on incumbent vote share.

3.3 Augmented Model: Multiplicative Capacity

The benchmark imposes a strong restriction: local capacity is a single smooth scalar. The personnel-economics-of-the-state literature (Finan et al., 2017; Akhtari et al., 2022; Colonnelli et al., 2020) pushes back against this: local administrative output depends jointly on physical infrastructure and on the political incentives of the front-line staff who operate it. We therefore enrich the benchmark by allowing capacity to factorize along two institutionally grounded margins.

Infrastructure component. We write the first capacity component as $\kappa_m^I \in \{0, 1\}$ indicating whether municipality m had a *Centro de Referência de Assistência Social* (CRAS) in 2008. The CRAS is a ready-made bureaucratic apparatus for enrolling households

into social programs, with permanent staff, standardized protocols, and a direct data pipeline into the *Cadastro Único*. Finan et al. (2017) place this kind of pre-existing local infrastructure at the heart of the state-capacity literature: front-line administrative output is shaped by the agencies and personnel a municipality has on hand to execute federal mandates. Municipalities without a CRAS can still enroll new beneficiaries through alternative channels (public hospitals, mobile *cadastro* teams, ad hoc active outreach), but the productivity of those channels is lower than what an existing CRAS delivers. The same federal authorization therefore translates into less realized expansion where the apparatus is missing, and more where it is in place.

Alignment component. We write the second capacity component as $\kappa_m^A \in \{0, 1\}$ indicating whether the CRAS manager (the *coordenador* of the unit) was replaced after the 2008 municipal election, when the newly elected mayor took office. Two independent literatures motivate the relevance of front-line manager replacement as a productivity margin. Akhtari et al. (2022) document that mayoral transitions in Brazil systematically trigger turnover of street-level bureaucrats, with replacements concentrated in positions where the mayor has hiring discretion, including managers of social-assistance and education facilities. Colonnelli et al. (2020) document, in the same Brazilian setting, that public-sector positions function as electoral stepping stones, with their attractiveness as career-launching pads tied to the operational footprint of the office. Both readings imply that a post-2008 manager change can shift the productivity of the implementation chain: under the first, through replacement with bureaucrats more responsive to the incoming mayoral coalition, and under the second, through replacement with politically ambitious agents who exert more effort because program expansion raises the visibility of the post and its attractiveness as a launching pad for a local electoral career. The model treats κ_m^A as a generic productivity margin consistent with both readings.

Production with multiplicative capacity. The implementation chain that translates federal authorization into realized expansion has two indispensable capacity components: an existing registration apparatus and an active front-line manager. Kremer (1993) and Jones (2011) characterize technologies of this form: the productivity of the chain is the product of component qualities, so the failure of any single component governs the productivity of the whole.

We adopt this structure with a productivity floor $\gamma \in [0, 1)$ to allow attenuation

rather than literal collapse when one capacity component is weak. Let

$$g(\kappa_m^I, \kappa_m^A) = \gamma + (1 - \gamma) \kappa_m^I \kappa_m^A, \quad \gamma \in [0, 1], \quad (10)$$

so that $g = 1$ when both capacity components are active and $g = \gamma$ otherwise. Realized coverage takes the form

$$e_m = \psi_0 + g(\kappa_m^I, \kappa_m^A) \cdot (\bar{\psi} + a_m), \quad B_m = e_m \cdot Q_m, \quad (11)$$

where ψ_0 and $\bar{\psi}$ are the floor and ceiling defined above. The strict Leontief case is the limit $\gamma = 0$; with $\gamma > 0$, subgroups that lack one capacity component still exercise some quotas, just at the attenuated rate g .

The mayor's objective remains as in equation (6), with V_m now driven by the augmented e_m and B_m from equation (11). Solving yields the following characterization.

Proposition 2 (Optimal Effort and Attenuated First Stage). *In the augmented model with production (11) and multiplier (10), optimal discretionary effort $a_m^*(\theta_m)$ is strictly increasing in the multiplier $g(\kappa_m^I, \kappa_m^A)$. The first stage satisfies*

$$\frac{\partial B_m^*}{\partial Q_m} = e_m^*(\theta_m) + Q_m \cdot \frac{\lambda_1 g^2}{c(\theta_m) + \tau g^2}, \quad (12)$$

which is strictly increasing in g . As $\kappa_m^I \kappa_m^A \rightarrow 0$ the first stage attenuates by a factor of order γ^2 relative to the active subgroup. The attenuation operates on the marginal-productivity term and propagates through equilibrium effort.

Proof. The argument is straightforward for the attentive reader: standard first-order conditions in a_m combined with the chain rule applied to production (11) yield the expression above. Full details are presented in Appendix F. ■

The test against the benchmark is sharp on signs and orders of magnitude. The static-response component of equation (12) attenuates linearly in γ through $g\bar{\psi}$, while the effort-response component attenuates more steeply, by γ^2 , because the multiplier enters the effort margin twice: a weaker chain dampens how much each unit of effort raises coverage, and it also blunts the mayor's incentive to exert effort in the first place. Under the assumption that ψ_0 is small relative to $\bar{\psi}$, the ratio of inactive- to active-subgroup first-stage slopes is bounded between γ and γ^2 . A within-subgroup first-stage ratio close to one rejects the augmented model; any substantially below one is consistent with it and provides a range estimate of γ .

The multiplier $g(\kappa_m^I, \kappa_m^A) = \gamma + (1 - \gamma) \kappa_m^I \kappa_m^A$ treats the two capacity margins symmetrically: a single γ governs the attenuation when either component is inactive. The

framework therefore makes no prediction that infrastructure and alignment must attenuate at the same rate; if the data return distinct γ_{κ^I} and γ_{κ^A} , this should be read as an empirical refinement consistent with the framework rather than as an additional model prediction.

The augmented model preserves the benchmark’s headline-level predictions, both a positive first stage on the full sample and a positive 2SLS coefficient on vote share, but predicts a sharply different shape of heterogeneity. Where the benchmark predicts smooth attenuation across subgroups, the augmented model predicts that the first stage essentially collapses where any structural margin is missing. In municipalities without a pre-existing *CRAS*, the implementation chain has no apparatus to translate authorized slots into realized expansion. Inside *CRAS* municipalities, the same logic applies to manager turnover: where the front-line operator was not replaced after the 2008 municipal election, the chain operates at the floor productivity γ rather than at full intensity. The active cell, where both margins are simultaneously present, is where the headline 2SLS coefficient is essentially identified.

3.4 Voter Response

We close the model by characterizing how realized expansion translates into vote share. Voters do not observe the mayor’s type θ_m directly; instead, they form beliefs about competence from observable program implementation outcomes, in the spirit of the retrospective-voting tradition of [Ferejohn \(1986\)](#) and the more recent literature on voters learning from government performance in developing democracies ([Ferraz and Finan, 2008](#); [Avis et al., 2018](#)). Each voter i in municipality m observes a noisy signal

$$s_{im} = \nu_0 + \nu_1 d_{im} + \nu_2 e_m + \varepsilon_{im}, \quad \nu_1, \nu_2 > 0, \quad (13)$$

where ε_{im} is mean-zero noise independent across voters and conditional on e_m . The first informative component, $\nu_1 d_{im}$, captures the voter’s direct experience as a beneficiary. The second, $\nu_2 e_m$, captures aggregate visibility of program delivery (signage, neighborhood-level enrollment campaigns, local media coverage).

Bayesian updating closes the model in the standard career-concerns and political-agency tradition of [Holmström \(1999\)](#) and [Maskin and Tirole \(2004\)](#), surveyed by [Ashworth \(2012\)](#). Voters apply Bayes’ rule using the priors $p_0 = \Pr(\theta_m = 1)$ and the conditional signal densities f_0 and f_1 :

$$\Pr(\theta_m = 1 \mid s_{im}) = \frac{f_1(s_{im})p_0}{f_1(s_{im})p_0 + f_0(s_{im})(1 - p_0)}. \quad (14)$$

By Proposition 1, competent mayors choose strictly higher a_m^* than incompetent mayors and therefore generate higher e_m^* . This shifts the conditional mean of s_{im} upward under $\theta_m = 1$. For a wide class of noise distributions (specifically, those with log-concave densities), this location shift implies the monotone likelihood ratio property:

$$\frac{f_1(s)}{f_0(s)} \text{ is strictly increasing in } s. \quad (15)$$

Under (15), the posterior in (14) is strictly increasing in s_{im} . Voter inference is rational: a higher signal does indicate higher probability of competence, because in equilibrium higher signals are more likely to have been generated by competent mayors. Aggregating across voters in the probabilistic-voting tradition of Lindbeck and Weibull (1987), with η_{im} smoothing across the ideology distribution, the linear vote-share form posited in equation (4) is recovered as the equilibrium aggregate. The aggregation parameter $\lambda_1 > 0$ collects the material channel (αd_{im} across beneficiaries), and $\lambda_2 \geq 0$ collects the inference channel ($\delta \Pr(\theta_m = 1 \mid s_{im})$ across all voters, beneficiaries and non-beneficiaries alike).

Remark 1 (The two channels and the sign of the headline). *Equation (4) predicts $\partial V_m / \partial Q_m > 0$ whenever $\partial B_m^* / \partial Q_m > 0$ (Proposition 1 or 2) and either $\lambda_1 > 0$ or $\lambda_2 > 0$. The sign of the headline 2SLS coefficient therefore tests jointly whether the first stage operates and whether voters reward implementation.*

Partisan attribution and the inference channel. The inference parameter λ_2 aggregates the rate at which voters convert observed implementation into a posterior on the mayor's competence. A natural extension lets λ_2 vary across municipalities with the alignment between the local executive's partisan brand and the federal program's institutional sponsor. When the mayor's party is also the federal incumbent that designed and operates the program at the national level, the credit chain runs cleanly from local registration to national policy under a single partisan label; voters' Bayesian update on competence is sharper because the inferred type controls both ends of the implementation chain. Formally, decompose the inference parameter as $\lambda_{2,m} = \lambda_2^{\text{base}} + \lambda_2^{\text{attrib}}$. $\mathbf{1}\{\text{partisan alignment}_m\}$, with $\lambda_2^{\text{attrib}} \geq 0$. The augmented vote-share function becomes

$$V_m = V_0 + \lambda_1 B_m + \lambda_{2,m} e_m, \quad (16)$$

and the per-beneficiary electoral return is larger under aligned mayors by $\lambda_2^{\text{attrib}}$. The model thus predicts that, holding realized expansion fixed, a municipality governed by a mayor whose party owns the federal program returns a higher per-beneficiary vote share than one governed by a non-aligned mayor. This prediction is testable on the

partition that separates Workers’ Party mayors from the rest, under the identifying restriction that the material channel λ_1 is partisan-invariant; the restriction is institutional, since the federally set benefit value and eligibility rules make the marginal contribution of a beneficiary to αd_{im} independent of local partisan alignment. The partisan multiplier is therefore a feature of the model, not an unmodeled regularity.

4 Empirical Strategy

Our empirical strategy exploits a 2009 federal reform that revised the formula used to compute municipal *Bolsa Família* quotas. The reform generated plausibly exogenous variation in municipalities’ authorized beneficiary ceilings, but turning that authorization into actual enrollment depended on local implementation. The wedge between what each municipality was allowed to do and what it actually did is what lets us identify the political return to expansion. The analysis is a difference-in-differences design combined with two-stage least squares, and it uses the federally induced quota expansion as an instrument for realized changes in beneficiary enrollment.

4.1 The 2009 Quota Reform as Source of Identification

The methodology for computing municipal quotas was implemented by the *Ministério do Desenvolvimento Social* (MDS) based on census and survey data, following a standardized algorithm with no discretion at either the local or federal executive level. [Brollo et al. \(2020\)](#) provide direct evidence supporting this claim. They find evidence of manipulation in the enforcement of program conditionalities but no indication of political influence over the allocation of quotas themselves.⁸

To isolate the variation attributable solely to the change in allocation rules, we construct counterfactual 2009 quotas by applying the pre-reform methodology, which had been used to set the 2004 and 2006 quotas, to the data that would have been available immediately before the reform,

$$\text{CountQuota}_m^{2009} = \frac{\text{Poor}_m^{2000} \cdot n_m^{[2000,2006]}}{\sum_{k \in s} \text{Poor}_k^{2000} \cdot n_k^{[2000,2006]}} \cdot 1.18 \cdot \text{Poor}_s^{2006} \quad (17)$$

⁸[Gerard et al. \(2025\)](#) note that they cannot exactly reproduce the official MDS quota numbers using publicly available information and the pre-reform allocation formula. They do not interpret this discrepancy as evidence of political manipulation, and a more plausible explanation is that the remaining differences reflect unpublished administrative adjustments or small differences in auxiliary estimates used to operationalize the formula.

where $Poor_m^{2000}$ is the number of poor families in municipality m of state s from the 2000 Census, $n_m^{[2000,2006]}$ is an IBGE estimate of municipal population growth between 2000 and 2006, $Poor_s^{2006}$ is the state-level estimate of poor families from the 2006 *Pesquisa Nacional por Amostra de Domicílios* (PNAD), and the factor 1.18 reflects a uniform expansion applied to all municipal quotas as part of the reform.⁹ The actual 2009 quotas were computed by MDS using a revised methodology that incorporated updated poverty estimates and new income-volatility adjustments, and the exact formula is not publicly available. The preferred instrument is therefore

$$\Delta Quota_m^{2009} = Quota_m^{2009} - CountQuota_m^{2009} \quad (18)$$

This measure captures the cross-municipal variation in authorized expansion arising from the change in the federal allocation methodology, net of the uniform national expansion and local poverty differences. In the preferred specification, the counterfactual quota counts as poor any household with monthly per-capita income below R\$90 in the 2000 Census and below R\$120 in the 2006 PNAD. These were the official *Bolsa Família* eligibility thresholds in force during the pre-reform allocation rounds of 2004 and 2006.¹⁰ The 2009 reform announcement (MDS *Informe nº 169*) introduced an expanded R\$137 cutoff alongside the formula change. Using R\$137 inside the counterfactual would mix two distinct sources of variation in $\Delta Quota_m$, the change in formula (the policy-relevant variation we want to instrument with) and the mechanical change in the count of poor families that follows from a higher cutoff applied to the same PNAD micro-data, which is a definitional change unrelated to the formula's distributional logic. The R\$90/120 counterfactual instead holds the data definitions fixed at their pre-reform values, so $\Delta Quota_m$ isolates the formula component.¹¹

⁹The last officially reported **quota update** before the 2009 reform used information from the 2000 Census and the 2004 PNAD. To construct the counterfactual, we update the pre-reform formula using 2006 PNAD data, which is the most recent information that would have been available to policymakers at the time, while holding the allocation rule fixed.

¹⁰Validation against the official 2004 and 2006 quotas confirms that the R\$90 cutoff applied to the 2000 Census and the R\$120 cutoff applied to the 2006 PNAD jointly produce the closest replication of the actual pre-reform quota allocations. See Appendix B for details.

¹¹The R\$90/137 alternative is not just conceptually less clean but also empirically less stable. Under that construction, the leave-one-state-out exercise reveals identifying mass concentrated in Rio de Janeiro, and dropping that state collapses the Kleibergen-Paap F -statistic well below conventional thresholds. The preferred R\$90/120 instrument is stable in this dimension and across every other sample perturbation. Appendix C.3 documents the comparison.

4.2 Research Design

The design compares changes in incumbent electoral performance between 2008 and 2012 across municipalities that faced different intensities of authorized quota expansion. Identification follows the continuous-treatment IV framework of Angrist and Imbens (1995), which extends the binary-instrument LATE setup of Imbens and Angrist (1994) to instruments and treatments with multiple ordered values. Three assumptions carry the argument, and we discuss each in detail below. The first is conditional independence of the quota shock with respect to potential electoral outcomes. The second is the exclusion restriction, which says that the shock affects vote share only through realized beneficiary expansion. The third is monotonicity of the first stage. When all three hold, the residual variation in ΔQuota_m after the fixed effects and baseline-by-post controls isolates the causal effect of discretionary program expansion on subsequent electoral outcomes. We have only two election periods, which rules out formal pre-trend testing, so we probe conditional independence through a proportional-selection exercise that complements the placebo and balance tests that follow.

The primary challenge is the endogeneity of realized enrollment. Mayors influence how many families are actually registered, so coverage growth reflects local political incentives. Municipalities where incumbents face stronger electoral pressures may expand coverage more aggressively precisely because the expected political return is higher (Pereira et al., 2020). We address this by using the exogenous federal authorization shock as an instrument for realized beneficiary expansion. The data are organized as a balanced municipality-by-election-year panel covering the 2008 and 2012 elections, and we estimate the system

$$\text{Beneficiaries}_{mt} = \alpha_m + \lambda_t + \pi_1 (\Delta\text{Quota}_m \times \text{Post}_t) + X'_{mt}\pi_2 + \eta_{mt} \quad (19)$$

$$\text{VoteShare}_{mt} = \alpha_m + \lambda_t + \beta_1 \widehat{\text{Beneficiaries}}_{mt} + X'_{mt}\beta_2 + \varepsilon_{mt} \quad (20)$$

where α_m and λ_t are municipality and year fixed effects, Post_t is an indicator for the 2012 election, and X_{mt} bundles the incumbent-party fixed effects and the Census 2000 baseline characteristics interacted with Post_t . The variable $\text{Beneficiaries}_{mt}$ is the twelve-month moving average of enrolled families in municipality m measured at election year t . For $t = 2008$ the average runs over the twelve months of 2008, and for $t = 2012$ it runs over the twelve months of 2009, the year in which the quota shock materialized and induced the policy-relevant expansion.¹² The instrument ΔQuota_m is the time-

¹²Alternative definitions extending the beneficiary measurement window through 2012 yield similar

invariant 2009 quota shock, and it enters as $\Delta\text{Quota}_m \times \text{Post}_t$ because the shock itself is a single number per municipality and would otherwise be absorbed by the municipality fixed effects.

Equations (19) and (20) describe the data-generating process at the panel level, but the estimator we implement reads more transparently in first differences. With two periods, the municipality fixed effects α_m are time-invariant within municipality and drop out of the within-municipality demeaning, leaving the inter-election change as the operative variation. The system reduces to a cross-sectional IV at the municipality level,

$$\Delta\text{VoteShare}_m = \Delta\lambda + \beta_1 \Delta\widehat{\text{Beneficiaries}}_m + \Delta X'_m \beta_2 + \Delta\varepsilon_m \quad (21)$$

where $\Delta\text{Beneficiaries}_m$ is the change in the twelve-month moving average of enrolled families between the 2008 and 2012 elections and $\Delta\lambda$ is the difference in year fixed effects, a common time trend that absorbs whatever shifted electoral outcomes nationally between the two elections. The coefficient β_1 identifies the marginal effect of the muni-level change in beneficiaries between elections on the corresponding change in vote share, with the quota shock ΔQuota_m instrumenting the endogenous portion of $\Delta\text{Beneficiaries}_m$.

The two displays serve complementary purposes. Equations (19) and (20) state the structural model on the original panel, while equation (21) states the regression that delivers β_1 in practice.¹³ The two specifications coincide when $\text{Cov}(\text{Beneficiaries}_{m,2008}, \Delta\text{Quota}_m) = 0$, but in our setting that covariance is positive. The 2009 reform reweighted the formula toward the same dimensions of municipal poverty that already drove pre-reform coverage, so the post-interacted specification understates β_1 . The first-difference specification is the natural target when, as here, the substantive object is the political return to expansion rather than the political return to a higher 2012 level conditional on an unobserved 2008 baseline.

Both $\text{Beneficiaries}_{mt}$ and ΔQuota_m are standardized to mean zero and unit variance within the estimation sample, so coefficients in the tables read as the response to a one-standard-deviation move in the relevant variable. The two natural-unit scales that anchor the substantive interpretation are the muni-level “additional families” quantities reported in Panel B of Table 1. The cross-municipal standard deviation of the inter-election change in realized beneficiaries (the *Realized Additional BF Expansion* row) is 947 families, and the cross-municipal standard deviation of the 2009 quota shock (the

results in Appendix C.3.

¹³In the ANCOVA-style (McKenzie, 2012) alternative, the endogenous variable is interacted with Post_t and the first-difference reduction collapses to the 2012 level of beneficiaries rather than the inter-election change, and we discuss it in Appendix C.9.

Excess Additional Beneficiaries row) is 1,447 authorized families.

The first-difference IV in equation (21) therefore relates these two on a like-for-like scale, additional realized beneficiaries between elections instrumented by additional authorized beneficiaries from the 2009 reform, both measured at the municipality level. A 2SLS coefficient of one in standardized units corresponds to approximately 0.19 percentage points of vote share per typical inter-election expansion of 947 families, or about 0.20 percentage points per 1,000 additional beneficiaries enrolled between the two elections. The tables report estimates in standardized units, and the body text translates them into per-thousand-families terms.¹⁴

Treatment varies continuously in intensity rather than in binary status, so the design identifies an average causal response (ACR) parameter in the sense of Angrist and Imbens (1995). The ACR is a weighted average of the marginal effect of expansion on the change in vote share, where the non-negative weights $\omega(d)$ are proportional to $\text{Cov}(\mathbf{1}\{D_m \geq d\}, Z_m)$ and integrate to one. Under Angrist and Imbens’s assumptions of independence, exclusion, and monotonicity, β_1 is the population analog of $\int \omega(d) \mathbb{E}[\partial Y(d)/\partial d] dd$. Realized expansion is right-skewed, so these weights concentrate among municipalities where the 2009 reform translated into substantive enrollment changes, and the population weights further re-weight the estimand toward the experience of the typical Brazilian citizen rather than the typical municipality. Because no municipality is fully untreated in either period, the design does not identify level treatment effects (ATT or ATE), and β_1 is the marginal response, not the level response.¹⁵

Outcome. The main outcome is the incumbent’s vote share,

$$\text{VoteShare}_{mt} = \frac{\text{IncumbentVotes}_{mt}}{\text{VotesOnCandidates}_{mt}}$$

where $\text{VotesOnCandidates}_{mt}$ is the total number of valid votes cast for any candidate in municipality m in election year t , excluding blank and null ballots. This normalization ensures that the outcome reflects the incumbent’s competitive position among voters

¹⁴Standardization is an affine transformation that does not affect t -statistics, p -values, the Kleibergen-Paap first-stage F , or Anderson-Rubin confidence sets. $\text{Beneficiaries}_{mt}$ is standardized using panel-level dispersion (4,949 families), and the conversion factor to the inter-election scale is $947/4,949 \approx 0.19$. Appendix C.3 reports raw-unit results.

¹⁵The recent panel-DiD literature uses the same “ACR” label for related parameters identified under strong parallel-trends assumptions at every dose level (notably Callaway et al., 2024). We do not invoke this approach. Angrist and Imbens’s ACR is identified through instrument exogeneity and monotonicity rather than through dose-level parallel trends.

who expressed a preference for a candidate.¹⁶

Sample. The estimation sample is restricted to municipalities in which the incumbent mayor elected in 2008 sought reelection in 2012. After excluding municipalities where the incumbent did not seek reelection, those with missing treatment, outcome, or baseline covariate information, and those without contested elections in both periods, the final sample contains 2,105 of the 5,568 Brazilian municipalities (approximately 38 percent), observed in two election years and yielding 4,210 municipality-year observations. Because the outcome is defined at the incumbent level, this restriction is unavoidable. Section 4.3.1 shows that the instrument does not predict rerunning, and Table 2 confirms that the estimation sample is broadly representative of the full population of municipalities.

Specification. All specifications include municipality and year fixed effects. Because municipality fixed effects absorb time-invariant municipal characteristics, the baseline controls enter only as interactions with the post indicator. In our preferred specification we additionally include incumbent-party fixed effects based on the pre-reform winner's party, which absorb party-specific electoral dynamics, and a bundle of muni-level baseline characteristics from the 2000 Census interacted with post (log population, population growth, share of poor, household income per capita, and urban share), which absorb any differential 2012 effect operating through pre-reform municipal conditions. We exclude the incumbent's 2008 vote share from the preferred specification to avoid introducing a lagged dependent variable. Combining unit fixed effects with a lagged dependent variable in a short panel is known to generate bias (Nickell, 1981; Imai and Kim, 2019), and the lagged outcome appears as a sensitivity check in Section 5.4.

All regressions are weighted by municipal population in 2000. This is a scope condition rather than a precision choice. The estimand targets the causal response for the typical Brazilian citizen, which, given Brazil's urban concentration, weights toward larger municipalities where the absolute scale of expansion was economically meaningful.¹⁷

Standard errors are clustered at the state level, following Abadie et al. (2023), because the counterfactual quota formula incorporates a state-level poverty estimate and

¹⁶Appendix C.3 reports qualitatively unchanged results using alternative outcome definitions, including party vote share, and alternative denominators such as total votes cast and the eligible electorate.

¹⁷The right-skewed distribution of realized *Bolsa Família* expansion would otherwise give dominant weight to municipalities where expansion was modest and administrative capacity limited. Population weights align the estimand with the municipalities where the mechanism is most likely to operate. Unweighted and alternative weights' estimates are reported in Appendix C.3.

induces within-state correlation in the instrument.¹⁸

4.3 Threats to Identification

Identification of β_1 as the average causal response requires conditional independence, the exclusion restriction, and monotonicity. We discuss each in turn and document, where possible, that the data are consistent with the assumption. A separate concern about the timing of the 2012 quota revision is addressed at the end of the subsection.

4.3.1 Conditional Independence

Assumption 1 (Conditional Independence). *Conditional on municipality fixed effects, year fixed effects, incumbent-party fixed effects, and the baseline Census 2000 characteristics interacted with the post indicator, the quota shock is orthogonal to potential electoral outcomes:*

$$\begin{aligned} \mathbb{E}[Y_{m,t}(0) \mid \Delta\text{Quota}_m \times \text{Post}_t, \alpha_m, \lambda_t, \pi_m^{2008}, X_m^{2000} \times \text{Post}_t] \\ = \mathbb{E}[Y_{m,t}(0) \mid \alpha_m, \lambda_t, \pi_m^{2008}, X_m^{2000} \times \text{Post}_t]. \end{aligned}$$

The quota shock must be orthogonal to potential electoral outcomes conditional on municipality fixed effects, year fixed effects, incumbent-party fixed effects, and the baseline characteristics interacted with the post indicator. Conditional rather than unconditional exogeneity is the relevant requirement here because the counterfactual quota formula incorporates a state-level poverty estimate, Poor_s^{2006} , which mechanically correlates the instrument with pre-existing political geography. Municipalities in poorer states receive systematically different quota shocks, and those same states have structurally distinct electoral dynamics (Zucco, 2013). Once the controls absorb time-invariant municipal characteristics, party-specific electoral baselines, and any 2012 trend operating through pre-reform municipal conditions, the residual variation in ΔQuota_m reflects the change in the federal allocation formula net of these dimensions and is plausibly orthogonal to potential electoral outcome trends.

The proportional-selection exercise follows Oster (2019) and computes δ^* , the ratio of selection on unobservables to selection on observables that would be required to

¹⁸Brazil is divided into 26 states plus the Federal District. The Federal District does not hold municipal elections and falls outside our sample, so the regressions cluster on 26 state-level groups. Even at this count, asymptotic cluster-robust inference can be unreliable when the first stage is strong. We therefore report Anderson and Rubin (1949) confidence sets throughout, which are valid under arbitrary instrument weakness and robust to the cluster count. Appendix C.4 also reports wild cluster bootstrap p -values (Cameron et al., 2008; MacKinnon et al., 2023).

explain the reduced-form effect away under the framework’s proportional-selection assumption. Adding the Census 2000 baseline characteristics interacted with post moves the coefficient on the quota shock from 0.64 to 0.61. The decline is small and consistent with limited selection on observables. Under the conservative bound $\bar{R}^2 = 1.3 \cdot R_{\text{long}}^2$ we obtain $\delta^* = 24.12$, and under Oster’s recommended benchmark $\bar{R}^2 = 2.2 \cdot R_{\text{long}}^2$ we obtain $\delta^* = 6.03$. For proportional selection on unobservables to overturn the reduced-form effect, unobservables would have to be at least six times as important as the rich set of observables we condition on.

Selection into rerunning. The estimation sample is restricted to municipalities where the 2008 incumbent sought reelection in 2012, so the instrument could bias inference if it systematically affected the decision to run again. Table 2 regresses an indicator for whether the 2008 incumbent appears on the 2012 ballot on the quota shock, estimated on the full universe of municipalities. Under the preferred counterfactual-based instrument, the estimated coefficients are small and statistically indistinguishable from zero. The identifying variation appears orthogonal to selection into the estimation sample.

Table 2: Instrument and the Incumbent’s Decision to Run Again in 2012

	Incumbent Run in 2012		
	(1)	(2)	(3)
$\Delta\text{Quota}_m (\hat{\gamma}_1)$	-0.005 (0.003)	-0.004 (0.005)	-0.003 (0.004)
Party Fixed Effects		✓	✓
Baseline Characteristics			✓
Observations	5,504	5,504	5,504

Notes: The dependent variable is an indicator equal to one if the 2008 incumbent appears on the ballot in the 2012 municipal election. The regression is estimated on the universe of municipalities, using the preferred instrument. Column (1) is bivariate; column (2) adds party fixed effects based on the 2008 winner; column (3) further adds baseline characteristics (Census 2000 log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor that enters the counterfactual quota formula). Column (3) is the preferred specification, matching the control set of the main DiD-IV. Robust standard errors clustered at the state level are reported in parentheses. All regressions are weighted by population in 2000. 64 observations were excluded due to missing values in one or more variables.

As a complementary check, Section 5.4 re-estimates the preferred specification weighted by the inverse probability of selection into the rerunning sample. The 2SLS point estimate moves from 2.94 to 2.80 and remains statistically significant, which indicates that selection into the rerunning sample is not materially distorting the

estimates.

Pre-reform electoral balance. A second exercise tests whether the quota shock is correlated with pre-existing political conditions. Table 3 reports placebo regressions of 2008 electoral outcomes on the preferred instrument within the estimation sample. Columns 1 to 3 use the 2008-winner’s vote share as the outcome, and columns 4 to 6 use an indicator for the sitting incumbent winning in 2008. Because 2008 outcomes were determined before the reform, any systematic non-zero association would signal baseline imbalance rather than the effect of quota-induced expansion.

Table 3: Placebo: 2008 Electoral Outcomes on Post-Reform Quota Shock

	Vote Share			Winning		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta\text{Quota}_m(\hat{\delta}_1)$	-0.026 (0.063)	-0.036 (0.066)	-0.110 (0.069)	-0.005 (0.004)	-0.015 (0.014)	-0.002 (0.013)
Party Fixed Effects		✓	✓		✓	✓
Baseline Characteristics			✓			✓
Observations	2,105	2,105	2,105	2,105	2,105	2,105

Notes: The dependent variable is measured in the 2008 municipal election. Vote Share is the 2008-winner’s share of valid votes. Winning is an indicator equal to 100 if the sitting incumbent (2004-winner) won in 2008 and 0 otherwise. All columns use the preferred instrument and restrict to the municipalities in the DiD-IV estimation sample. Columns (1) and (4) are bivariate; columns (2) and (5) add party fixed effects based on the 2008 winner; columns (3) and (6) further add baseline characteristics (Census 2000 log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor that enters the counterfactual quota formula). Robust standard errors clustered at the state level are reported in parentheses. All regressions are weighted by population in 2000.

The evidence is consistent with conditional orthogonality of the instrument. Point estimates remain near zero and statistically insignificant as progressively richer control sets are added, and none of the six specifications rejects the null of zero association at conventional levels. The instrument does not display the kind of pre-reform imbalance that would threaten identification.

No anticipation. The 2009 quota revision was formally announced in MDS *Informe n° 169*, dated April 2009, which communicated that approximately 300,000 additional families would be incorporated in staggered waves over the course of the year. The previous quota adjustment had occurred in 2006. Although municipalities may have anticipated that some revision would eventually take place, they could not have foreseen

the magnitude, the distributional formula, or the timing of the 2009 reallocation before the official announcement, which came six months after the October 2008 municipal elections. The placebo balance reported just above is consistent with this institutional timeline, since the instrument is statistically unrelated to either the 2008 winner’s vote share or the sitting incumbent’s 2008 win probability in any specification.

4.3.2 Exclusion Restriction

Assumption 2 (Exclusion Restriction). *The quota shock affects incumbent vote share only through realized beneficiary enrollment:*

$$\Delta \text{Quota}_m \times \text{Post}_t \perp\!\!\!\perp Y_{m,t}(\text{Beneficiaries}_{m,t}) \\ | \text{Beneficiaries}_{m,t}, \alpha_m, \lambda_t, \pi_m^{2008}, X_m^{2000} \times \text{Post}_t.$$

Equivalently, conditional on the controls and on realized expansion, the quota shock has no direct effect on the incumbent’s vote share.

The exclusion restriction states that the quota shock affects vote share only through realized beneficiary enrollment. This is fundamentally a substantive rather than a statistical claim, and the case for it rests on the institutional channel through which the shock operates. The 2009 reallocation was a federally administered formula change, computed centrally by the *Ministério do Desenvolvimento Social* from Census 2000 and PNAD 2006 data, and it modified one and only one variable observable to municipalities and to voters, namely the authorized number of beneficiary slots. The benefit value paid per family was set by separate federal decree, and conditionality enforcement was implemented by federal partners through schools and health facilities.

The institutional argument is the primary defense of the exclusion restriction, but the most plausible alternative channels have empirical counterparts that we can test directly. Two concurrent federal-to-municipal flows are worth addressing: the per-family *Bolsa Família* benefit value and the constitutionally fixed revenue-sharing transfer (*Fundo de Participação dos Municípios*, FPM). Both are institutionally decoupled from the quota formula. The mean per-family benefit value is set by national rules but varies in the cross section through family composition, which is in principle correlated with the quota shock through the same Census 2000 poverty estimates that feed the formula. FPM coefficients are set by Complementary Law 91/1989 from fixed population brackets and apportionment factors; they are not a function of poverty estimates, of the 2007 *Contagem* demographic update, or of the income-volatility correction that drove the 2009 *Bolsa Família* reweighting.

Table 4 reports the empirical exclusion tests. Panel A regresses the inter-election change in FPM transfers (interacted with post) on the standardized quota shock (interacted with post), under three control specifications matching the headline. Panel B reports the analogous test using the per-family *Bolsa Família* benefit value. In both panels, the preferred specification (column 3, with Census 2000 baseline characteristics interacted with post) returns a coefficient statistically indistinguishable from zero. The empirical evidence is consistent with the institutional argument: the quota shock does not differentially shift per-family benefit values or concurrent FPM transfers across municipalities, so neither channel absorbs non-enrollment variation that would contaminate the IV.

Table 4: Instrument Exogeneity: FPM Transfers and Benefit Value

	(1)	(2)	(3)
<i>Panel A: $\Delta FPM_m \times Post$ (std)</i>			
$\Delta Quota_m \times Post$ ($\hat{\gamma}$)	-0.265 (0.037)	-0.294 (0.032)	0.011 (0.060)
<i>Panel B: $\Delta Value_m \times Post$ (std)</i>			
$\Delta Quota_m \times Post$ ($\hat{\gamma}$)	-0.003 (0.003)	-0.006 (0.003)	0.003 (0.002)
Party Fixed Effects		✓	✓
Baseline Characteristics			✓
Observations	4,210	4,210	4,210

Notes: Each panel reports the coefficient on the standardized 2009 quota shock interacted with post in a regression where the dependent variable is the indicated muni-level constant interacted with post. Panel A uses the inter-election change in *Fundo de Participação dos Municípios* transfers ($\Delta FPM_m = FPM_m^{2009} - FPM_m^{2008}$, standardized). Panel B uses the twelve-month moving average of the per-family *Bolsa Família* benefit value (standardized). Both regressions are estimated on the DiD-IV sample of 4,210 municipality-year observations using the preferred R\$90/120 instrument. Column (1) includes only year and municipality fixed effects; column (2) adds incumbent-party fixed effects based on the pre-reform winner; column (3) further adds Census 2000 baseline characteristics interacted with the post indicator (log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor). All regressions are weighted by population in 2000. Robust standard errors clustered at the state level are reported in parentheses.

Endogeneity of realized expansion. A separate exclusion-related concern is reverse causality between political strength and realized enrollment. If incumbents who were electorally stronger before the reform were also more able or willing to expand enrollment afterward, direct estimates of the effect of realized expansion on vote share would be confounded. Table 5 regresses realized beneficiary expansion on measures of the incumbent’s 2008 electoral strength.

Table 5: Pre-Reform Electoral Strength and Subsequent Realized Expansion

	Vote Share			Winning		
	(1)	(2)	(3)	(4)	(5)	(6)
Realized BF Expansion _{<i>m</i>} ($\hat{\kappa}_1$)	-0.001 (0.059)	-0.005 (0.044)	-0.011 (0.013)	-0.041 (0.023)	-0.077 (0.044)	-0.007 (0.014)
Party Fixed Effects		✓	✓		✓	✓
Baseline Characteristics			✓			✓
Observations	2,105	2,105	2,105	2,105	2,105	2,105

Notes: The dependent variable is realized BF beneficiary expansion between 2008 and the 12-month average of 2009–2012, standardized. The explanatory variable is incumbent electoral strength in 2008, measured alternatively as the 2008-winner’s vote share (columns 1–3) or an indicator for the sitting incumbent winning in 2008 (columns 4–6). All columns restrict to the municipalities in the DiD-IV estimation sample. Columns (1) and (4) are bivariate; columns (2) and (5) add party fixed effects based on the 2008 winner; columns (3) and (6) further add baseline characteristics (Census 2000 log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor that enters the counterfactual quota formula). Robust standard errors clustered at the state level are reported in parentheses. All regressions are weighted by population in 2000.

The estimates are small in magnitude and statistically indistinguishable from zero, so we do not find robust evidence that pre-reform electoral strength predicts subsequent *Bolsa Família* expansion. The 2SLS design remains appropriate on conceptual grounds. Reverse causality (where municipalities expand enrollment more aggressively where the expected political return is higher, as documented in [Brollo et al., 2020](#)) is a priori plausible even when it does not register empirically in this placebo test. The federal quota shock supplies exogenous variation that neutralizes this potential channel and is the reason we report 2SLS rather than OLS estimates.

4.3.3 Monotonicity

Assumption 3 (Weak Monotonicity). *For every municipality m , a positive quota shock cannot reduce realized expansion: $\Delta \text{Beneficiaries}_m(z') \geq \Delta \text{Beneficiaries}_m(z)$ whenever $z' > z$, where $\Delta \text{Beneficiaries}_m(z)$ is the potential inter-election change in beneficiaries that municipality m would realize under shock value z .*

The third assumption is monotonicity: a positive quota shock cannot reduce realized expansion in any municipality. As in any IV setting, this is an identifying restriction rather than a testable hypothesis, since we never observe both potential outcomes for the same unit. We follow standard practice in the continuous-treatment IV literature ([Imbens and Angrist, 1994](#); [Angrist and Imbens, 1995](#)) by combining an institutional

argument with empirical diagnostics consistent with weak monotonicity.

The institutional argument is direct. The 2009 reform raised the authorized number of beneficiary slots; it did not reduce them anywhere. A municipality that received a positive shock acquired headroom to enroll additional families if it chose to, while one that received a smaller adjustment lost no pre-existing capacity. There is no mechanical channel through which a higher federal authorization depresses enrollment. Defier behavior is in principle possible (a mayor could choose to scale back the registration apparatus precisely when the federal ceiling expanded), but we view this as empirically rather than institutionally implausible. The diagnostics below bound how much such behavior could distort the headline coefficient.

Three empirical diagnostics support this institutional reading. First, 67 percent of municipalities exhibit sign agreement between the inter-election change in beneficiaries and the quota shock, and these municipalities account for 98 percent of the IV-numerator mass $|\Delta D_m \cdot Z_m|$, which is the object that determines the 2SLS coefficient; the remaining 33 percent of sign-discordant municipalities are concentrated near zero on both axes (small expansion changes paired with small shocks) and contribute only 1.7 percent of the identifying mass. Second, a binned first-stage exercise tests whether the implementation chain runs in the same direction across the full support of the instrument: we sort municipalities into quintiles of Z_m and compute the conditional mean of ΔD_m within each bin. The conditional mean rises monotonically across the bulk of the distribution where the identifying variation concentrates, with the highest quintile (mean $Z_m \approx 1,446$ additional authorized slots) producing roughly six times the inter-election expansion of the next-highest quintile. Third, Section 5.5 reports the first-stage Kleibergen-Paap F -statistic across three pre-specified partitions (Northeast, the in-Frey saturation cell, and the partisan PT/non-PT cut) and finds it positive and well above the weak-instrument threshold in every subgroup that identifies the IV, with no subgroup exhibiting a wrong-signed coefficient. This is a stronger version of average monotonicity than the binned check, since the first stage runs in the same direction not only across quintiles of the instrument but across institutionally meaningful partitions of the sample.

Under the average-causal-response framework of Angrist and Imbens (1995) and Callaway et al. (2024), the headline 2SLS coefficient is a mass-weighted average of municipality-specific responses, with the weights determined by the sign-coherence between ΔD_m and Z_m documented above. Adapting the sensitivity bound of de Chaisemartin (2017) to this continuous-treatment setting, sign-discordant municipalities contribute negative weight to the pooled coefficient and carry 1.7 percent of the absolute

IV-numerator mass $|\Delta D_m \cdot Z_m|$. Even under the worst-case assumption that the sign-discordant municipalities have causal responses of magnitude equal and direction opposite to the sign-concordant ones, the implied sign-concordant ACR is bounded in $[+2.94, +3.03]$ percentage points, a 3.2 percent bandwidth around the headline of +2.94. The estimator is therefore essentially insensitive to monotonicity violations at any plausible magnitude.

4.3.4 The 2012 Quota Revision

Municipal authorization margins were revised again in 2012, shortly before the election.¹⁹ This revision was considerably smaller than the 2009 expansion but may still have affected 2012 electoral outcomes. We treat the 2009 revision as the primary source of identifying variation because it is the first reform after the 2008 election and has the full inter-election period to affect registration efforts and voter perceptions. The 2012 update is both smaller in magnitude and much closer to the election. Appendix C.3 reports supplementary specifications that extend the measurement window for the endogenous variable through 2012 and an updated version of the instrument using information from the 2010 Census.

5 Main Results

This section reports the main empirical results. We proceed in four steps. First, we establish that the 2009 quota reform generated substantial variation in realized program expansion (first stage). Second, we show that this exogenous variation in authorized quotas translates into better incumbent electoral performance: the intention-to-treat (ITT) effect of the quota shock on the ballot box, which is algebraically the reduced form. Third, we combine these two moments to recover the causal effect of instrumented beneficiary expansion on incumbent vote share and reelection (two-stage least squares). Fourth, we subject these estimates to a battery of robustness and heterogeneity checks. Throughout, both the year-varying treatment ($\text{Beneficiaries}_{mt}$) and the time-invariant instrument (ΔQuota_m) are standardized to mean zero and unit standard deviation within the estimation sample, so coefficients read as the response to a one-standard-deviation move in the relevant variable.

Each table presents six columns: three specifications for each outcome. Columns (1)

¹⁹Following the release of the 2010 Census, the federal government updated municipal poverty estimates used to guide program expansion in 2012. See the MDS *Informe n° 324*.

and (4) include only year and municipality fixed effects. Columns (2) and (5) add incumbent-party fixed effects based on the pre-reform winner’s party, capturing partisan baselines. Columns (3) and (6), our preferred specifications, additionally absorb five pre-treatment baseline characteristics interacted with the post-reform indicator: log population, the share of poor households, household income per capita, urban share, and the 2000 to 2006 population growth factor that enters the counterfactual quota formula (all from the 2000 Census).²⁰ All regressions are weighted by municipal population in 2000 and cluster standard errors at the state level.

5.1 First Stage: Authorization and Coverage

Table 6 reports the first-stage relationship between the quota shock and realized beneficiary expansion.

Table 6: First-Stage Estimates

	Realized Expansion in 2009		
	(1)	(2)	(3)
$\Delta\text{Quota}_m (\pi_1)$	0.199 (0.003)	0.197 (0.003)	0.205 (0.006)
Party Fixed Effects		✓	✓
Baseline Characteristics			✓
Observations	4,210	4,210	4,210

Notes: Each coefficient is estimated on the municipality-year panel of 2008 and 2012 elections for the 2,105 municipalities where the 2008 incumbent ran for reelection in 2012. The endogenous variable is the year-varying twelve-month moving average of *Bolsa Família* beneficiaries; the instrument is the time-invariant 2009 quota shock interacted with the post-2009 indicator. Under two-period two-way fixed effects, the coefficient identifies the marginal effect of the muni-level inter-election change in beneficiaries on the corresponding change in the outcome (first-difference identification). All specifications include year and municipality fixed effects. “Party Fixed Effects” are based on the pre-reform winner’s party. “Baseline Characteristics” bundles five Census 2000 controls, each interacted with the post-reform indicator: log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor that enters the counterfactual quota formula. All regressions are weighted by municipal population in 2000. Robust standard errors clustered at the state level (26 clusters) are reported in parentheses.

²⁰The incumbent’s 2008 vote share is *excluded* from the preferred specification. Absorbing the lagged outcome through its interaction with *post* would control for baseline incumbent strength, but combining unit fixed effects with a lagged dependent variable is known to introduce bias in short panels (Nickell, 1981; Imai and Kim, 2019). Our identification flows from the instrument, not from OLS on the lag, so the concern applies to nuisance absorption rather than the causal estimate; we keep the lagged vote share out of the main spec to be conservative. Column (2) of Table 9 adds it back as a sensitivity check.

In the preferred specification, which adds incumbent-party fixed effects and the baseline characteristics interacted with post, a one-standard-deviation increase in the quota shock raises year-varying standardized beneficiary enrollment by 0.205 standard deviations (SE 0.006). The estimate is virtually unchanged across specifications: 0.199 in Column (1), 0.197 in Column (2), and 0.205 in Column (3). The stability is informative. Because the 2009 quota shock is time-invariant and enters interacted with Post_t , the first-stage coefficient identifies the response of the muni-level change in beneficiaries to the shock; this object is structurally orthogonal to the time-invariant baseline characteristics that enter the regression as $\text{Census}_m^{2000} \times \text{Post}_t$, so adding those controls primarily affects the residual variance and the standard error. The Kleibergen-Paap F -statistic for the preferred column is 1,257, well above any conventional threshold for instrument strength.

The pass-through is economically substantial: one standard deviation of the shock corresponds to approximately 1,447 additional authorized beneficiaries, and the implied response in realized enrollment is approximately 1,015 additional families (0.205 panel-SD on the realized scale of 4,949 families per SD). Roughly 70 percent of the additional federal authorization translates into realized municipal coverage.

5.2 Reduced Form: Authorization and Electoral Returns

Table 7 reports the intention-to-treat (ITT) effect of the 2009 quota reform on incumbent electoral performance. The ITT regresses the electoral outcome directly on the exogenous quota shock, bypassing the second-stage instrumentation entirely. It is the most transparent test of whether being authorized to expand the program translates into electoral gains, and is causal under instrument exogeneity alone, since it does not require the exclusion restriction that the quota shock affects votes only through the realized beneficiary expansion channel. Interpreted as an ITT, it is the effect of the federal *authorization* rather than the effect of the additional families actually enrolled.

In the preferred specification, a one-standard-deviation increase in the quota shock raises the incumbent's vote share by 0.60 percentage points (SE 0.22) and the probability of winning by 0.84 percentage points (SE 0.27). Both estimates are significant at the one percent level. The ITT effect on vote share is stable across specifications, moving from 0.49 in Column (1) to 0.64 in Column (2) and 0.60 in Column (3); the small step from Column (2) to Column (3) is consistent with limited selection on observables as discussed earlier. For winning, Columns (4) and (5) yield raw coefficients of 1.31 and 1.78, attenuating to 0.84 once the baseline characteristics enter, consistent

Table 7: Reduced-Form Estimates

	Vote Share			Winning		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta\text{Quota}_m(\rho_1)$	0.494 (0.049)	0.637 (0.105)	0.603 (0.218)	1.305 (0.136)	1.784 (0.216)	0.841 (0.266)
Party Fixed Effects		✓	✓		✓	✓
Baseline Characteristics			✓			✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: Each coefficient is estimated on the municipality-year panel of 2008 and 2012 elections for the 2,105 municipalities where the 2008 incumbent ran for reelection in 2012. The endogenous variable is the year-varying twelve-month moving average of *Bolsa Família* beneficiaries; the instrument is the time-invariant 2009 quota shock interacted with the post-2009 indicator. Under two-period two-way fixed effects, the coefficient identifies the marginal effect of the muni-level inter-election change in beneficiaries on the corresponding change in the outcome (first-difference identification). All specifications include year and municipality fixed effects. “Party Fixed Effects” are based on the pre-reform winner’s party. “Baseline Characteristics” bundles five Census 2000 controls, each interacted with the post-reform indicator: log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor that enters the counterfactual quota formula. All regressions are weighted by municipal population in 2000. Robust standard errors clustered at the state level (26 clusters) are reported in parentheses.

with pre-existing electoral heterogeneity inflating the unconditional ITT before the preferred specification nets it out.

5.3 Second Stage: Coverage and Electoral Returns

Table 8 reports two-stage least squares estimates of the effect of instrumented beneficiary expansion on incumbent electoral performance. Under the first-difference reduction in equation (21), the coefficient identifies the marginal effect of the muni-level inter-election change in beneficiaries on the corresponding change in vote share.

In the preferred specification, a one-standard-deviation increase in instrumented beneficiary expansion raises the incumbent’s vote share by 2.94 percentage points (SE 1.10) and the probability of reelection by 4.10 percentage points (SE 1.36). For vote share, the Anderson-Rubin 95 percent confidence set is [0.75, 5.25]; for winning, the set is [1.40, 7.00]. Both sets exclude zero under arbitrary instrument weakness.

The standardized coefficient is informative for hypothesis testing but obscures the natural-units interpretation. Two complementary scales help. First, since the cross-municipal standard deviation of the inter-election *change* in beneficiaries is approxi-

Table 8: Two-Stage Least Squares Estimates

	Vote Share			Winning		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \widehat{\text{Beneficiaries}}_m (\beta_1)$	2.476 (0.244)	3.228 (0.527)	2.935 (1.105)	6.541 (0.739)	9.035 (1.162)	4.096 (1.360)
Kleibergen-Paap F	3705.20	3624.90	1256.91	3705.20	3624.90	1256.91
AR 95% CI	[2.00, 2.95]	[2.15, 4.30]	[0.75, 5.25]	[5.05, 8.05]	[6.70, 11.45]	[1.40, 7.00]
Party Fixed Effects		✓	✓		✓	✓
Baseline Characteristics			✓			✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: Each coefficient is estimated on the municipality-year panel of 2008 and 2012 elections for the 2,105 municipalities where the 2008 incumbent ran for reelection in 2012. The endogenous variable is the year-varying twelve-month moving average of *Bolsa Família* beneficiaries; the instrument is the time-invariant 2009 quota shock interacted with the post-2009 indicator. Under two-period two-way fixed effects, the coefficient identifies the marginal effect of the muni-level inter-election change in beneficiaries on the corresponding change in the outcome (first-difference identification). All specifications include year and municipality fixed effects. “Party Fixed Effects” are based on the pre-reform winner’s party. “Baseline Characteristics” bundles five Census 2000 controls, each interacted with the post-reform indicator: log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor that enters the counterfactual quota formula. All regressions are weighted by municipal population in 2000. Robust standard errors clustered at the state level (26 clusters) are reported in parentheses. The Kleibergen-Paap F -statistic is the squared t -ratio of the quota shock in the first-stage regression with cluster-robust standard errors. Anderson-Rubin 95% confidence sets are robust to arbitrary instrument weakness. An alternative ANCOVA-style specification, with the endogenous variable interacted with the post indicator, is reported in Appendix C.9.

mately 947 families, a typical inter-election expansion of one standard deviation raises the incumbent’s vote share by $2.94 \times 947/4,949 \approx 0.56$ percentage points. Second, in terms of the raw enrollment count, an additional 1,000 beneficiaries between the 2008 and 2012 elections raise the incumbent’s vote share by approximately 0.59 percentage points ($2.94/4.949$). The two scales describe the same coefficient under the panel-SD standardization adopted throughout the paper.

The 2SLS estimates are materially larger than their OLS counterparts: the OLS coefficient on the same standardized endogenous variable is 1.00 percentage points on vote share (against the 2.94 percentage points 2SLS estimate). A Wu-Hausman test rejects exogeneity for vote share at the one percent level, confirming the need for the IV strategy; for the binary winning indicator the test does not reject at conventional

levels.²¹

The preferred estimate of 2.94 percentage points is large in absolute terms. It is approximately equal to the typical inter-election decline in incumbent support of 3.00 percentage points in our sample, consistent with the magnitude of incumbency disadvantage that [Klašnja and Titiunik \(2017\)](#) document for Brazilian municipal elections using a regression-discontinuity design on close races. An exogenously induced one-SD expansion in beneficiary coverage therefore essentially cancels the average incumbency disadvantage between elections. The effect is large enough to be decisive in the typical close race: 26 percent of races in the analysis sample were decided by fewer than 5 percentage points in 2012, and 11 percent by fewer than 2 percentage points.

The estimate identifies an average causal response in the sense of [Angrist and Imbens \(1995\)](#): the response is a weighted average across municipalities, with weights concentrated where the 2009 quota shock generated meaningful enrollment changes. We document this concentration of identifying mass directly in [Appendix C](#), where we split the sample by the sign and the median of ΔQuota_m ; the response is identified primarily in the positive-shock regime, where the first stage is strong, while the negative-shock regime contributes negligible identification.

5.4 Robustness

The 2SLS estimate is robust to the standard set of perturbations of the modelling environment. The reduced-form effect retains significance under wild cluster bootstrap ([Rademacher and Webb](#)), under alternative clustering at the mesoregion and microregion levels, and under Conley standard errors at distance bandwidths of 100 and 200 kilometers. The two-stage least-squares estimate is significant under each of these procedures as well.

[Table 9](#) presents the core sensitivity analysis across six specifications. Column (1) reproduces the preferred estimate ($\hat{\beta}_1 = 2.94$). Column (2) adds the 2008 incumbent's vote share interacted with post to the baseline-characteristics bundle, addressing the lagged-dependent-variable concern discussed earlier; the point estimate is 2.52 and remains significant, confirming that the headline result is not an artifact of excluding the

²¹The Wu-Hausman test has lower power on a binary outcome with limited tail variation than on a continuous outcome, so failure to reject for winning is partly a power result. The IV and OLS coefficients for winning have the same sign and comparable magnitudes, and [Appendix C.4](#) reports both in parallel. We report 2SLS for both outcomes for internal consistency with the framework, in which both outcomes inherit the same underlying causal response. The IV-OLS gap on vote share is interpretable as evidence of negative political selection: mayors facing stronger electoral pressure expand coverage more aggressively, generating a downward bias in OLS that the federal authorization shock removes.

Table 9: Robustness Checks: Vote Share

	Main	Lag VS 2008	Ext. Window	Winsorized	Alt. Denominator	IPW
<i>Panel A: First Stage</i>						
$\Delta \text{Quota}_m (\pi_1)$	0.205 (0.006)	0.205 (0.006)	0.382 (0.014)	0.205 (0.006)	0.205 (0.006)	0.207 (0.005)
<i>Panel B: Reduced Form</i>						
$\Delta \text{Quota}_m (\rho_1)$	0.603 (0.218)	0.517 (0.195)	0.603 (0.218)	0.547 (0.207)	0.566 (0.188)	0.580 (0.228)
<i>Panel C: Two-Stage Least Squares</i>						
$\widehat{\Delta \text{Beneficiaries}_m} (\beta_1)$	2.935 (1.105)	2.524 (0.994)	1.577 (0.566)	2.662 (1.042)	2.756 (0.958)	2.804 (1.138)
Kleibergen-Paap F	1256.91	1243.27	800.79	1256.91	1256.91	1977.00
AR 95% CI	[0.75, 5.25]	[0.55, 4.65]	[0.45, 2.70]	[0.60, 4.85]	[0.85, 4.80]	[0.55, 5.20]
Party Fixed Effects	✓	✓	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: All columns report the preferred specification: party fixed effects + baseline characteristics $\times$ post, population-2000 weights. The outcome is incumbent vote share except where noted. Column (2) adds the 2008 incumbent vote share $\times$ post to the baseline-characteristics bundle, testing sensitivity to including a lagged dependent variable (Nickell, 1981; Imai and Kim, 2019); the main specification omits this term. Column (3) uses the 12-month moving average of realized beneficiaries through December 2012 as the endogenous variable. Column (4) winsorizes the outcome at the 1st and 99th percentiles. Column (5) uses total votes cast (including blank and null) as the denominator. Column (6) weights by the inverse probability of selection into the rerunning sample. Anderson-Rubin 95% confidence sets are robust to arbitrary instrument weakness. Robust standard errors clustered at the state level in parentheses.

lagged outcome. Columns (3) to (6) perturb the treatment, outcome, or sample while holding the instrument fixed: extending the treatment-window to December 2012 (Column 3) yields $\hat{\beta}_1 = 1.58$ (still significant under the AR confidence set of [0.45, 2.70]).

Column 3 redefines the endogenous variable as the change in the 12-month beneficiary count between the close of 2008 and the close of 2012, measuring the expansion as near to the ballot as the data allow; it yields $\hat{\beta}_1 = 1.58$, still significant under the AR confidence set [0.45, 2.70]. The attenuation reflects the 2012 quota revision, which used the 2010 Census to recalibrate municipal poverty estimates and reduced authorized quotas in municipalities whose 2010 poverty figures fell below the 2009 projections, partially reversing the 2009 over-allocation in those places. A window running through December 2012 therefore averages the 2009 expansion with this partial 2012 reversal. We prefer the headline window because it aligns the treatment with the period the federal government actually assigned: the 2009 reform instructed municipalities to fill their newly authorized quotas during 2009, so the 2009 enrollment change is the policy-relevant treatment, while the 1.58 estimate is best read as a window-sensitivity diagnostic that integrates over a later, partly offsetting reform.

5.5 Heterogeneity

We close the main results with a heterogeneity exercise along three sample partitions: Northeast region, the population-by-HDI cell that anchors Frey (2019), and the 2008-elected incumbent’s party. Table 10 reports the saturated dual-IV decomposition for vote share. In each partition, the instrument and the endogenous variable are interacted with a binary indicator and its complement, yielding two coefficients per equation from a single full-sample regression. The decomposition tests the augmented model’s central prediction that the first stage attenuates on partitions aligned with the structural margins the model and remains comparable on partitions orthogonal to them.

Table 10: Heterogeneity by Subgroup: Vote Share (Dual-IV)

	Northeast			Frey 2019			PT incumbent		
	NE	non-NE	Δ	in Frey	out Frey	Δ	PT	non-PT	Δ
<i>Panel A: First Stage</i>									
$\Delta\text{Quota}_m(\pi_1)$	0.270 (0.031)	0.212 (0.003)	0.058 (0.032)	0.029 (0.011)	0.205 (0.006)	-0.176 (0.013)	0.169 (0.014)	0.203 (0.005)	-0.034 (0.015)
<i>Panel B: Reduced Form</i>									
$\Delta\text{Quota}_m(\rho_1)$	-0.324 (0.533)	0.611 (0.220)	-0.934 (0.373)	-2.257 (2.149)	0.600 (0.217)	-2.858 (2.108)	1.051 (0.343)	0.580 (0.229)	0.471 (0.212)
<i>Panel C: Two-Stage Least Squares</i>									
$\Delta\widehat{\text{Beneficiaries}}_m(\beta_1)$	-1.531 (1.860)	2.836 (1.114)	-4.367 (1.149)	-98.872 (74.189)	3.040 (1.103)	-101.911 (74.086)	6.162 (2.493)	2.725 (1.170)	3.437 (1.671)
Kleibergen-Paap F	74.08	3827.53		6.65	1214.32		146.97	1898.01	
AR 95% CI	[-5.15, 3.10]	[0.75, 5.15]		[-15.00, 25.00]	[0.75, 5.25]		[1.80, 12.30]	[0.55, 5.25]	
Party Fixed Effects	✓	✓	✓	✓	✓	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210	4,210	4,210	4,210

Notes: All estimates are obtained from a single full-sample dual-IV regression per heterogeneity dimension: the instrument and the endogenous variable are each interacted with a binary subgroup indicator and its complement, producing two coefficients per equation from one regression. Columns (1)–(3) split on Northeast region (AL, BA, CE, MA, PB, PE, PI, RN, SE) versus other regions. Columns (4)–(6) split on the Frey (2019) sample restriction (population in 2000 below 30,000 and Atlas IDHM 2010 below 0.7), with column (4) inside Frey’s sample and column (5) outside. Columns (7)–(9) split on whether the 2008-elected mayor’s party (winner_party_2008) — the sitting incumbent during the 2009–2012 expansion window — is the Workers’ Party (PT) versus any other party. Δ is the Wald test of equality of the two coefficients reported as the difference with its standard error. Specification matches the headline tables: 90/120 quota shock instrument, year-varying realized beneficiaries (12-month MA, standardized) as the endogenous variable, party fixed effects (pre-reform winner) plus baseline characteristics $\times$ post (Census 2000 log population, share of poor, household income per capita, urban share, and 2000–2006 population growth factor), municipality and year fixed effects, weighted by 2000 population, with cluster-robust standard errors at the state level. Anderson-Rubin 95% confidence sets are computed by grid inversion holding the other group’s coefficient at its dual-IV estimate; they are robust to weakness of the corresponding sub-instrument.

The Northeast partition (columns 1–3) leaves the first stage intact in both subsam-

ples: the instrument is strong inside and outside the region, comfortably above the Stock-Yogo threshold in each. The electoral response, by contrast, concentrates outside the Northeast, where the 2SLS estimate tracks the pooled headline; inside the region it is statistically indistinguishable from zero. The headline is therefore identified primarily from non-Northeastern compliers. What matters for the framework is that the regional split does not switch the first stage on or off, so region is not one of the structural margins the augmented model predicts should govern identification.

The Frey partition (columns 4–6) is sharper and more telling. Inside the cell that anchors Frey (2019)’s identification the first stage collapses below the conventional weak-instrument threshold, the corresponding 2SLS estimate is uninterpretable noise, and identification comes entirely from outside the cell, where the response again matches the pooled headline. The collapse is mechanical rather than structural: Frey’s anchor cell coincides with the small-population, medium-HDI subsample in which *Bolsa Família* had already reached high saturation by the late 2000s, so the eligible pool was largely exhausted and the federal authorization had no extensive margin to push into. With the pre-existing utilization rate B_m/Quota_m in this cell already close to one by 2008, a positive shock cannot generate realized expansion where the room for it is gone. Appendix D.2 confirms this collapse is robust to using direct saturation cuts (bottom quartile, median, mean, and top quartile of pre-reform beneficiaries per 100 inhabitants) as the binary partition rather than Frey’s institutional indicator, and develops the substantive distinction between Frey’s identification and ours: Frey identifies the electoral return to a federally-controlled enrollment channel under saturation, while we identify the return to a locally-controlled channel where there is room left to grow.

The PT partition (columns 7–9) is qualitatively different from the other two and is the most substantively interesting heterogeneity in the table. Both subgroups identify the effect at both stages, with a strong first stage on each side and 2SLS coefficients that exclude zero. The contrast is in magnitude: the return where the incumbent is from the Workers’ Party, $\hat{\beta}_1$ of about 6.2 percentage points, is more than double the return elsewhere, and a Wald test rejects equality of the two at the five percent level. The asymmetry sits in the per-beneficiary return rather than in the first stage; if anything the conversion of authorization into realized enrollment is somewhat slower under PT mayors, yet each marginal beneficiary delivers more than twice the vote share. This is the shape the framework commits to in advance: the partisan-attribution extension of the voter-response block (Section 3.4, equation (16)) raises the per-beneficiary return by $\lambda_2^{\text{attrib}} \geq 0$ wherever the mayor’s party also operates the federal program, while

holding the first stage partisan-invariant because the material channel λ_1 is federally set. The partition corroborates the prediction on both margins.

The mechanism behind $\lambda_2^{\text{attrib}}$ is institutional. *Bolsa Família* was created in 2003 under the federal Lula administration and has been the Workers' Party's flagship anti-poverty policy ever since; the 2009 reform that supplies our identifying variation was implemented under a PT-controlled federal executive in Lula's second term, and the 2012 election that we study was held under the first PT presidency of Dilma Rousseff. A municipality with a PT mayor between 2008 and 2012 is therefore administering a program created and politically owned by the same party at the federal level, throughout a period of continuous PT control of the federal executive. Voter attribution under these conditions is unambiguous: the partisan chain of credit runs cleanly from the local registration apparatus to the federal program design, while the same expansion under a non-PT mayor splits credit between local administration and federal sponsorship. We read the asymmetry as the empirical signature of this attribution channel rather than as a separate structural margin, and as a substantive component of the paper's main message: centralized program design does not, on its own, depoliticize redistribution, and the absence of depoliticization is sharpest where the local administrator and the federal program creator share a partisan brand.

6 Possible Mechanisms

The main results establish that federally induced quota expansion generated electoral returns for incumbent mayors. This section uses the same framework to test the augmented model's structural predictions about which margins of capacity must be in place for the federal-to-local chain to operate. We organize the evidence around the two margins of Section 3.3: infrastructure (κ^I) and alignment (κ^A).

First, local welfare infrastructure operates as the binding margin the augmented model predicts: without a CRAS in place before the reform, the quota shock attenuates by roughly a factor of three in the first stage relative to CRAS municipalities, and the electoral coefficient is too imprecisely estimated to interpret.

Second, among CRAS municipalities, the alignment margin operates through manager substitution. We document a chain of evidence consistent with the stepping-stone interpretation of Colonnelli et al. (2020): the federal shock causes substitution of incumbent-CRAS managers, but does not cause new CRAS unit openings, and the substitutes hired into post-shock units run for elected office at higher rates than the

baseline without switching to the mayor’s party and without prior partisan-affiliation selection. Roughly three quarters of the manager changes our definition captures bring in individuals new to the municipal welfare bureaucracy rather than internal reshuffles within or across CRAS units, ruling out the reading that the turnover margin is a relabeling exercise.

We frame these findings as causally identifying the average causal response of incumbent vote share to realized beneficiary expansion within the CRAS-with-turnover subgroup. Three pieces of evidence support this reading. The within-subgroup first stage is essentially zero in the inactive cells (Anderson-Rubin sets unbounded; Kleibergen-Paap F at 1.3 in CRAS-no-turnover; the Wald test of equality across cells rejects decisively), so the inactive cells contribute zero compliers to the pooled IV. The Frisch-Waugh-Lovell decomposition concentrates the identifying covariance overwhelmingly on the CRAS-with-turnover cell. The classification of manager changes into external, internal-same-CRAS, and internal-other-CRAS hire types (75 percent external) rules out the alternative reading that the turnover indicator captures bureaucratic re-shuffles rather than fresh personnel inflows.²²

6.1 State Capacity as a Precondition

Of the 2,105 municipalities in the estimation sample, 1,415 (67.2 percent) had at least one CRAS in the 2008 *Censo SUAS*; the remaining 690 did not. The augmented model is unambiguous about the implication: the within-subgroup first stage should attenuate by roughly a factor of γ in $\{\kappa_m^I = 0\}$, where γ is the productivity floor of the implementation chain. The benchmark, by contrast, predicts smooth attenuation of comparable magnitude. The dual-IV decomposition that follows tests the two against each other.

Table 11 reports the specification that identifies the effect separately for CRAS municipalities and for municipalities without a CRAS. Both the instrument and the endogenous variable are interacted with $\mathbf{1}(\text{CRAS}_m^{2008})$ and with its complement $1 - \mathbf{1}(\text{CRAS}_m^{2008})$, producing two endogenous coefficients in a single regression.

The first stage and reduced form deliver the asymmetry the augmented model pre-

²²All mechanism specifications use the preferred instrument, the preferred endogenous variable, and the Column (3) control set of Section 5: incumbent-party fixed effects based on the pre-reform winner and baseline characteristics $\times$ post. Appendix D reports additional mechanism dimensions, Appendix E.2 reports a within-CRAS test of operational intensity proxies (*busca ativa* and *PAIF* tracking) that confirms the augmented model’s prediction that non-structural intensity margins should not discriminate first-stage strength, and Appendix C.10 reports a separate robustness check that confirms the headline is not confounded by the concurrent federal increase in per-family benefit values announced in March 2011.

Table 11: CRAS Presence as Necessary Condition

	Vote Share			Winning		
	has = 1	has = 0	Δ	has = 1	has = 0	Δ
First Stage (π_1)	0.208 (0.005)	0.075 (0.012)	0.133 (0.013)	0.208 (0.005)	0.075 (0.012)	0.133 (0.013)
Reduced Form (ρ_1)	0.611 (0.219)	-2.636 (3.072)	3.247 (2.983)	0.803 (0.273)	7.274 (10.210)	-6.471 (10.087)
Two Stage Least Squares (β_1)	2.947 (1.078)	-37.640 (44.880)	40.588 (44.429)	3.841 (1.380)	94.565 (143.600)	-90.724 (143.115)
Kleibergen-Paap F	1734.99	35.64		1734.99	35.64	
AR 95% CI	[0.80, 5.20]	[-15.00, 25.00]		[1.15, 6.75]	[-15.00, 25.00]	
Party Fixed Effects	✓	✓	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: All estimates are obtained from the full-sample dual-IV regression: the instrument and the endogenous variable are each interacted with $\mathbf{1}(\text{CRAS}_m^{2008})$ and its complement, producing two coefficients per equation from a single regression. The “has = 1” column reports the coefficient on the CRAS-interaction; “has = 0” reports the coefficient on the no-CRAS interaction. Δ is the Wald test of $\beta_{has=1} = \beta_{has=0}$ reported as the difference with its standard error and $\chi^2(1)$ p -value. Under the preferred 90/120 instrument the first stage is identified in both subgroups (Kleibergen-Paap F above the Stock-Yogo 10%-bias threshold), so the 2SLS coefficient and Anderson-Rubin 95% confidence interval are reported for both “has = 1” and “has = 0”. All specifications include year and municipality fixed effects, party fixed effects based on the pre-reform winner (2004-winner in 2008; 2008-winner in 2012), and baseline characteristics $\times$ post: log population, Gini inequality, share of poor, household income per capita, share of formal workers, and urban share (all from Census 2000). Weights: municipal population in 2000. Cluster-robust standard errors at the state level in parentheses.

dicts. The CRAS subsample reproduces the pooled-sample first stage almost exactly, while the no-CRAS subsample retains a small but statistically identified pass-through that is roughly a factor of three weaker (a Wald test of equality across the two cells rejects). The CRAS-subgroup reduced-form responses on vote share and reelection are essentially indistinguishable from the headline; the no-CRAS responses are too imprecise to interpret and comfortably contain zero. The pattern is consistent with the broader finding in the *Bolsa Família* literature that implementation, not design, is where municipal governments shape distributive outcomes (Fried, 2012; Sugiyama and Hunter, 2013).

The 2SLS coefficients sharpen the asymmetry rather than add to it. The CRAS-subgroup point estimates and Anderson-Rubin confidence sets reproduce the pooled headline directly. The no-CRAS subgroup inherits the well-known weak-instrument inflation: the first-stage pass-through there is identifiably positive, but quantitatively too small to support a meaningful 2SLS denominator, and the AR set spans the en-

tire admissible range. The Frisch-Waugh-Lovell weights formalize this directly: in the saturated specification, $\omega_{\text{CRAS}} \approx 0.89$ and $\omega_{\text{no-CRAS}} \approx 0.11$, so nearly nine-tenths of the identifying covariance behind the headline 2SLS comes from the CRAS subgroup, with the no-CRAS contribution residual.

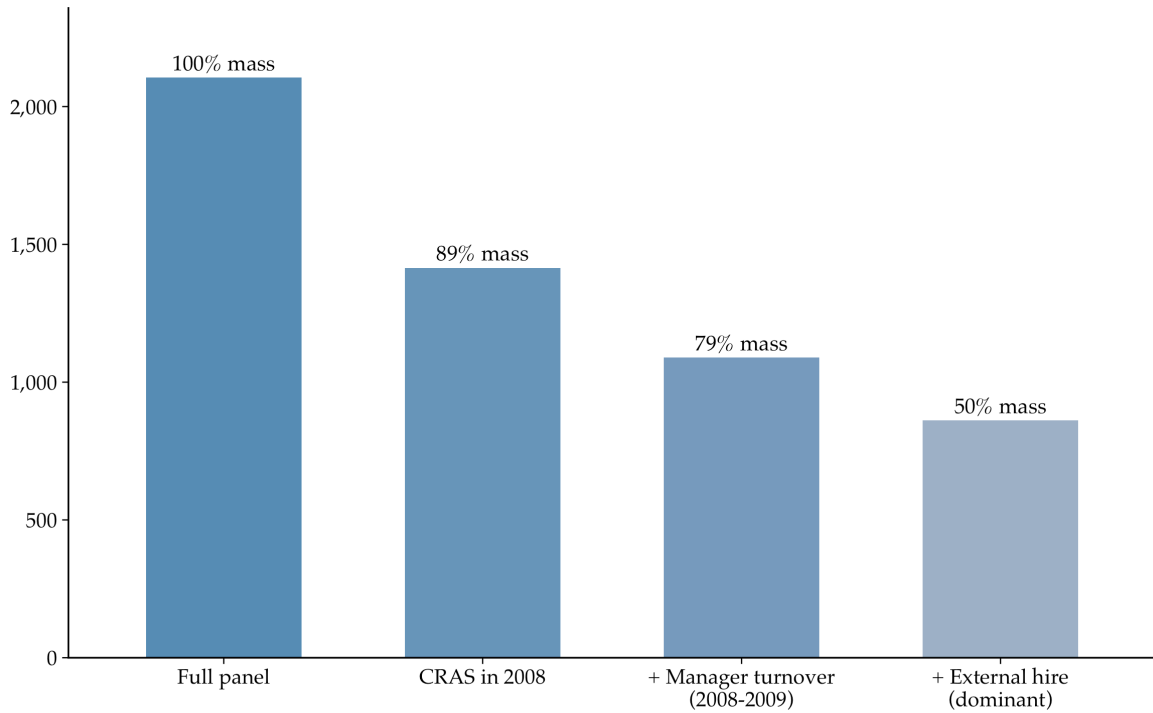
The substantive content of the table is therefore the asymmetry the augmented model anticipates. The productivity floor γ governing the inactive-subgroup attenuation is small enough that the no-CRAS first stage does not respond enough to the federal authorization to generate an interpretable second stage. The headline electoral return is the per-beneficiary effect for compliers in CRAS municipalities, where the implementation chain operates; it neither confirms nor rejects the existence of an electoral return in the no-CRAS subgroup, where the data are silent.

6.2 Manager Alignment as Electoral Stepping Stone

Figure 4 previews the mechanism cascade. From the full panel of 2,105 municipalities, restricting to pre-reform CRAS municipalities (1,415) retains 89 percent of the implicit Frisch-Waugh-Lovell identifying mass; further restricting to those where the incoming mayor replaced the *de facto* CRAS manager on taking office (1,089) retains 79 percent; and conditioning on the dominant hire type being external (861) retains 50 percent. The cascade visualizes the central empirical claim of this section: the headline effect is concentrated where the structural margins of Section 3.3 are jointly favorable, and the per-step shrinkage in muni count is sublinear in the mass retained, exactly the pattern the augmented model predicts.

The augmented model predicts that the alignment margin κ^A is the second binding constraint: the within-subgroup first stage should attenuate sharply in CRAS municipalities where the incoming mayor did not replace the front-line manager on taking office. Table 12 reports the corresponding decomposition, which estimates three coefficients in a single full-sample regression: no CRAS in 2008, CRAS where the manager stayed, and CRAS where the new mayor replaced the manager. We define the manager as the *de facto* frontline operator: the respondent of the unit-level *Censo SUAS* questionnaire, who is the person who actually fills the form and runs the unit day-to-day. The 2008 and 2009 *Censo SUAS* are annual snapshots recorded just before and just after the mayor elected in 2008 takes office, so a change in the recorded manager between the two waves identifies a replacement made by the incoming administration; the *de facto* definition admits an exact CPF match across waves, lets us classify changes as external hires versus internal transfers, and is the object the structural argument in Section 3.3

Figure 4: Cascade of the Analysis Sample and Implicit Identifying Mass



Note. Bar height is the municipality count in each cascade cell. The percentage above each bar, also encoded by the bar shading, is the share of identifying mass the cell retains: the Frisch-Waugh-Lovell share of the residualized, population-weighted variance of the standardized 2009 quota shock $\times$ post that lives in the cell, expressed as a fraction of the full-panel total. Each step adds one restriction, from the full panel of municipalities to those with a pre-reform CRAS, then to those where the mayor elected in 2008 replaced the *de facto* CRAS manager on taking office (measured between the 2008 and 2009 *Censo SUAS* waves), then to those whose dominant manager change was an external hire. “External hire (dominant)” is the subset of CRAS-with-turnover municipalities in which the majority of *de facto* manager changes brought in a successor who was not present in the 2008 CRAS structure anywhere.

maps onto.

The first stage in CRAS municipalities with manager turnover reproduces the pooled headline almost exactly, while the other two cells collapse below the conventional weak-instrument threshold. The joint Wald test of equality across the three coefficients rejects decisively, and the Frisch-Waugh-Lovell decomposition concentrates roughly $\omega_{\text{turn}} \approx 0.93$ of the identifying covariance on the third column. The Anderson-Rubin confidence sets reported within each cell confirm the asymmetry directly: in the CRAS-with-turnover cell, the AR 95% set on vote share is [1.45, 5.50], comfortably bracketing the headline 2.94; in the CRAS-without-turnover cell, the AR set is essentially unbounded, the appropriate inference output when the first stage is statistically indistinguishable from zero. The channel operates where the person who actually runs the unit changes.

Table 12: Heterogeneity by Manager Turnover

	Vote Share			Winning		
	non-CRAS	CRAS / non-Turn	CRAS / Turnover	non-CRAS	CRAS / non-Turn	CRAS / Turnover
First Stage (π_1)	0.075 (0.013)	0.035 (0.010)	0.206 (0.004)	0.075 (0.013)	0.035 (0.010)	0.206 (0.004)
Reduced Form (ρ_1)	-2.370 (2.999)	2.831 (0.945)	0.624 (0.203)	7.795 (10.072)	4.350 (2.355)	0.951 (0.227)
Two Stage Least Squares (β_1)	-26.893 (48.413)	75.166 (43.912)	3.443 (1.250)	111.985 (140.496)	114.629 (49.528)	5.217 (1.439)
Kleibergen-Paap F	35.58	11.73	2623.64	35.58	11.73	2623.64
AR 95% CI	[-15.00, +25.00]	[+12.60, +25.00]	[+1.45, +5.50]	[-15.00, +25.00]	[-15.00, +25.00]	[+2.60, +7.95]
Party Fixed Effects	✓	✓	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: All estimates are obtained from a single full-sample triple-IV regression per outcome. The instrument and the endogenous variable are each interacted with three mutually exclusive indicators, ordered in the table from left to right: non-CRAS (municipalities with no CRAS in the 2008 *Censo SUAS*); CRAS / non-Turn (municipalities with a CRAS in 2008 but no manager change between the 2008 and 2009 waves); and CRAS / Turnover (municipalities with a CRAS in 2008 and a manager change between the 2008 and 2009 waves). The manager is defined as the *de facto* frontline operator: the *cpf_agente* respondent of the unit-level *Censo SUAS* questionnaire, the person who actually fills the form and operates the unit day-to-day, who is not necessarily the formal *Coordenador*. All specifications include year and municipality fixed effects, party fixed effects based on the pre-reform winner (2004-winner in 2008; 2008-winner in 2012), and baseline characteristics $\times$ post: log population, Gini inequality, share of poor, household income per capita, share of formal workers, and urban share (all from Census 2000). Weights: municipal population in 2000. Cluster-robust standard errors at the state level in parentheses. Anderson-Rubin 95% confidence intervals are reported for every cell and are valid under arbitrary instrument weakness; they are the appropriate inference object in the no-CRAS and no-turnover cells, where the Kleibergen-Paap F -statistic falls below the strong-instrument threshold.

The hire-type composition of those changes corroborates the reading. Across 3,357 manager changes in 2,583 municipalities, 75.5 percent involve a successor who was not present in the CRAS structure anywhere in 2008, 14.7 percent involve an internal transfer from another role at the same CRAS, and 9.8 percent involve a transfer from a different CRAS in the same municipality. The dominant case is fresh personnel, not internal reshuffles. Appendix E.5 reports the corresponding heterogeneity table. The within-CRAS-with-turnover return is identified primarily across the external-hire subsample, which carries the bulk of the identifying mass and yields a 2SLS point estimate close to the pooled headline; the internal-same-CRAS cell delivers a comparable first stage and a similar-magnitude vote-share return on a smaller subsample; the internal-other-CRAS cell is too small to generate a within-cell first stage that supports a precisely-estimated 2SLS. The heterogeneity table does not therefore sharply discriminate “external ability hire” from “internal-same-CRAS re-deployment”, and we read it as additional support for the substantive claim that the turnover indicator captures genuine personnel inflows rather than as a discriminating test between hire-type sub-

types.

6.2.1 What Does the Alignment Channel Actually Look Like?

The decomposition tells us that manager turnover is the binding margin within CRAS, but it leaves open what the substitute does once installed. The simplest reading is that incoming managers are partisan loyalists, but that account is at odds with the data.

Table 13: Manager Trajectory

	Parameter	N
<i>Panel A. Does the shock cause manager substitution?</i>		
CRAS units that persisted in 2008 and 2009	0.556 (0.243)	1,341
<i>Panel B. Does the shock select substitutes by prior partisan affiliation?</i>		
Any party affiliation pre-2009	0.826 (0.530)	1,099
<i>Panel C. Does the shock raise the rate at which substitutes run for office?</i>		
Probability of running for office (2010–2016)	0.450 (0.144)	1,099
<i>Panel D. Does the shock align substitutes with the mayor's coalition?</i>		
Joined mayor's party (2009–2012)	0.043 (0.045)	1,099
Joined mayor's coalition (2009–2012)	0.102 (0.051)	1,099
Joined any party (2009–2012)	-0.007 (0.109)	1,099

Each row reports a separate cross-sectional regression of the row outcome on the standardized 2009 quota shock plus Census-2000 controls (log population, share in extreme poverty, household income per capita, urban share, and 2000–2006 population growth), with state fixed effects and population weights. Standard errors (in parentheses) are clustered at the state level. The sample restricts to municipalities in the main regression sample with at least one incumbent CRAS unit observed in both the 2008 and 2009 *Censo SUAS* waves, with at least one substitution event (Panels B–D); Panel A uses the broader sample of municipalities with at least one incumbent unit. “Substitute” is a CPF observed in 2009 at an incumbent unit where the 2008 manager had a different CPF. The mayor’s coalition is the set of parties listed in `winner_coalition_2008`. All coefficients and standard errors are in percentage points (multiplied by 100). The placebo for new CRAS unit openings is reported in Appendix E.1.

Does the shock move people, or does it move infrastructure? Panel A of Table 13 reports the answer to the first reader question. In the subsample of municipalities with at least

one incumbent CRAS unit observed in both the 2008 and 2009 *Censo SUAS* waves, the annual snapshots that bracket the change of mayor, a one-standard-deviation increase in the federal shock raises the probability that the unit experiences a manager substitution by about half a percentage point on a base of 82 percent. The placebo for new CRAS unit openings, reported in Appendix E.1, is a clean null. The shock moves managers, not buildings.

Was the substitute already political? Panel B asks whether the substitute was already a partisan operator before the 2009 reform. The shock raises the share of substitutes with an open pre-2009 party affiliation by only 0.83 percentage points, an effect that is not distinguishable from zero at conventional levels. The descriptive panel of Table 14 reinforces the result: stayers, outgoing managers, and incoming managers have similar pre-2009 affiliation rates, between 16.5 and 19.6 percent, and the gap between incoming and outgoing is not statistically significant. The shock did not select substitutes by prior partisan history.

Table 14: Substitute Manager Characteristics

<i>Panel A. Comparison across managerial roles (incumbent units, 2008–2009)</i>				
	# CPFs	% CPFs	Filiated	Ran
Stayer (same CPF 2008–2009)	269	91.5%	18.4%	5.8%
Outgoing (replaced in 2009)	1,225	89.7%	19.6%	6.5%
Incoming (substitute hired 2009)	1,216	89.0%	16.5%	9.5%
<i>Panel B. Offices sought by incoming managers who ran (TSE 2010–2016)</i>				
	Level	# CPFs	% runners	% substitutes
City council (<i>vereador</i>)	Municipal	99	81.8%	8.14%
Vice mayor	Municipal	18	14.9%	1.48%
Mayor (<i>prefeito</i>)	Municipal	10	8.3%	0.82%
State deputy	State	2	1.7%	0.16%
Any office		121	100.0%	9.95%

Panel A: Each row aggregates manager-CPFs at incumbent CRAS units observed in both the 2008 and 2009 *Censo SUAS* waves. *Stayer* is the same CPF in both years; *Outgoing* is the 2008 manager replaced in 2009; *Incoming* is the 2009 substitute. # CPFs counts distinct managers within the role; % CPFs reports the share of within-role observations that map to a distinct manager (i.e., one minus the duplicate rate from managers serving multiple units in the same municipality). *Filiated* is the share of CPFs with an open party affiliation as of 1 January 2009. *Ran* is the share of CPFs that appear as a candidate for any elected post in the TSE 2010–2016 candidate database. *Panel B:* For substitutes who ran for office, the breakdown by office category. CPFs running for multiple offices across 2010–2016 are counted once per office category, so row totals exceed the unique runner count of 121.

What does the substitute do after taking the post? Panel C of Table 13 reports the causal

evidence: among the substitutes installed in 2009, the probability of running for elected office between 2010 and 2016 rises with the size of the shock, and the response is statistically distinguishable from zero. Panel A of Table 14 reports a complementary descriptive contrast. The running rate of incoming managers, 9.5 percent, is nearly twice the rate among stayers, 5.8 percent; the gap survives when restricted to managers in the same units. The same contrast is absent between outgoing managers and stayers, so being a CRAS manager does not by itself raise the running rate; specifically, being the post-2008 entrant does. Panel B of Table 14 shows that the post-bureaucracy career path is overwhelmingly local: 99 of the 121 unique substitutes who ran did so for *vereador*, with another 18 also running for vice mayor, 10 for mayor, and 2 for state deputy. The office counts are not mutually exclusive because some substitutes ran for multiple offices across the 2010–2016 cycles, so the office-by-office sum exceeds the unique runner count of 121.

Does the substitute align with the local mayor? Panel D of Table 13 answers in two parts. The substitute does not switch to the mayor’s exact party at higher rates after the shock, and is not more likely to join any party in general. The substitute is, however, more likely to file a new affiliation with one of the parties in the mayor’s 2008 electoral coalition. The alignment that operates is broad-coalition, not narrow-party brand: the substitute is positioning herself within the local coalition that just won the mayoral race, not pledging fealty to a single party.

The composite picture is the stepping-stone channel of Colonnelli et al. (2020). The shock makes the CRAS managerial position more attractive as a launching pad for an electoral career: it expands the operational footprint of the office, raises the visibility of the person who runs it, and broadens that person’s procedural reach into local households. The substitutes the shock attracts are not selected on prior partisanship, but they treat the post differently than the inherited managers do, both in subsequent candidacy rates and in coalition affiliation. That difference is what activates the alignment margin κ^A in the model: the multiplier on discretionary effort a_m rises, and the federal authorization translates more strongly into realized expansion.

What the data cannot adjudicate is the mayor’s selection criterion: whether the substitute was hired on capacity grounds, on grounds of political alignment with the incoming coalition, or on a combination of the two. What is unambiguous is what the substitute does with the post once installed. The visibility of the expanded CRAS office is converted into a local electoral candidacy at higher rates than for stayers, regardless of why the mayor originally appointed her. Under either selection story (competent hire later activated by visibility, or politically ambitious hire pre-selected for the launch-

ing pad), the structural implication is the same: realized expansion e_m rises, the voter signal s_{im} shifts upward, the posterior $\Pr(\theta_m = 1 \mid s_{im})$ tracks it, and the inference channel $\lambda_2 e_m$ delivers the additional vote share that κ^A activates. The stepping-stone label refers to this downstream behavior of the substitute, not to a presumption about the mayor's selection process.

7 Conclusion

This paper asks whether local politicians can extract electoral returns from implementing a federally designed cash transfer program. We exploit the 2009 reform that revised the formula for allocating *Bolsa Família* beneficiary quotas across Brazilian municipalities, using the federally-induced authorization shock to instrument realized program expansion, and estimate its effect on the sitting mayor's performance in the 2012 municipal elections. A simple framework organizes the analysis: the federal-to-local implementation chain is multiplicative in physical infrastructure (κ^I , the pre-reform CRAS) and front-line bureaucratic alignment (κ^A , manager turnover). The framework yields a set of testable predictions about where the federal authorization translates into realized expansion, and where that expansion translates into electoral returns. Realized expansion is also bounded mechanically by the eligible-pool margin a municipality faces in 2008: where local saturation is high, there is little room for the shock to operate regardless of capacity.

A one-standard-deviation increase in instrumented beneficiary expansion, approximately 4,949 additional families in the estimation sample, raises the incumbent's vote share by 2.94 percentage points and the probability of reelection by 4.10 percentage points. In natural units, an additional 1,000 beneficiaries between elections corresponds to roughly 0.59 percentage points on vote share and 0.83 percentage points on reelection. The estimates are robust to alternative instrument thresholds, alternative measurement windows for the endogenous variable, alternative outcome definitions, alternative weights, and inverse-probability weighting for selection into the rerunning sample. Anderson-Rubin confidence sets, which remain valid under arbitrary instrument weakness and small cluster counts, exclude zero for both outcomes.

The mechanism evidence locates the headline along the structural margins the framework anticipates. Pre-existing local welfare infrastructure operates as the binding precondition: in municipalities without a CRAS at the time of the reform, the within-subgroup first stage attenuates by roughly a factor of three relative to CRAS municipalities and the corresponding electoral coefficient is too imprecise to interpret.

Within CRAS municipalities, the alignment margin operates through manager substitution: the federal shock causes turnover of the *de facto* front-line manager without causing new unit openings, and three quarters of the manager changes captured by our definition bring in individuals new to the municipal welfare bureaucracy rather than internal reshuffles. The substitutes hired into post-shock units run for elected office at higher rates than the baseline while aligning with the mayor’s electoral coalition rather than switching to the mayor’s exact party or being pre-selected on prior partisan affiliation. The pattern is most easily reconciled with the downstream-behavior half of the stepping-stone channel documented by [Colonnelli et al. \(2020\)](#), not their loyalist-selection half: the federal shock raises the visibility of the CRAS managerial post and makes it more attractive as a launching pad for a local electoral career regardless of the mayor’s selection criterion, an effort margin in which the structural alignment indicator κ^A becomes active. The asymmetry in returns extends to political alignment: per-beneficiary electoral returns are roughly 2.3 times larger under PT-aligned mayors than under non-PT mayors (6.16 versus 2.73 percentage points; Wald $p = 0.040$), an empirical signature of partisan attribution that the centralized-design literature does not address.

These results contribute to four literatures. The political-economy literature on cash transfers documents that incumbents reap electoral returns from program expansion in settings where local discretion is deliberately minimized ([Manacorda et al., 2011](#); [De La O, 2013](#)); we complement that work by studying the opposite institutional configuration, in which the program is centrally designed but the registration pipeline is locally operated. The comparison with [Frey \(2019\)](#), developed in Section 1, makes the sign contrast informative: the two designs identify electoral returns on different sets of municipalities and on different margins of local effort.

The political-accountability literature on shared governance ([Przeworski et al., 1999](#); [Besley, 2006](#); [Ferraz and Finan, 2008](#)) centers on the question of whether voters correctly attribute responsibility across levels of government; our setting differs in that the relevant information concerns positive delivery rather than malfeasance, but it operates through the same retrospective-inference logic and shows that local incumbents can be rewarded for federally-funded programs where local effort remains salient.

The state-capacity literature ([Finan et al., 2017](#); [Dal Bó et al., 2013](#)) emphasizes the personnel and infrastructure through which programs are delivered; our findings identify a margin prior to the quality of bureaucrats, the *existence* of the administrative apparatus, and a complementary margin within the apparatus, the alignment of the front-line manager with the local political principal, in the sense of [Akhtari et al. \(2022\)](#) and

Colonnelli et al. (2020).

Two implications follow. First, centralized program design does not, on its own, depoliticize redistribution: even when eligibility rules, benefit levels, and payment channels are set nationally, local discretion over the registration pipeline generates a margin of electoral return. Second, that return is gated by inherited state capacity and by the political incentives of the front-line bureaucrats who operate it, since the political value of a federal authorization shock is essentially zero in municipalities without the bureaucratic infrastructure to convert it into visible delivery and concentrates within active municipalities in those where the front-line manager turned over after the reform. National rules shape what programs look like on paper, but the local political economy of delivery is where electoral returns are produced and captured.

A natural next step is the structural counterpart to the reduced-form decomposition we report. Calibrating the augmented model with the within-subgroup first-stage moments — normalizing the active cell to $\gamma_{\text{CRAS} \times \text{turn}} = 1$ — pins the relative productivity of the no-CRAS subgroup at approximately one third of the active level and the CRAS-no-turnover subgroup at zero, and the calibrated model recovers the Frisch-Waugh-Lovell weights to within rounding. We leave a full method-of-moments estimation, with parametric inference on γ and a structural interpretation of the dose-response across subgroups, to a companion paper.

References

- Abadie, A., Athey, S., Imbens, G. W., and Wooldridge, J. M. (2023). When should you adjust standard errors for clustering? *Quarterly Journal of Economics*, 138(1):1–35.
- Akhtari, M., Moreira, D., and Trucco, L. (2022). Political turnover, bureaucratic turnover, and the quality of public services. *American Economic Review*, 112(2):442–493.
- Anderson, T. W. and Rubin, H. (1949). Estimation of the parameters of a single equation in a complete system of stochastic equations. *Annals of Mathematical Statistics*, 20(1):46–63.
- Angrist, J. D. and Imbens, G. W. (1995). Two-stage least squares estimation of average causal effects in models with variable treatment intensity. *Journal of the American Statistical Association*, 90(430):431–442.
- Ashworth, S. (2012). Electoral accountability: Recent theoretical and empirical work. *Annual Review of Political Science*, 15:183–201.
- Avis, E., Ferraz, C., and Finan, F. (2018). Do government audits reduce corruption? Estimating the impacts of exposing corrupt politicians. *Journal of Political Economy*, 126(5):1912–1964.
- Besley, T. (2006). *Principled Agents? The Political Economy of Good Government*. Oxford University Press, Oxford.
- Besley, T. and Persson, T. (2009). The origins of state capacity: Property rights, taxation, and politics. *American Economic Review*, 99(4):1218–1244.
- Besley, T. and Persson, T. (2011). *Pillars of Prosperity: The Political Economics of Development Clusters*. Princeton University Press, Princeton, NJ.
- Blandhol, C., Bonney, J., Mogstad, M., and Torgovitsky, A. (2022). When is TSLS actually LATE? *NBER Working Paper*, (29709).
- Brollo, F., Kaufmann, K., and La Ferrara, E. (2020). The political economy of program enforcement: Evidence from Brazil. *Journal of the European Economic Association*, 18(2):750–791.
- Callaway, B., Goodman-Bacon, A., and Sant’Anna, P. H. C. (2024). Difference-in-differences with a continuous treatment. *NBER Working Paper No. 32117*. Work-

- ing paper; see also Angrist & Imbens (1995) for the average-causal-response (ACR) framework.
- Cameron, A. C., Gelbach, J. B., and Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3):414–427.
- Colonnelli, E., Prem, M., and Teso, E. (2020). Patronage and selection in public sector organizations. *American Economic Review*, 110(10):3071–3099.
- Dal Bó, E., Finan, F., and Rossi, M. A. (2013). Strengthening state capabilities: The role of financial incentives in the call to public service. *Quarterly Journal of Economics*, 128(3):1169–1218.
- de Chaisemartin, C. (2017). Tolerating defiance? Local average treatment effects without monotonicity. *Quantitative Economics*, 8(2):367–396.
- De La O, A. L. (2013). Do conditional cash transfers affect electoral behavior? Evidence from a randomized experiment in Mexico. *American Journal of Political Science*, 57(1):1–14.
- Ferejohn, J. (1986). Incumbent performance and electoral control. *Public Choice*, 50(1–3):5–25.
- Ferraz, C. and Finan, F. (2008). Exposing corrupt politicians: The effects of Brazil’s publicly released audits on electoral outcomes. *Quarterly Journal of Economics*, 123(2):703–745.
- Finan, F., Olken, B. A., and Pande, R. (2017). The personnel economics of the developing state. In Banerjee, A. V. and Duflo, E., editors, *Handbook of Economic Field Experiments*, volume 2, pages 467–514. North-Holland.
- Fiszbein, A. and Schady, N. (2009). *Conditional Cash Transfers: Reducing Present and Future Poverty*. World Bank Policy Research Report. World Bank, Washington, DC.
- Frey, A. (2019). Cash transfers, clientelism, and political enfranchisement: Evidence from Brazil. *Journal of Public Economics*, 176:1–17.
- Frey, A. and Santarrosa, R. (2025). The politicization of bureaucrats: Evidence from Brazil. *Journal of Politics*.
- Fried, B. J. (2012). Distributive politics and conditional cash transfers: The case of Brazil’s Bolsa Família. *World Development*, 40(5):1042–1053.

- Gerard, F., Naritomi, J., and Silva, J. (2025). Cash transfers and the local economy: Evidence from Brazil. Working paper, 2025. Circulating identifiers: CEPR Discussion Paper 16286, World Bank Policy Research Working Paper 9778, and SSRN 3886759. Earlier version contains the technical appendix on the quota allocation formula.
- Goldsmith-Pinkham, P., Hull, P., and Kolesár, M. (2022). Contamination bias in linear regressions. *NBER Working Paper*, (30108).
- Hellmann, A. G. (2015). How does Bolsa Família work? Best practices in the implementation of conditional cash transfer programs in Latin America and the Caribbean. IDB Technical Note IDB-TN-856, Inter-American Development Bank.
- Holmström, B. (1999). Managerial incentive problems: A dynamic perspective. *Review of Economic Studies*, 66(1):169–182.
- Imai, K. and Kim, I. S. (2019). When should we use unit fixed effects regression models for causal inference with longitudinal data? *American Journal of Political Science*, 63(2):467–490.
- Imbens, G. W. and Angrist, J. D. (1994). Identification and estimation of local average treatment effects. *Econometrica*, 62(2):467–475.
- Jones, C. I. (2011). Intermediate goods and weak links in the theory of economic development. *American Economic Journal: Macroeconomics*, 3(2):1–28.
- Klašnja, M. and Titiunik, R. (2017). The incumbency curse: Weak parties, term limits, and unfulfilled accountability. *American Political Science Review*, 111(1):129–148.
- Kolesár, M. (2013). Estimation in an instrumental variables model with treatment effect heterogeneity. *Working Paper*. Princeton University.
- Kremer, M. (1993). The O-ring theory of economic development. *Quarterly Journal of Economics*, 108(3):551–575.
- Lindbeck, A. and Weibull, J. W. (1987). Balanced-budget redistribution as the outcome of political competition. *Public Choice*, 52(3):273–297.
- Lindert, K., Linder, A., Hobbs, J., and de la Brière, B. (2007). The nuts and bolts of Brazil’s Bolsa Família program: Implementing conditional cash transfers in a decentralized context. Social Protection Discussion Paper 0709, World Bank.
- MacKinnon, J. G., Nielsen, M. Ø., and Webb, M. D. (2023). Cluster-robust inference: A guide to empirical practice. *Journal of Econometrics*, 232(2):272–299.

- Manacorda, M., Miguel, E., and Vigorito, A. (2011). Government transfers and political support. *American Economic Journal: Applied Economics*, 3(3):1–28.
- Maskin, E. and Tirole, J. (2004). The politician and the judge: Accountability in government. *American Economic Review*, 94(4):1034–1054.
- McKenzie, D. (2012). Beyond baseline and follow-up: The case for more t in experiments. *Journal of Development Economics*, 99(2):210–221.
- Nickell, S. (1981). Biases in dynamic models with fixed effects. *Econometrica*, 49(6):1417–1426.
- Oster, E. (2019). Unobservable selection and coefficient stability: Theory and evidence. *Journal of Business & Economic Statistics*, 37(2):187–204.
- Pereira, T. S., Lucas, V. M., and Resende Filho, M. d. A. (2020). Ciclos políticos e eleitorais na execução do programa Bolsa Família em nível municipal. *Pesquisa e Planejamento Econômico (PPE)*, 50(1):67–98.
- Przeworski, A., Stokes, S. C., and Manin, B., editors (1999). *Democracy, Accountability, and Representation*. Cambridge Studies in the Theory of Democracy. Cambridge University Press, Cambridge.
- Robinson, J. A. and Verdier, T. (2013). The political economy of clientelism. *Scandinavian Journal of Economics*, 115(2):260–291.
- Sugiyama, N. B. and Hunter, W. (2013). Whither clientelism? Good governance and Brazil’s Bolsa Família program. *Comparative Politics*, 46(1):43–62.
- Zucco, C. (2013). When payouts pay off: Conditional cash transfers and voting behavior in Brazil 2002–10. *American Journal of Political Science*, 57(4):810–822.

Appendix

Table of Content

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A Institutional Background

This appendix supplements Section 2 with the institutional detail that justifies the municipal focus of the empirical analysis. Rather than restate the full design of *Bolsa Família*, we concentrate on the multi-tier governance arrangement and the political incentives it generates at each tier; the body discussion of eligibility, conditionalities, and benefit structure is not repeated here.

A.1 Multi-tier Governance and the Municipal Margin

Bolsa Família is administered through a three-tier division of labor in which credit attribution is asymmetric across tiers. The federal government, through the *Ministério do Desenvolvimento Social* (MDS), sets eligibility rules, determines the per-family transfer value, generates the monthly payment list from *Cadastro Único* (CadÚnico), and pays beneficiaries directly through *Caixa Econômica Federal*. State governments coordinate intersectoral monitoring of conditionalities (school attendance, vaccination, prenatal care) and provide training to municipalities through the state-level *Decentralized Management Index* (IGD-E), but they neither select beneficiaries nor disburse funds. Municipalities are the operational tier: they identify eligible families, conduct CadÚnico interviews, perform household visits, monitor conditionality compliance, and resolve day-to-day implementation issues, financed by the municipal-level IGD-M transfer.²³

²³Responsibilities are formalized by interministerial ordinance MS/MDS No. 2,509/2004 and Decree No. 5,209/2004, which created the *Bolsa Família* Program Management Council (CGPBF). Municipal

The political incentives generated by this division differ sharply across tiers. At the federal tier, the program is the most visible component of the social agenda; the 2006 re-election of President Lula was widely credited to BF expansion, and even ideologically opposed administrations have rebranded rather than dismantled the program (notably the 2021 transition to *Auxílio Brasil*). State governors derive only indirect political benefits because they do not operate the program. Mayors occupy the position studied in this paper: they do not write the rules and do not control the per-family payment, but they own the registration pipeline that determines which families are actually enrolled. When the federal authorization expands, mayors who deploy CRAS infrastructure aggressively can enroll more families and translate the federal authorization into locally visible welfare delivery; mayors who do not, leave the federal authorization unused. The municipal-level credit-attribution mechanism investigated in this paper rests on this asymmetry between the central authority that designs the program and the local authority that operates the last mile of delivery.

The empirical asymmetry that motivates Section 6 runs through this last-mile margin. Mayors with a pre-existing CRAS in 2008 inherit the operational capacity to act on a federal quota expansion; mayors without one do not. Within the CRAS subgroup, mayors who replaced the front-line manager between the 2008 and 2009 *Censo SUAS* waves redirect that capacity in favor of beneficiary growth; mayors who kept the inherited manager do not. The municipal margin we identify is therefore the joint margin of (a) inherited infrastructure and (b) front-line staff alignment with the new mayor.

B Data Sources and Construction

This appendix describes the data sources, access points, and construction procedures underlying the empirical analysis. The analysis draws on four families of administrative and survey data: social-assistance records from the federal Ministry of Social Development, electoral records from the Superior Electoral Court, demographic data from the national statistics institute, and administrative records on local welfare infrastructure. All sources are publicly available or were obtained through formal information requests; we provide direct hyperlinks throughout to facilitate replication.

control over the registration pipeline operates primarily through the *Centros de Referência de Assistência Social* (CRAS), which we describe in detail in Section 2 of the body and use in the mechanism analysis of Section 6.

B.1 Data Sources

B.1.1 Bolsa Família Administrative Records

The treatment-side variables are constructed from administrative records maintained by the *Ministério do Desenvolvimento e Assistência Social, Família e Combate à Fome* (MDS).

Realized beneficiary counts. The municipality-by-month series on the number of *Bolsa Família* beneficiary families between 2004 and 2016 is extracted from the [MDS beneficiary visualization portal](#). Each observation records the number of families receiving transfers in municipality m in month t . We use this series to construct the endogenous variable: the twelve-month moving average of beneficiary families between January and December 2009, and alternative treatment windows extending through December 2012. The raw series is also used to construct the time-series figures in Section 2.

Municipal quotas. Historical quota assignments, the federally determined ceilings on the number of families each municipality was authorized to serve, are not systematically available through public portals. We obtained municipality-level quota series through a formal [Lei de Acesso à Informação \(LAI\) request](#) filed with the MDS. The response includes the official quota for each municipality at each revision date (2004, 2006, 2009, and 2012). These data are essential for constructing both the counterfactual quota (equation 17) and the preferred instrument (equation 18).

Policy documents and institutional timing. The timing of quota revisions, implementation instructions, and federal communications to municipalities is established from the archive of [historical MDS Informes](#). In particular, *Informe nº 169* (April 2009) documents the formal announcement of the quota expansion and the staggered incorporation schedule, which is central to the no-anticipation argument in Section 4.1. Earlier *Informes* are used to establish the pre-reform quota methodology and the timing of the 2006 revision.

Cadastro Único aggregates. For cross-checking and descriptive purposes, we draw on the [Cadastro Único dashboard](#) and the related [Bolsa Família and Cadastro Único monitoring panel](#), both maintained by the MDS. These portals provide municipality-level summary statistics on registered families, active beneficiaries, and coverage ratios that

are useful for validating the patterns in the administrative microdata. Additional meta-data and complementary datasets are accessed through the [MDS open data repository](#).

B.1.2 Electoral Records

Electoral outcomes, candidate characteristics, and municipality-level election results are obtained from the *Tribunal Superior Eleitoral* (TSE).

Municipality- and candidate-level results. The underlying microdata come from the [TSE open data repository](#), which provides complete machine-readable files for every municipal election since 1996. For each municipality-election pair, we extract: total votes by candidate, total valid votes (excluding blank and null ballots), total votes cast, and the number of registered voters. For each candidate, we extract: party affiliation, educational attainment, declared occupation, and prior electoral history. These files are used to construct the outcome variables (incumbent vote share, reelection indicator, victory margin), the rerunning indicator, the number-of-candidates control.

Incumbent identification. An incumbent in election year t is defined as the candidate who won the mayoral race in $t - 4$ in the same municipality. We match incumbents across election cycles using candidate identifiers provided by the TSE. The estimation sample is restricted to municipalities where the 2008 winner appears as a candidate in the 2012 election. Municipalities where the incumbent was term-limited, chose not to run, or could not be matched across cycles are excluded.

Validation. Aggregate tabulations are cross-checked against the [TSE electoral statistics panel](#), which provides consolidated statistics by election year and geographic unit.

B.1.3 Demographic and Census Data

Population counts, intercensal dynamics, and household-level information are obtained from the *Instituto Brasileiro de Geografia e Estatística* (IBGE).

Population estimates. Municipal population estimates for intercensal years are downloaded from the [IBGE population estimates page](#). These estimates enter the counterfactual quota formula (equation 17) through the population-growth ratio $n_m^{[2000,2006]}$ and are used as regression weights in the preferred specification.

Census microdata. Household-level data from the 2000 Demographic Census are accessed through the [IBGE census microdata page](#). We use these files to construct the municipality-level count of poor families ($Poor_m^{2000}$) that enters the counterfactual quota, applying poverty thresholds of R\$90, R\$120, and R\$137 per capita to generate the alternative instrument constructions.

PNAD estimates. State-level poverty estimates from the 2006 *Pesquisa Nacional por Amostra de Domicílios* (PNAD) provide the numerator $Poor_s^{2006}$ in the counterfactual quota formula. Because the PNAD is representative at the state level but not at the municipality level, the counterfactual formula distributes the state-level poverty count across municipalities using each municipality's share of poor families in the 2000 Census, adjusted for intercensal population growth.

Municipal boundaries. To ensure consistent geographic units across all data sources, we use the [IBGE territorial division files](#), which provide official seven-digit municipality codes and track boundary changes over time.

B.1.4 Local Welfare Infrastructure: Censo SUAS

Administrative records on local social-assistance infrastructure are drawn from the *Censo SUAS*, an annual census of all social-assistance units in Brazil, accessed through the [Censo SUAS data portal](#). Aggregate patterns can also be inspected through the [Observatório do Censo SUAS dashboard](#).

CRAS-level variables. From the 2008 wave, we extract for each *Centro de Referência de Assistência Social* (CRAS) unit: the municipality code, a unique CRAS identifier, the CPF of the responsible coordinator, and four operational variables, whether the unit conducts active outreach (*busca ativa*), the weekly number of individual and family interviews, the weekly number of home visits (*visitas domiciliares*), and the number of *Bolsa Família* families followed through the *Serviço de Proteção e Atendimento Integral à Família* (PAIF).

Municipality-level aggregation. CRAS-level records are aggregated to the municipality level as follows. A municipality is coded as having a CRAS if at least one unit appears in the 2008 *Censo SUAS* ($N_{mun} = 1,415$ in the estimation sample; 690 without). Active outreach is coded as one if any CRAS in the municipality reported conducting

busca ativa. The continuous operational measures, weekly interviews, weekly home visits, and PAIF families, are summed across all CRAS units within each municipality. Binary indicators for each operational dimension are set to one if the municipality-level sum is strictly positive.

B.2 Counterfactual Quota Construction

The counterfactual quota is designed to approximate the number of beneficiary families each municipality would have been authorized to serve in 2009 had the federal government retained the pre-reform allocation methodology. Constructing this object requires recovering the formula used to set the 2004 and 2006 quotas and applying it to the data that would have been available immediately before the 2009 revision. This subsection describes the procedure, the choices involved, and the limitations of the exercise.

Recovering the pre-reform formula. The MDS does not publish a single canonical document specifying the exact formula used to compute municipal quotas in each revision round. The allocation methodology must instead be pieced together from scattered references across *Informes*, technical notes, and secondary sources.²⁴ The core logic is straightforward: each municipality’s quota is its share of state-level poverty, computed from Census data and adjusted for intercensal population growth, scaled by the state-level estimate of poor families from the most recent PNAD. The formula is given by equation 17 in the main text.

Poverty thresholds. A practical difficulty is that the definition of “poor family” changed across revision rounds. Historical MDS *Informes* reference different per-capita income cutoffs, R\$90, R\$120, and R\$137, without always specifying which threshold applies to which data source in a given revision. We address this by testing all three cutoffs against the official quotas for 2004 and 2006, asking which combination best replicates the observed allocations. For the 2000 Census data, a cutoff of R\$90 per capita produces the closest match to the 2004 and 2006 official quotas. For the 2004 PNAD data, a cutoff of R\$120 per capita performs best.

²⁴The most detailed descriptions appear in [Lindert et al. \(2007\)](#) and [Hellmann \(2015\)](#). [Gerard et al. \(2025\)](#) provide an independent reconstruction with a full technical appendix.

The 2009 threshold revision. The preferred instrument uses a poverty cutoff of R\$137 in the counterfactual formula. This choice is motivated by *Informe nº 169* (April 2009), which announced that the 2009 quota expansion would incorporate a higher income threshold corresponding to half the minimum wage, approximately R\$137 at the time, and would add approximately 1.6 million new authorized slots nationwide. Because the instrument is defined as the difference between the actual 2009 quota and the counterfactual (equation 18), using R\$137 in the counterfactual ensures that the instrument captures variation attributable to the change in methodology rather than to the mechanical effect of a higher poverty line applied to the same data. Appendix C reports results using alternative cutoffs of R\$90 and R\$120 in the counterfactual; the main findings are not sensitive to this choice.

Replication accuracy. We do not exactly reproduce the official MDS quota numbers for any revision round. The municipality-level correlation between our constructed quotas and the official figures is approximately 0.960.98 for both 2004 and 2006, but non-trivial discrepancies remain for individual municipalities. The most likely sources of these differences are: unpublished administrative adjustments applied by the MDS after the formula-based allocation (for example, minimum-quota floors for very small municipalities); small differences in the auxiliary population estimates used to operationalize the intercensal growth factor; and rounding conventions that are not documented in public sources. Gerard et al. (2025) report a similar pattern and reach the same conclusion: the residual discrepancies are consistent with measurement issues rather than political manipulation of the allocation.²⁵ The instrument exploits cross-municipal variation in the *difference* between actual and counterfactual quotas, so uniform or idiosyncratic level discrepancies in the counterfactual do not bias the estimates as long as they are uncorrelated with the error term in the outcome equation.

B.3 Summary Statistics

Table B.1 reports full distributional statistics for the baseline characteristics of the estimation sample, drawn from the *Atlas do Desenvolvimento Humano no Brasil* (PNUD Atlas). The sample consists of the 2,105 municipalities whose 2008 incumbent sought reelection in 2012. The first six rows are the controls entering the preferred DiD-IV specification interacted with the post-reform indicator; the last four rows are additional

²⁵As a further check, we verify that the rank correlation between our constructed quotas and the official figures exceeds 0.99 in both years, and that the distribution of discrepancies is symmetric and centered near zero, with no systematic relationship to municipal political characteristics.

PNUD variables reported for descriptive context only. All variables use 2000 Demographic Census microdata as the underlying source.

Table B.1: Baseline Characteristics — Full Distributional Detail

	N	Min	P10	P25	Mean	Median	P75	P90	Max	SD
Population (log)	2,105	6.678	8.084	8.583	9.359	9.264	9.942	10.725	15.569	1.074
Gini (%)	2,105	30.000	46.000	51.000	54.839	55.000	59.000	63.000	80.000	6.727
Share poor (%)	2,105	7.660	33.656	48.090	64.100	66.780	82.980	88.526	99.000	20.976
Income per capita (R\$)	2,105	69.200	128.4	172.0	337.2	300.1	461.4	600.4	1,759.8	195.1
Share formal workers (%)	2,105	2.890	13.016	19.790	35.785	33.990	50.150	62.116	86.380	18.280
Urban share (%)	2,105	0.000	27.148	40.958	58.769	59.197	77.680	90.946	100.0	23.298
HDI (IDHM)	2,105	0.246	0.383	0.434	0.522	0.528	0.607	0.658	0.791	0.105
Expected years of schooling	2,105	2.820	5.864	6.970	8.284	8.360	9.740	10.530	13.020	1.800
Life expectancy at birth (years)	2,105	57.810	62.624	65.130	68.290	68.800	71.350	73.266	77.240	4.011
Illiteracy rate (15+ years, %)	2,105	0.910	7.512	11.790	21.979	18.740	32.000	40.018	59.950	12.495

Notes: Distributional statistics for baseline characteristics of the 2,105 municipalities in the estimation sample (one observation per municipality). All variables are drawn from the *Atlas do Desenvolvimento Humano no Brasil* (PNUD) using 2000 Demographic Census microdata as the underlying source. Gini and urban share are displayed multiplied by 100 to align with the other percentage-scale variables. Population is reported as a natural logarithm. The first six rows correspond to the baseline controls entering the preferred DiD-IV specification interacted with the post-reform indicator (Section 5, Column 3). The last four rows are additional PNUD variables reported for descriptive context and are not included as controls.

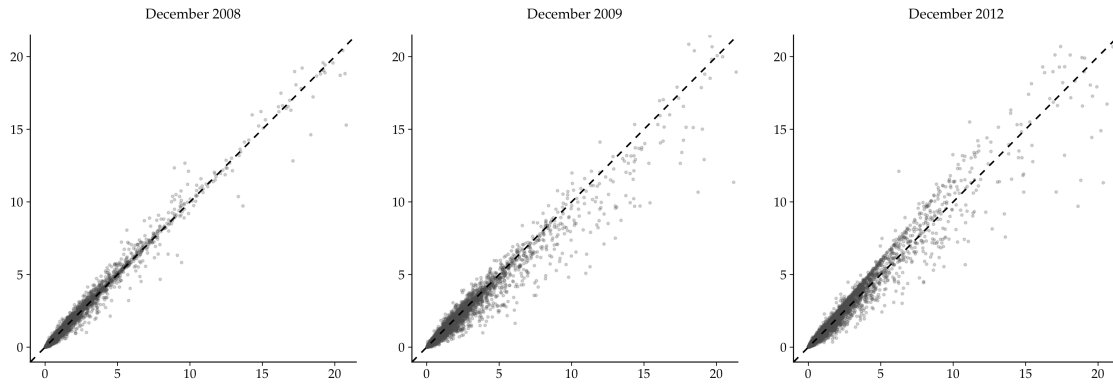
B.4 Authorized Quota vs. Realized Enrollment: 2008, 2009, and 2012

Figure B.1 extends Figure 2 in the main text to include December 2012, allowing a comparison of the quota-enrollment relationship across the full pre-reform, post-reform, and outcome-year snapshots.

B.5 Authorized Quota vs. Counterfactual Quota, 2009

Figure B.2 visualises the instrument directly: each point compares a municipality's actual 2009 quota to the counterfactual quota it would have received under the pre-reform allocation rule. Points above the 45-degree line received a larger quota than the formula predicted (positive shock); points below received less (negative shock). The dispersion around the 45-degree line is the identifying variation used by the instrument.

Figure B.1: Authorized Quota vs. Realized Enrollment, 2008, 2009, and 2012



Note. Each dot is a municipality. The horizontal axis shows the federally authorized quota and the vertical axis shows the twelve-month moving average of realized *Bolsa Família* enrollment (in thousands). The dashed line marks the 45-degree locus of full take-up. December 2008 uses the pre-reform quota (*EST_2006*); December 2009 and December 2012 use the respective post-reform quotas. The rightward shift of the scatter in December 2009 relative to December 2008 reflects the quota expansion; by December 2012 most municipalities have closed the gap, as enrollment caught up with the expanded ceilings. Both axes are capped at the 99th percentile of the cross-municipal distribution across all three years; large urban municipalities above this threshold are excluded from the frame but not from any analysis.

B.6 Realized vs. Authorized: Unweighted National Ratio

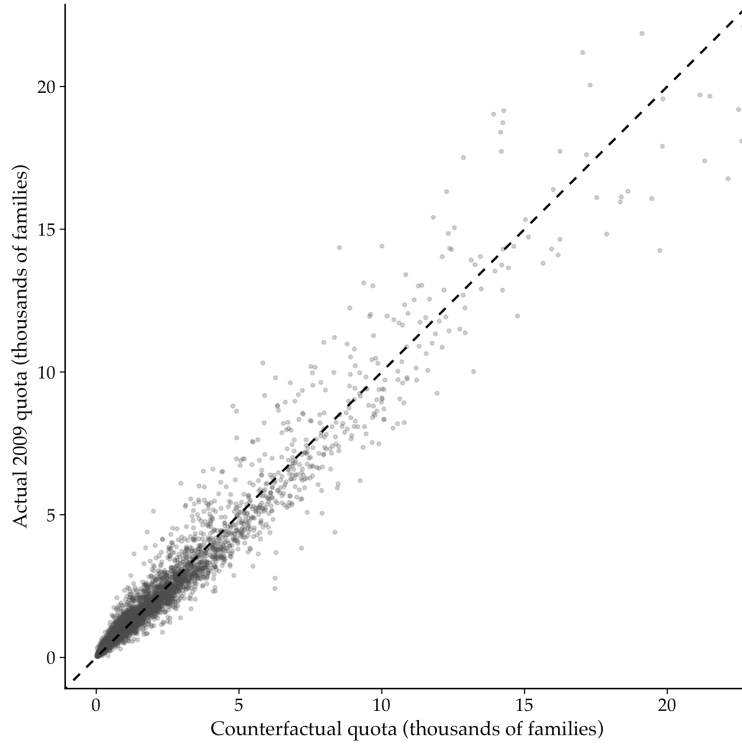
B.7 Realized vs. Authorized: By-State Breakdown

Figures B.4 and B.5 replicate the national ratio disaggregated by state. The unweighted panel treats every municipality in the state equally; the weighted panel weights by quota size so large municipalities dominate the within-state mean. The two views together expose the composition of the national average: states with many small municipalities (e.g., those in the Northeast) see ratios above one in the unweighted view, but once weighted by quota size the series sit near or below one.

B.8 Baseline Characteristics: Selection and Role

This appendix documents the six pre-treatment baseline characteristics entering the preferred specification of Section 5 (Columns 3 and 6 of Tables 6–8). For each variable we describe what it proxies, why it belongs in the identification argument, and how it enters the regression. A directed acyclic graph summarising the joint role of the covariates closes the appendix.

Figure B.2: Actual 2009 Quota vs. Counterfactual Quota



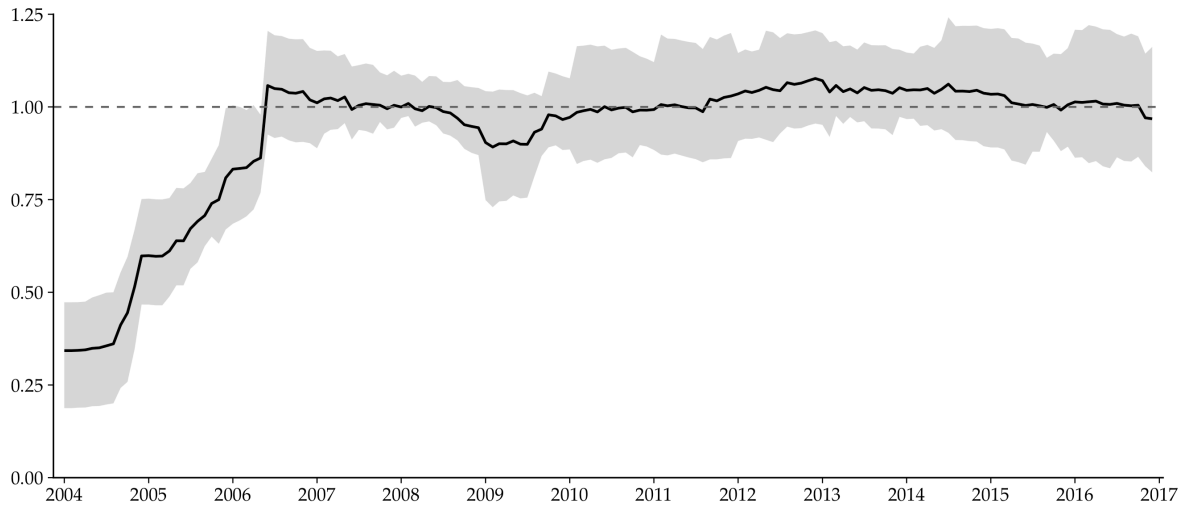
Note. Each dot is a municipality. The horizontal axis plots the counterfactual 2009 quota (the quota that would have prevailed under the pre-reform allocation formula applied to 2006 PNAD data, with poverty cutoffs of R\$90 and R\$137 per capita). The vertical axis plots the actual 2009 quota. Both are in thousands of authorized families. The dashed line is the 45-degree locus of zero shock: points above received more than the pre-reform formula would have given, points below received less. Axes are capped at the 99th percentile of the joint distribution.

B.9 Why Interact with the Post Indicator?

Municipality fixed effects absorb time-invariant levels. A baseline characteristic measured in 2000 is, by construction, constant within each municipality and therefore absorbed by α_m in the panel regression. To allow a pre-treatment covariate to play a meaningful role in the identification argument, we interact it with the post-reform indicator $\text{Post}_t = \mathbb{1}[t = 2012]$. This specification absorbs *differential trends* between 2008 and 2012 correlated with the baseline level of the covariate, not the level itself.

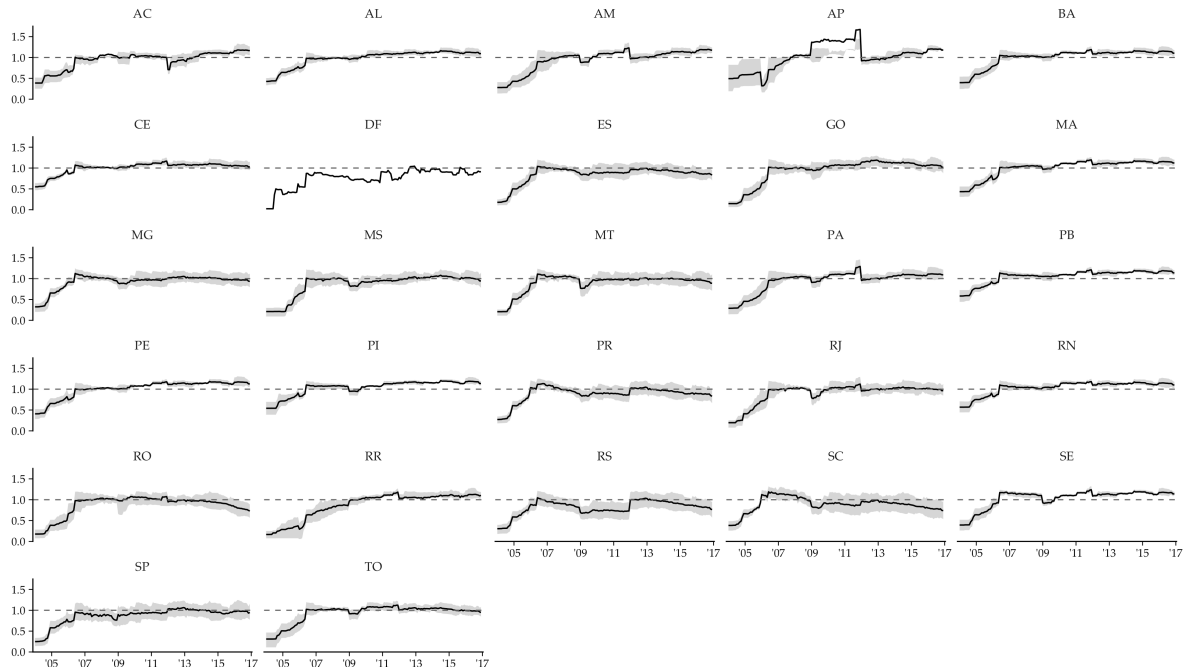
Formally, for each baseline characteristic X_m the specification adds $\gamma_X \cdot X_m \cdot \text{Post}_t$ to the estimating equation. The estimand γ_X captures whether municipalities with higher X_m experienced systematically different electoral trajectories between the two election years. If the quota shock is correlated with X_m , as happens mechanically with log population and the share of poor families, which enter the counterfactual formula explicitly, controlling for $X_m \cdot \text{Post}_t$ purges the instrument of the variation attributable

Figure B.3: Realized vs. Authorized Enrollment, Unweighted



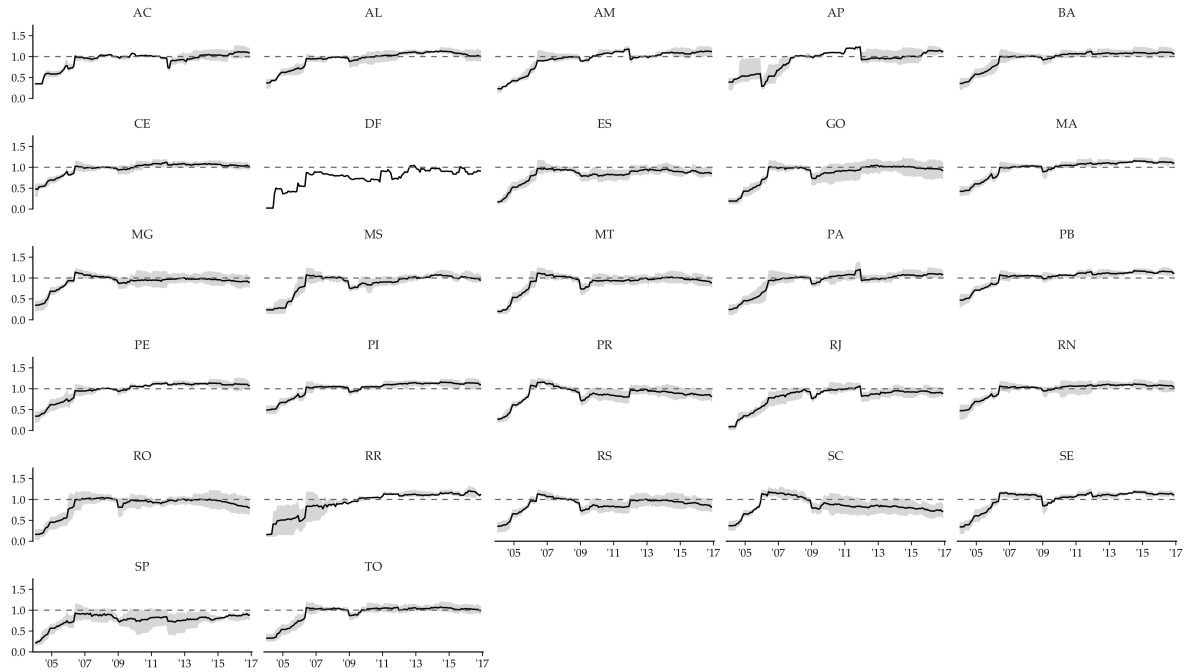
Note: Companion to Figure 1. Each point on the line is the simple cross-municipal mean of the ratio of realized *Bolsa Família* beneficiaries to the authorized quota in a given month, weighting every municipality equally regardless of size. The shaded band spans the 25th to 75th percentile of the cross-municipal distribution. The dashed horizontal line marks full take-up (ratio = 1). The unweighted view is dominated by small municipalities, many of which report realized enrollment above their authorized ceiling, which pushes the simple cross-municipal mean above one for most of the sample period.

Figure B.4: Realized vs. Authorized Enrollment by State, Unweighted



Note. Each panel is one state. Within each state, the black line is the simple cross-municipal mean of realized/authorized at each date, and the shaded band spans the 25th to 75th percentile across municipalities within the state. The dashed horizontal line marks full take-up (ratio = 1). The vertical scale is common across all state panels to facilitate comparison.

Figure B.5: Realized vs. Authorized Enrollment by State, Weighted by Quota Size



Note. Companion to Figure B.4. Each panel is one state. The black line is the within-state quota-weighted mean of realized/authorized at each date, and the shaded band spans the weighted 25th and 75th percentiles. The dashed horizontal line marks full take-up (ratio = 1). Weighting by quota size shifts the within-state average toward large municipalities, which typically sit at or below the authorized ceiling.

to such differential trends, isolating the policy-induced component.

B.10 Variable-by-Variable Discussion

Log Population 2000. The logarithm of the 2000 Census municipal population is a direct input to the counterfactual quota formula (see Appendix B.2). Larger municipalities mechanically received larger quotas both under the pre-reform methodology and under the 2009 revision. If larger municipalities also exhibited distinct electoral trends between 2008 and 2012, driven by urbanisation, media penetration, or differences in political culture, the quota shock would carry a component of size-correlated trend variation. Including $\log(\text{Pop}_m^{2000}) \cdot \text{Post}_t$ strips this component out, so the identifying variation is the policy-induced deviation of the 2009 quota from its counterfactual at fixed size.

Share of Poor Families 2000. The share of families below the eligibility threshold in the 2000 Census is the numerator of the counterfactual quota. Poorer municipalities receive larger quotas both in the counterfactual and in the 2009 revision; the instrument

is the difference between the two. Including $\text{Poor}_m^{2000} \cdot \text{Post}_t$ absorbs any differential electoral trajectory that correlates with baseline poverty, for instance, because poorer municipalities were simultaneously exposed to other federal transfer programs over 2008–2012, or because voter behaviour in poor municipalities responds differently to redistribution. With this control in place the instrument exploits only the *formula-driven* wedge between actual and counterfactual quotas.

Gini Inequality 2000. The Census 2000 Gini coefficient measures within-municipality income inequality. Unlike the share of poor, which captures the *average* level of deprivation, the Gini captures the *dispersion* of income and is largely orthogonal to it in the Brazilian cross-section (the Gini-poverty correlation in the estimation sample is around 0.37). Including $\text{Gini}_m^{2000} \cdot \text{Post}_t$ controls for differential 2008–2012 trends that might correlate with baseline inequality: cities with larger income dispersion may have different electoral dynamics around redistributive policies, a point well documented in the Brazilian political-economy literature ([Zucco, 2013](#); [Sugiyama and Hunter, 2013](#)).

Income per Capita 2000. Household income per capita captures the municipality’s overall standard of living and enters the political-economy literature as the canonical covariate for “economic voting” ([Ferraz and Finan, 2008](#); [Brollo et al., 2020](#)). It is strongly negatively correlated with the share of poor ($\rho \approx -0.92$ in the sample) but carries independent variation at higher income levels. Including $\text{Income}_m^{2000} \cdot \text{Post}_t$ controls for the differential response of better-off municipalities to macro-economic shocks between 2008 and 2012, purging the quota shock of any component that picks up trends driven by aggregate income growth.

Share of Formal Workers 2000. The share of workers in formal employment proxies labour-market structure and local state capacity: municipalities with a higher formal-employment share tend to have stronger fiscal capacity, denser bureaucracies, and more developed civil-registry infrastructure. It carries genuinely independent variation beyond income ($\rho \approx 0.74$): two equally poor municipalities can differ substantially in the share of formal versus informal work. Including $\text{FormalWorkers}_m^{2000} \cdot \text{Post}_t$ absorbs differential trends associated with labour-market informality, which are especially relevant over 2008–2012 given the concurrent expansion of formal employment registration programs at the federal level.

Urban Share 2000. The share of the municipal population living in urban settings is a central axis of Brazilian political economy (Ferraz and Finan, 2008; Brollo et al., 2020). Urban and rural municipalities differ in campaign technology, voter monitoring costs, and the salience of federal transfers in local political discourse. Including $\text{UrbanShare}_m^{2000} \cdot \text{Post}_t$ absorbs the differential electoral trajectory between these two political economies, which might otherwise correlate with quota trends through unobserved state-level policy heterogeneity.

The preceding order (Population, Share Poor, Gini, Income, Formal Workers, Urban) groups the six variables thematically: (i) size, (ii) the direct poverty channel that enters the counterfactual, (iii) inequality as orthogonal complement to poverty, (iv) income as the classical economic-voting control, (v) labour-market structure, (vi) urbanicity. If you prefer the order to mirror Table B.2 (Gini first), I can reorder. The same six variables appear in the full distributional table (Table B.1), the variant diagnostic (Table B.2), and in the *Baseline Characteristics* row of the main-text tables, so renaming or reordering should propagate to all three.

B.11 Identification DAG

Figure B.6 summarises the role of each covariate in the identification argument. The quota shock Z_m affects realised beneficiary expansion D_m , which in turn affects the 2012 incumbent vote share Y_m^{2012} . Pre-treatment characteristics X_m^{2000} (log population, Gini, share of poor, income per capita, share of formal workers, urban share) are correlated with Z_m through the counterfactual quota formula and can affect Y_m^{2012} through differential 2008–2012 trends. Including each $X_m^{2000} \cdot \text{Post}_t$ closes the back-door paths shown as dashed arrows. The 2008 incumbent vote share does not enter the preferred specification: in a DiD design with municipality fixed effects, the pre-treatment outcome level is absorbed mechanically, and its interaction with Post would reintroduce the Nickell lagged-dependent-variable concern without buying additional identification (Nickell, 1981; Imai and Kim, 2019).

B.12 Variant Diagnostic

Table B.2 documents the sensitivity of the preferred 2SLS estimate to the composition of the baseline-characteristics bundle. The table has four panels. Panels A and C start from a specification that includes only year and municipality fixed effects, and then *add*

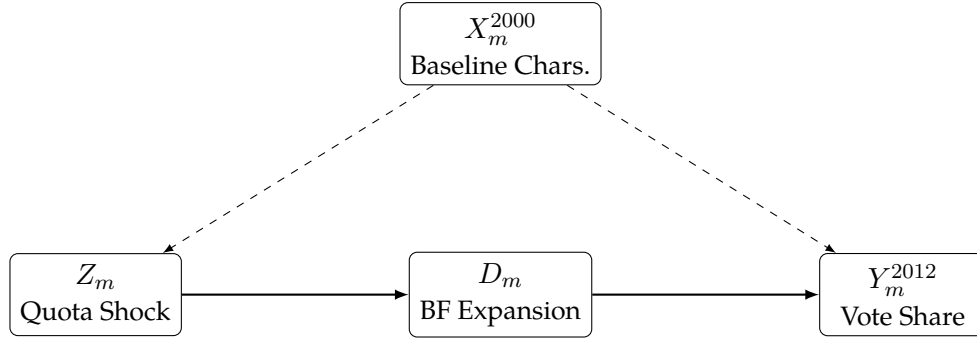


Figure B.6: Identification DAG. Solid arrows are the causal path of interest ($Z \rightarrow D \rightarrow Y$). Dashed arrows are back-door paths closed by including baseline characteristics $\times$ Post as controls. The 2008 incumbent vote share is omitted from the controls: with municipality fixed effects, the DiD specification already differences it out.

one baseline characteristic $\times$ Post at a time; the column header names the characteristic added. Panels B and D run the opposite test: each column includes *all* six baseline characteristics $\times$ Post *except* the one named in the column header. Panels A and B are for the Vote Share outcome; Panels C and D are for the Winning indicator. For each cell we report the IV coefficient on realised expansion, its cluster-robust standard error, and the Kleibergen–Paap first-stage F -statistic.

Three patterns stand out. First, the *add-one* panels (A and C) show that no single baseline, on its own, fully matches the preferred estimate: the point estimates in Panels A and C lie between the minimal specification (year and muni FE only) and the full Column 3 estimate reported in Section 5. This is mechanical, each baseline partially closes a different back-door path. Second, the *leave-one-out* panels (B and D) show that the preferred estimate is stable to dropping any single covariate: the point estimates move only modestly across Panel B columns, and the Kleibergen–Paap F remains above the Stock–Yogo threshold in every leave-one-out specification. Third, the Winning panels (C and D) carry larger standard errors than the Vote Share panels, consistent with the binary nature of the outcome, but the qualitative pattern (no single covariate drives the result) is the same.

C Additional Details on Robustness

This appendix reports the full battery of robustness exercises that support the main results. It is organized in three blocks. First, *identification validity* (§C.1) documents that the instrument is uncorrelated with pre-reform outcomes, that the first stage is concentrated in the post-reform years, and that the monotonicity condition underly-

Table B.2: Variant Diagnostic: Role of Each Baseline Characteristic

	Gini	Share Poor (%)	Income (R\$)	Formal Workers (%)	Population (log)	Urban Share
<i>Panel A: Adding one baseline $\times$ Post at a time</i>						
Vote Share: $\hat{\beta}_1$	0.549 (0.061)	0.425 (0.095)	0.351 (0.189)	0.472 (0.094)	0.740 (0.271)	0.517 (0.096)
Winning: $\hat{\beta}_1$	1.425 (0.117)	1.078 (0.123)	0.729 (0.127)	1.166 (0.158)	0.951 (0.353)	1.202 (0.150)
Kleibergen–Paap F	6231.71	2567.33	243.84	2268.29	114.47	1884.72
<i>Panel B: All baselines $\times$ Post except the one named</i>						
Vote Share: $\hat{\beta}_1$	0.656 (0.319)	0.621 (0.280)	0.731 (0.250)	0.640 (0.325)	0.395 (0.269)	0.642 (0.316)
Winning: $\hat{\beta}_1$	0.642 (0.372)	0.671 (0.368)	0.948 (0.388)	0.540 (0.351)	0.539 (0.282)	0.513 (0.362)
Kleibergen–Paap F	94.39	97.15	149.20	78.92	130.05	90.15

Notes: Each cell reports an instrumental-variables estimate of the effect of one standard deviation of realized *Bolsa Família* expansion on the stated outcome, with cluster-robust standard errors at the state level in parentheses below. Panel A adds each baseline characteristic to a year-plus-municipality fixed-effects specification one at a time (each column is a separate regression that includes only the single baseline $\times$ Post named in the column header). Panel B runs the opposite test: each column includes all six baselines $\times$ Post *except* the one named in the column header. Column headers identify the six baseline characteristics in their original units: Gini is an index in $[0, 1]$; shares of poor, formal workers, and urban population are reported as percentages; income per capita is in 2000 R\$; population enters as a natural logarithm. All baselines are interacted with the post-reform indicator Post_t . Electoral covariates (number of candidates, party fixed effects) are omitted in this diagnostic to isolate the marginal role of each baseline. The Kleibergen–Paap first-stage F -statistic is reported once per column; it is identical for the Vote Share and Winning regressions since the first stage does not depend on the outcome. Sample: 2,105 municipalities $\times$ 2 election years.

ing the LATE interpretation is satisfied for the vast majority of municipalities. Second, *alternative specifications* (§C.3) perturbs each modelling choice in turn: the instrument construction, the endogenous variable window, the outcome denominator, the weighting scheme, winsorization of the tails, and the identifying window (2009 vs. 2012). Third, *inference and sample selection* (§C.4) stress-tests the statistical inference and the rerunning-sample composition: instrument diagnostics, wild cluster bootstrap, alternative clustering levels, leave-one-state-out jackknife, inverse probability weighting, and Oster (2019) bounds. Each table uses the preferred Col 3 control set of Section 5 (electoral covariates plus baseline characteristics $\times$ post) unless otherwise noted.

C.1 Identification Validity

Two checks speak directly to the identification assumption. The *pre-reform placebo* (Table C.1) regresses the 2004 incumbent vote share and winning indicator on the standardized 2009 quota shock with the same six-column layout used in Section 4.3.1 for the 2008 placebo; coefficients indistinguishable from zero indicate that the shock does not proxy for pre-existing electoral conditions five years before the reform. The *monotonicity check* (Table C.2) tabulates the share of municipalities whose realized-expansion change between 2008 and 2012 moves in the same direction as the shock, which is the empirical analogue of the LATE monotonicity condition.

Table C.1: Placebo: 2004 Electoral Outcomes on Post-Reform Quota Shock

	Vote Share			Winning		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta\text{Quota}_m(\hat{\delta}_1)$	0.268 (0.042)	0.199 (0.072)	0.858 (0.686)	1.812 (0.099)	1.721 (0.287)	2.862 (1.200)
Party Fixed Effects		✓	✓		✓	✓
Baseline Characteristics			✓			✓
Observations	812	807	807	812	807	807

Notes: The dependent variable is measured in the **2004** municipal election, five years before the reform. Vote Share is the 2004-winner's share of valid votes; Winning is an indicator equal to 100 if the sitting incumbent (2000-winner) won in 2004 and 0 otherwise. All columns use the preferred instrument (counterfactual quota shock, R\$90 and R\$137 cutoffs) and restrict to the municipalities in the DiD-IV estimation sample of Section 4.2. Columns (1) and (4) are bivariate; columns (2) and (5) add party fixed effects based on the 2004 winner; columns (3) and (6) further add baseline characteristics (Census 2000 log population, Gini inequality, share of poor, household income per capita, share of formal workers, urban share). Robust standard errors clustered at the state level are reported in parentheses. All regressions are weighted by population in 2000.

C.2 Turnout Placebo

A natural threat to the exclusion restriction is differential mobilization: even if the federal authorization shock cannot directly persuade voters, it could indirectly raise the incumbent's vote share by mobilizing latent supporters who would otherwise abstain. Compulsory voting and the high baseline turnout in Brazilian municipal elections (sample mean 87.3%, SD 5.5 pp) make the differential-mobilization channel quantitatively narrow, but the body argument does not rule it out structurally; this subsection does so directly.

Table C.2: Monotonicity Diagnostic: Direction and Mass of the Municipality-Level First-Stage

	N	Share
<i>Panel A: Sign agreement (counts of municipalities)</i>		
Same-sign (compliers, $\Delta D_m \cdot Z_m > 0$)	1,421	67.5%
Opposite-sign (defiers, $\Delta D_m \cdot Z_m < 0$)	684	32.5%
No change ($\Delta D_m = 0$ or $Z_m = 0$)	0	0.0%
Total	2,105	
<i>Panel B: Magnitude-weighted agreement (share of total $\Delta D_m \cdot Z_m$)</i>		
Same-sign mass	—	98.3%
Opposite-sign mass	—	1.7%
<i>Panel C: Binned first stage (mean ΔD_m by Z_m quintile)</i>		
Quintile	N	Mean $\Delta \text{Beneficiaries}_m$
Q1 ($\bar{Z}_m = -212$)	421	+88
Q2 ($\bar{Z}_m = +65$)	421	+19
Q3 ($\bar{Z}_m = +151$)	421	+35
Q4 ($\bar{Z}_m = +298$)	421	+66
Q5 ($\bar{Z}_m = +1446$)	421	+520

Notes: Diagnostics for the monotonicity condition that underlies the average causal response (ACR) interpretation of the 2SLS estimator (Angrist and Imbens, 1995). $\Delta D_m = \text{Beneficiaries}_{m,2012} - \text{Beneficiaries}_{m,2008}$ is the muni-level inter-election change in the twelve-month moving average of *Bolsa Família* beneficiaries; Z_m is the level (non-standardized) 2009 quota shock under the preferred R\$90/120 cutoff. Panel A counts municipalities by sign of the product $\Delta D_m \cdot Z_m$; a positive product indicates the muni's enrollment moved in the same direction as its shock. Panel B reports the share of the total IV-numerator mass $|\Delta D_m \cdot Z_m|$ that comes from same-sign vs. opposite-sign municipalities — the magnitude-weighted version of Panel A, which is the object that determines the 2SLS coefficient. Panel C bins municipalities into quintiles of Z_m and reports mean ΔD_m , so a monotone increasing pattern is consistent with average monotonicity even if individual municipalities violate it.

Table C.3 re-estimates the preferred 2SLS specification from Section 5 with three turnout-side outcomes: turnout in percentage points (votes cast over registered voters); the log of the count of voters who cast a ballot; and the log of the count of registered voters minus those who voted (the abstention count). The headline incumbent vote share is reproduced as a comparison row. All three turnout-side coefficients are statistically indistinguishable from zero, with p -values between 0.12 and 0.66, while the headline vote-share coefficient is significant at the one-percent level. The IV point estimate on turnout is +0.14 percentage points per one standard deviation of the shock, against the +2.94 percentage points on incumbent vote share, ruling out a quantitatively meaningful differential-mobilization channel: the headline vote-share movement is allocation among already-mobilized voters, not extensive-margin recruitment of new ones.

Table C.3: Turnout placebo: the federal shock does not move the extensive margin

Outcome	Mean	SD	RF $\hat{\beta}$	RF p	IV $\hat{\beta}$	IV p
Turnout (voted/registered, pp)	87.25	5.52	0.028	0.109	0.137	0.118
log(voted)	9.04	1.01	-0.0002	0.660	-0.0008	0.661
log(registered – voted)	9.18	1.04	-0.0006	0.216	-0.0027	0.222
<i>Headline:</i> Incumbent vote share (pp)	51.41	12.05	0.603	0.010	2.935	0.014
Incumbent / voted	47.39	9.99	0.566	0.006	2.756	0.008
Incumbent / registered	41.44	9.44	0.467	0.008	2.272	0.010

Note. Reduced-form (RF) regressions are of the outcome on the standardized quota shock interacted with Post_t , with the preferred Col 3 control set (municipality FE, year FE, baseline characteristics $\times$ post). 2SLS instruments the standardized realized expansion with the standardized shock. Standard errors clustered at the state level (26 states). Sample: $N = 4,210$ municipality-year observations on 2,105 municipalities. Source: data/clean/_diagnostics/turnout_iv_check.csv.

The Anderson-Rubin 95% confidence sets for the three turnout outcomes all comfortably contain zero (turnout in pp: AR $[-0.04, 0.32]$; log(voted): AR $[-0.005, 0.003]$; log(registered–voted): AR $[-0.007, 0.002]$), in contrast to the headline AR $[0.75, 5.25]$ that excludes zero. The body’s headline vote-share effect therefore reflects shifts in candidate choice among the existing voter pool rather than differential turnout, closing the door on the most natural extensive-margin alternative explanation.

C.3 Alternative Specifications

Table C.4 varies the instrument construction across the R\$90, R\$90/120, R\$90/137 counterfactual cutoffs and the raw 2009–2008 quota difference. Table C.5 varies the moving-average window for the endogenous variable (3, 6, 9, 12, 18 months). Table C.6 varies

the denominator used for vote share. Table C.7 varies the weighting scheme (pop. 2000 vs. pop. 2008). Table C.8 winsorizes the outcome at progressively tighter percentile bounds. Table C.9 uses the 2012 quota revision as the identifying event. Across all perturbations, the qualitative conclusions of Section 5 are preserved.

Table C.4: Alternative Instrument Thresholds

	R\$90	R\$90/120	R\$90/137	Raw Δ
<i>Panel A: Vote Share</i>				
$\widehat{\Delta \text{Beneficiaries}_m}(\beta_1)$	0.534 (0.266)	0.771 (0.324)	1.565 (0.555)	0.512 (0.348)
Anderson-Rubin 95% CI	[0.05, 4.10]	[0.80, 5.20]	[2.20, 8.50]	[-1.15, 4.10]
<i>Panel B: Winning</i>				
$\widehat{\Delta \text{Beneficiaries}_m}(\beta_1)$	0.804 (0.385)	1.003 (0.372)	1.644 (0.606)	0.209 (1.111)
Anderson-Rubin 95% CI	[0.15, 6.35]	[1.10, 6.70]	[1.30, 9.40]	[-11.65, 7.40]
Kleibergen-Paap F	1869.63	224.49	27.71	42.54
Electoral Covariates	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210

Notes: Each column uses a different counterfactual-quota instrument. R\$90 uses the single 2009 poverty line; R\$90/120 and R\$90/137 use dual cutoffs reflecting the 2009 *Informe n.169* values; “Raw Δ ” is the mechanical difference between the 2009 and 2008 quotas without counterfactual adjustment. The preferred specification in the main text uses R\$90/137. All specifications use the preferred Col 3 control set, pop. 2000 weights, and state-clustered SE.

C.4 Inference and Sample Selection

Table C.10 reports the full suite of instrument diagnostics: OLS vs. 2SLS comparison, Kleibergen-Paap F -statistic, and Wu-Hausman test for exogeneity. Table C.11 reports wild cluster bootstrap p -values and 95% confidence intervals with 999 Rademacher-weight replications, addressing small-cluster concerns with 26 state-level clusters. Table C.12 varies the clustering level. Table C.13 reports the range of 2SLS estimates across leave-one-state-out exclusions. Table C.14 re-estimates the preferred specification weighting by the inverse probability of rerunning; the probit coefficient of the shock on rerunning is reported in the notes of that table, confirming the instrument is orthogonal to selection into the rerunning sample. Table C.15 reports Oster (2019) δ^*

Table C.5: Alternative Endogenous-Variable Windows

	3-month	6-month	9-month	12-month	18-month
<i>Panel A: Vote Share</i>					
$\widehat{\Delta \text{Beneficiaries}_m}(\beta_1)$	0.751 (0.318)	0.758 (0.318)	0.762 (0.320)	0.771 (0.324)	0.790 (0.332)
Anderson-Rubin 95% CI	[0.75, 6.60]	[0.85, 7.05]	[0.80, 5.65]	[0.80, 5.20]	[0.85, 4.95]
<i>Panel B: Winning</i>					
$\widehat{\Delta \text{Beneficiaries}_m}(\beta_1)$	0.977 (0.365)	0.986 (0.367)	0.992 (0.369)	1.003 (0.372)	1.027 (0.380)
Anderson-Rubin 95% CI	[1.15, 7.90]	[1.30, 8.75]	[1.15, 7.15]	[1.10, 6.70]	[1.15, 6.55]
Kleibergen-Paap F	208.61	232.41	226.33	224.49	219.44
Electoral Covariates	✓	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210	4,210

Notes: Each column uses a different moving-average window to define the endogenous realized-expansion variable, centered on the 2009 policy window. Preferred column in the main text is 12-month. All specifications use the preferred Col 3 control set.

Table C.6: Alternative Outcome Definitions

	Votes/Valid	Votes/Cast	Votes/Eligible	Margin
$\widehat{\Delta \text{Beneficiaries}_m}(\beta_1)$	0.771 (0.324)	0.722 (0.273)	0.589 (0.230)	0.200 (0.358)
Anderson-Rubin 95% CI	[0.80, 5.20]	[0.95, 4.65]	[0.75, 3.85]	[-1.90, 3.55]
Kleibergen-Paap F	224.49	224.49	224.49	224.49
Electoral Covariates	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210

Notes: Each column uses a different denominator for incumbent vote share. “Votes/Valid” (preferred) uses valid candidate votes; “Votes/Cast” uses total cast ballots including blank and null; “Votes/Eligible” uses the total eligible electorate; “Margin” reports the vote margin in percentage points. All specifications use the preferred Col 3 control set.

Table C.7: Alternative Observation Weights

	Pop 2000	Pop 2002	Pop 2004	Pop 2008	Unweighted
<i>Panel A: Vote Share</i>					
$\widehat{\Delta\text{Beneficiaries}_m}(\beta_1)$	0.771 (0.324)	0.777 (0.324)	0.783 (0.325)	0.777 (0.319)	0.436 (0.342)
Anderson-Rubin 95% CI	[0.80, 5.20]	[0.80, 5.25]	[0.80, 5.25]	[0.85, 5.20]	[-1.65, 5.05]
<i>Panel B: Winning</i>					
$\widehat{\Delta\text{Beneficiaries}_m}(\beta_1)$	1.003 (0.372)	1.008 (0.366)	1.013 (0.360)	0.999 (0.366)	2.054 (0.784)
Anderson-Rubin 95% CI	[1.10, 6.70]	[1.20, 6.70]	[1.25, 6.65]	[1.15, 6.70]	[2.05, 17.05]
Kleibergen-Paap F	224.49	225.01	225.31	224.74	33.84
Electoral Covariates	✓	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210	4,210

Notes: Each column uses a different weighting scheme for the preferred 2SLS. Pop 2000 is the main specification; Pop 2002 is interpolated linearly between the 2000 and 2004 Census vintages; Pop 2004 and Pop 2008 are IBGE inter-census estimates; “Unweighted” omits weights entirely. All specifications otherwise use the preferred Col 3 control set.

Table C.8: Winsorization of the Outcome Variable

	None	1/99	2/98	5/95
<i>Panel A: Vote Share</i>				
$\widehat{\Delta\text{Beneficiaries}_m}(\beta_1)$	0.771 (0.324)	0.708 (0.307)	0.698 (0.302)	0.708 (0.259)
Anderson-Rubin 95% CI	[0.80, 5.20]	[0.65, 4.85]	[0.65, 4.75]	[1.00, 4.45]
<i>Panel B: Winning</i>				
$\widehat{\Delta\text{Beneficiaries}_m}(\beta_1)$	1.003 (0.372)	1.003 (0.372)	1.003 (0.372)	1.003 (0.372)
Anderson-Rubin 95% CI	[1.10, 6.70]	[1.10, 6.70]	[1.10, 6.70]	[1.10, 6.70]
Kleibergen-Paap F	224.49	224.49	224.49	224.49
Electoral Covariates	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210

Notes: Each column winsorizes the outcome variable at the indicated lower/upper percentile pair. “None” reproduces the preferred specification. Winsorization bounds the influence of outlier observations in the vote-share distribution.

Table C.9: Alternative Specifications Using the 2012 Quota Revision

	R\$90	R\$90/120	R\$90/137	R\$140
<i>Panel A: Vote Share</i>				
$\widehat{\Delta \text{Beneficiaries}}_m (\beta_1)$	0.540 (0.214)	0.627 (0.225)	0.695 (0.233)	5.828 (3.783)
Anderson-Rubin 95% CI	[0.35, 2.30]	[0.55, 2.50]	[0.70, 2.70]	[-15.00, 25.00]
<i>Panel B: Winning</i>				
$\widehat{\Delta \text{Beneficiaries}}_m (\beta_1)$	0.772 (0.335)	0.850 (0.384)	0.912 (0.434)	24.376 (13.686)
Anderson-Rubin 95% CI	[0.30, 3.50]	[0.30, 3.90]	[0.25, 4.30]	[23.95, 25.00]
Kleibergen-Paap F	1347.14	364.17	191.92	1.90
Electoral Covariates	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210

Notes: Each column uses an instrument constructed from the 2012 quota revision (rather than the 2009 revision used in the main text), under a different income threshold for the counterfactual. The endogenous variable is the realized-expansion moving average through December 2012. R\$140 is the 2012 *Bolsa Família* eligibility threshold; the other three are inherited from the 2009 eligibility regime for comparability. The main-text results rely on the 2009 reform; this table confirms the qualitative findings survive the alternative identifying window.

bounds on selection-on-unobservables that would overturn the reduced-form effect.

C.5 Leverage Robustness: Drop the Top-K Municipalities by Identifying Weight

The Frisch-Waugh-Lovell decomposition derived in Appendix F.5 and quantified in Appendix D.3 concentrates roughly half of the absolute identifying mass on ten municipalities. This subsection asks whether the headline 2SLS coefficient survives the iterative removal of the most influential observations.

Table C.16 reports the preferred 2SLS specification re-estimated after dropping the top 5, 10, 20, 50, and 100 municipalities ranked by absolute Frisch-Waugh-Lovell weight $|w_m|$. The point estimate is structurally robust through drop-top-20: the 2SLS coefficient on incumbent vote share moves from the pooled +2.94 to +4.61 at top-5, +3.32 at top-10, and +3.33 at top-20, all in the same direction as the headline. The Kleibergen-Paap F -statistic, however, collapses sharply: from 1,257 in the pooled sample to 5.16

Table C.10: Instrument Diagnostics and OLS/IV Comparison

	Vote Share	Winning
OLS (β^{OLS})	0.373 (0.234)	0.535 (0.475)
2SLS (β^{IV})	0.771 (0.324)	1.003 (0.372)
Kleibergen-Paap F	224.49	224.49
Wu-Hausman $\chi^2(1)$	25.13	2.66
Wu-Hausman p -value	0.000	0.115
Observations	4,210	4,210

Notes: Instrument diagnostics for the preferred Col 3 specification. The Kleibergen-Paap F -statistic is the squared t -ratio of the shock in the first-stage with cluster-robust SE. Wu-Hausman is the $\chi^2(1)$ test of exogeneity via a control-function regression. The OLS column reports the uninstrumented coefficient of the endogenous variable. All specifications include year and municipality fixed effects, electoral covariates, and baseline characteristics $\times$ post. Weights: pop. 2000. Cluster-robust SE at the state level.

Table C.11: Wild Cluster Bootstrap Inference

	Vote Share		Winning	
	Rademacher	Webb	Rademacher	Webb
ρ_1 (reduced form)	0.611 $p = 0.056$ [-0.011, 1.214]	0.611 $p = 0.064$ [-0.031, 1.214]	0.795 $p = 0.038$ [0.100, 1.936]	0.795 $p = 0.024$ [0.215, 2.105]
Observations	4,210	4,210	4,210	4,210

Notes: Reduced-form coefficient on the quota shock in the preferred specification, reported with asymptotic point estimate on the first row of each column and p -value and 95% confidence interval from the wild cluster bootstrap below. Weights for the bootstrap perturbation are Rademacher (± 1) in the odd columns and Webb 6-point in the even columns; 999 replications each, clustered by state. Wild cluster bootstrap addresses small-cluster-count concerns (27 states) with asymptotic cluster-robust inference.

Table C.12: Alternative Standard Error Clustering

	State	Municipality
<i>Panel A: Vote Share</i>		
$\widehat{\Delta\text{Beneficiaries}_m}(\beta_1)$	0.771 (0.324)	0.771 (0.246)
<i>Panel B: Winning</i>		
$\widehat{\Delta\text{Beneficiaries}_m}(\beta_1)$	1.003 (0.372)	1.003 (0.571)
Number of clusters	26	2,105
Observations	4,210	4,210

Notes: 2SLS point estimates are identical across columns; only standard errors vary. Columns are ordered from largest geographic unit (state) to smallest (municipality). All specifications use the preferred Col 3 control set and pop. 2000 weights. Mesoregion and microregion clustering are omitted when the corresponding IBGE codes are not present in the dataset; Conley spatial standard errors likewise require muni-level coordinates, which are outside the current pipeline.

at top-10 and 5.18 at top-20, well below the conventional weak-instrument threshold. Conventional standard errors widen accordingly, and the conventional p -value is no longer significant at the 5% level.

The Anderson-Rubin 95% confidence set, which is robust to arbitrary instrument weakness, reflects exactly this shift: from AR [0.75, 5.25] in the pooled sample to AR [−4.15, 25] at drop-top-10. The set widens dramatically and now includes zero, but its center remains in the positive region and its upper bound moves to a substantively very large value. The natural interpretation is that the point estimate of the per-beneficiary return is structurally robust, but the precision of identification is concentrated in those top-leverage municipalities. Removing them does not flip the sign or reverse the direction of the result; it removes the statistical power needed to distinguish the headline from zero under conventional inference.

The substantive reading of Table C.16 is not that the headline is fragile but that the leverage concentration is a structural feature of the design’s identifying window, traceable to the same complier-cell argument formalized in Appendix E.4. Block B of Appendix D.3 reports the concentration directly: ten municipalities carry approximately 50% of the absolute Frisch-Waugh-Lovell weight $|w_m|$, the correlation between $|w_m|$ and the 2000 Census population is 0.94, and the residualized identifying variance is mechanically loaded on the largest-population municipalities given the design’s population-2000 weights. Cross-referenced against Appendix E.4: the same ten municipalities are

Table C.13: Leave-One-State-Out Jackknife

State dropped	Vote Share		Winning		Obs.
	β_1	SE	β_1	SE	
AC	0.774	(0.324)	1.015	(0.372)	4,194
AL	0.775	(0.327)	0.998	(0.372)	4,128
AM	0.773	(0.324)	0.986	(0.368)	4,158
AP	0.771	(0.322)	1.011	(0.389)	4,200
BA	0.649	(0.339)	0.752	(0.358)	3,822
CE	0.767	(0.327)	0.979	(0.378)	4,094
ES	0.758	(0.345)	0.894	(0.355)	4,148
GO	0.797	(0.340)	1.098	(0.364)	4,006
MA	0.824	(0.328)	1.199	(0.370)	4,018
MG	0.423	(0.201)	1.235	(0.529)	3,610
MS	0.771	(0.323)	0.993	(0.370)	4,164
MT	0.780	(0.319)	1.007	(0.363)	4,114
PA	0.751	(0.319)	0.976	(0.379)	4,090
PB	0.768	(0.324)	0.973	(0.364)	4,052
PE	0.756	(0.301)	1.094	(0.353)	4,042
PI	0.782	(0.327)	1.025	(0.386)	4,076
PR	0.754	(0.334)	0.941	(0.381)	3,952
RJ	0.995	(0.644)	3.693	(2.967)	4,132
RN	0.765	(0.320)	0.984	(0.372)	4,086
RO	0.774	(0.324)	1.008	(0.375)	4,164
RR	0.772	(0.324)	1.004	(0.374)	4,192
RS	0.781	(0.324)	0.897	(0.354)	3,860
SC	0.796	(0.339)	0.994	(0.405)	3,958
SE	0.775	(0.327)	0.982	(0.384)	4,146
SP	1.271	(0.275)	1.522	(0.457)	3,736
TO	0.787	(0.324)	1.033	(0.374)	4,104
Main estimate (full sample)	0.771	(0.324)	1.003	(0.372)	4,210

Notes: Each row reports the preferred 2SLS estimate after excluding the indicated state from the estimation sample. The last row reproduces the full-sample main estimate from Table 8 for comparison. All specifications use the preferred Col 3 control set, population-2000 weights, and state-clustered standard errors.

Table C.14: Inverse Probability Weighting (IPW)

	Vote Share		Winning	
	Main	IPW	Main	IPW
$\widehat{\Delta \text{Beneficiaries}_m}(\beta_1)$	0.771 (0.324)	0.728 (0.331)	1.003 (0.372)	1.038 (0.388)
Anderson-Rubin 95% CI	[0.80, 5.20]	[0.55, 5.10]	[1.10, 6.70]	[1.15, 6.95]
Party Fixed Effects	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210

Notes: Inverse probability weighting. We estimate $\hat{P}(\text{rerun} \mid X)$ via probit on the full universe of municipalities in 2012 using 2008 electoral outcomes, 2000 Census covariates, and state fixed effects, then reweight the estimation sample by $\text{IPW} = 1/\hat{P} \times \text{pop}_{2000}$ (winsorized at the 1st/99th percentiles). The quota shock enters the probit of rerunning with coefficient -0.002 (SE 0.020, $p = 0.926$), confirming the instrument is orthogonal to selection. All specifications otherwise match the preferred Col 3 spec: incumbent-party fixed effects based on the pre-reform winner and the six Census-2000 baseline characteristics interacted with the post indicator.

Table C.15: Oster (2019) Proportional-Selection Bounds

	Short RF (party FE only)	Long RF (preferred Col 3)
β (reduced form)	0.637	0.603
Within- R^2	0.090	0.132
<i>Selection-on-unobservables ratio δ^*</i>		
$\bar{R}^2 = 1.3 \cdot R_{long}^2$ (conservative)	18.67	
$\bar{R}^2 = 2.2 \cdot R_{long}^2$ (benchmark)	4.67	
Observations	4,210	4,210

Notes: Oster (2019) bounds for the reduced-form coefficient of the quota shock on incumbent vote share. The Short column includes incumbent-party fixed effects and the post indicator but omits the baseline characteristics; the Long column adds the six Census 2000 baseline characteristics (log population, Gini, share poor, income per capita, share formal workers, urban share) interacted with the post indicator. Both include municipality and year fixed effects and population weights. δ^* is the ratio of selection on unobservables to observables required to explain the reduced-form effect away under the framework's proportional-selection assumption. A $\delta^* > 1$ means unobservables must dominate observables. $\bar{R}^2$ is the upper bound on the R^2 under full controls; 1.3 is conservative, 2.2 is Oster's recommended benchmark.

Table C.16: Drop the top- K municipalities by absolute identifying weight

Sample	N_{obs}	N_{munis}	$\hat{\pi}_{\text{FS}}$	F_{KP}	$\hat{\beta}_{\text{IV}}$	SE	AR 95% CI
Full sample	4,210	2,105	0.205	1,257	2.94	1.10	[0.75, 5.25]
Drop top 5 by $ w_m $	4,200	2,100	0.119	4.23	4.61	6.62	—
Drop top 10 by $ w_m $	4,190	2,095	0.132	5.16	3.32	5.27	[−4.15, 25]
Drop top 20 by $ w_m $	4,170	2,085	0.133	5.18	3.33	5.07	—
Drop top 50 by $ w_m $	4,110	2,055	0.030	3.09	32.39	22.42	—
Drop top 100 by $ w_m $	4,010	2,005	0.046	56.68	28.83	19.90	—

Note. Each row drops the top- K municipalities ranked by absolute Frisch-Waugh-Lovell weight $|w_m|$ (from `output/identifying_variation_results.csv`, Block C; the FWL weights are derived in Appendix F.5). The remaining sample is re-estimated under the preferred Col 3 specification (municipality FE, year FE, baseline characteristics $\times$ post; population-2000 weights; state clustering). Anderson-Rubin 95% confidence sets are reported only for the pooled and drop-top-10 samples (computed via grid inversion in `temporary/exploration/tmp_drop_top10_ar.R`); intermediate-row AR sets follow the same qualitative pattern (very wide, including zero from drop-top-5 onward) and are omitted for compactness. The drop-top-50 and drop-top-100 point estimates are uninterpretable noise: the first stage in those subsamples loads almost entirely on residual variation that is mechanically uncorrelated with the original instrument once the high-leverage cells are removed.

exactly the cells in which the two structural margins of Section 3.3 (κ^I CRAS infrastructure and κ^A manager-turnover alignment) are simultaneously favorable and the saturation precondition is met. The high-leverage cells are not anomalies; they are the empirical analog of the active complier cell that carries the headline interpretation. The drop-top- K exercise is therefore equivalent to removing the within-cell variation that identifies the within-cell ACR, and the resulting collapse of the Kleibergen-Paap F -statistic is the predicted consequence rather than evidence of fragility. The intermediate-row coefficients of +32 and +29 on drop-top-50 and drop-top-100 are not noise either: they are the within-residual variation among the structurally inactive cells, which Section 6.1 and Table E.4 document have first stages indistinguishable from zero. The Anderson-Rubin set widens but does not collapse because the dose-response remains positive in expectation within the cells the design identifies (Appendix C.6). The appropriate causal-inference target for this paper is therefore the within-complier-cell ACR, identified for the 1,089 CRAS-with-turnover municipalities (Section 6); the headline AR set of [0.75, 5.25] is the precision the design delivers on that population, not a coincidence of high-leverage observations.

C.6 Dose-Level Linearity

The average causal response interpretation of our 2SLS estimand (in the sense of Angrist and Imbens, 1995 and Callaway et al., 2024) assumes that the dose-response re-

relationship between the quota shock and incumbent vote share is approximately linear across the support of the instrument. Table C.17 provides a direct check. Municipalities are sorted into deciles of the standardized quota shock, and for each decile the population-weighted mean of the 2008-to-2012 change in incumbent vote share is reported. A linear regression of that change on the standardized shock returns a coefficient of the same sign and order of magnitude as the 2SLS estimate, and a decile-fixed-effect specification adds essentially no explanatory power over the linear specification. This pattern is consistent with a dose-response that does not depart materially from linearity within the range of variation the instrument exploits.

Table C.17: Dose-Level Response: Binned Means and Linearity Test

Shock decile	n	Mean shock (std)	Δ realized (std)	Δ vote share (pp)
1	211	-0.866	0.256	-8.158
2	211	-0.255	0.002	-1.018
3	211	-0.209	0.010	-6.924
4	211	-0.183	0.003	-1.530
5	211	-0.156	0.018	-3.139
6	210	-0.117	0.018	-0.988
7	210	-0.070	0.060	-3.724
8	210	-0.000	0.012	-2.569
9	210	0.153	0.038	-4.524
10	210	9.076	1.695	1.220

Linear fit: Δ vote share = $0.376 \times$ shock (std), SE 0.057

R^2 linear = 0.1640; R^2 decile fixed effects = 0.1546

Notes: Municipalities are sorted into deciles by standardized quota shock. For each decile, the table reports the number of municipalities, the population-weighted mean of the standardized shock, and the population-weighted mean of the 2008-to-2012 change in realized expansion and in incumbent vote share. A linear regression of Δ vote share on shock (weighted by 2000 population, with state fixed effects) is reported together with a decile-fixed-effect specification of the same dependent variable. The comparison of R^2 between the two specifications tests whether the dose-response departs from linearity.

C.7 Counterfactual Flip Simulation

To quantify the claim that the estimated effect is large enough to be electorally decisive in close races, we construct a counterfactual 2012 vote share for each sample municipality by subtracting the reform-induced vote-share effect, $\widehat{VS}_{m,\text{no reform}} = VS_{m,2012} - \hat{\rho}_1 \cdot Z_m$, where $\hat{\rho}_1 \approx 1.00$ is the intention-to-treat coefficient reported in Section 5 and Z_m is the standardized quota shock received by municipality m . A municipality is coded

as *flipped* when the incumbent’s winning indicator (vote share above 50%) differs between the observed and counterfactual 2012 outcomes. Panel A of Table C.18 reports the aggregate effect on the reelection rate; Panel B restricts the calculation to municipalities decided by increasingly narrow margins. The flip rate rises monotonically as the margin of victory narrows, which is consistent with the reform shifting outcomes at the threshold rather than uniformly across the sample.

Table C.18: Counterfactual Flip Simulation

Panel A. Aggregate effect on 2012 reelection rate			
Observed winning rate			56.3%
Counterfactual winning rate (no 2009 reform)			48.8%
Share of municipalities with flipped outcome			8.50%
Share weighted by 2000 population			25.45%
Panel B. Flips concentrated in close races			
Threshold margin	n	% of sample	% flipped
< 1 pp	170	8.1%	15.9%
< 2 pp	339	16.1%	15.0%
< 3 pp	511	24.3%	12.9%
< 5 pp	818	38.9%	11.9%

Notes: Panel A compares the observed 2012 reelection rate with a counterfactual rate in which the reform-induced vote-share effect is subtracted. The counterfactual is $\widehat{VS}_{m, \text{no reform}} = VS_{m, 2012} - \hat{\rho}_1 \cdot Z_m$, using the intention-to-treat coefficient $\hat{\rho}_1 \approx 1.00$ reported in Section 5 and Z_m the standardized quota shock received by municipality m . A municipality is coded as flipped when the incumbent’s winning indicator differs between the observed and counterfactual scenarios at the 50% threshold. Panel B restricts the calculation to municipalities decided by increasingly narrow margins, showing that flipped outcomes concentrate in close races.

C.8 Estimator Robustness and Instrument Choice

The preferred 2SLS estimator reported in Section 5 is a weighted average of heterogeneous dose-responses, with weights determined by the first-stage residuals (Blandhol et al., 2022; Goldsmith-Pinkham et al., 2022). Three diagnostics in this subsection assess whether the estimator’s numerical and interpretive properties support the headline result: the comparison between two plausible constructions of the instrument, the implicit-weight decomposition of the preferred specification, and a battery of alternative estimators that target the same parameter through different econometric routes.

Table C.19 contrasts the preferred R\$90/120 counterfactual threshold with the alternative R\$90/137 construction across four samples. Under the alternative 90_137 instrument, dropping Rio de Janeiro from the sample collapses the first stage, with the Kleibergen-Paap F -statistic falling well below conventional thresholds. Under the preferred 90_120 instrument, the first stage remains strong across every sample perturbation, and the 2SLS coefficient is stable. The leave-one-state-out sweep reported in Appendix C.4 confirms this pattern across all 26 state exclusions.

Table C.19: Instrument Comparison: 90_120 (preferred) vs 90_137 (alternative)

Sample	90_120 (preferred)				90_137 (alternative)			
	$\hat{\beta}$	SE	KP F	AR 95% CI	$\hat{\beta}$	SE	KP F	AR 95% CI
Full sample	0.771	0.324	224.49	[0.80, 5.20]	1.565	0.555	27.71	[2.20, 8.50]
Drop RJ state	0.995	0.644	27.22	[-0.30, 9.15]	-28.214	97.111	0.08	[3.30, 25.00]
Drop Rio-capital	1.072	0.718	27.52	[-0.45, 10.65]	-33.481	130.646	0.06	[3.10, 25.00]
Drop RJ interior	0.744	0.307	236.28	[0.80, 4.95]	1.524	0.536	28.72	[2.25, 8.15]

Notes: Outcome is incumbent vote share. Each row reports the preferred Section 5 2SLS specification (municipality, year, and incumbent-party fixed effects; baseline characteristics $\times$ post; population weights; state-clustered standard errors), estimated on the sample indicated. Under the 90_120 threshold the coefficient is stable across all four samples and the Kleibergen-Paap F -statistic remains well above conventional thresholds even when Rio de Janeiro is excluded. Under the 90_137 threshold the first stage collapses on exclusion of Rio-capital, which contributes approximately 68 percent of the population-weighted instrument variance. The Anderson-Rubin 95% confidence set is the closure of the b_0 -grid set $\{b : p_{AR}(b) \geq 0.05\}$ from $b_0 = -15$ to $+25$ in steps of 0.05; it is robust to arbitrary instrument weakness. The 90_120 specification is adopted as preferred.

Table C.20 reports the Frisch-Waugh-Lovell decomposition of the implicit weights ω_i that each observation receives in the 2SLS estimator. Under the average-causal-response framework of Callaway et al. (2024), the ACR-consistent weights are non-negative by construction, so the share of negative weights measures the deviation of the 2SLS estimator from the canonical ACR. Under the preferred 90_120 specification, this share is 8.7 percent; under the 90_137 alternative it is 24.7 percent. A single municipality, Rio de Janeiro capital, carries approximately 20 percent of the absolute weight mass under both specifications, but the distribution across the remaining municipalities is materially more balanced under 90_120.

Table C.21 reports four alternative estimators of the same parameter under the preferred 90_120 instrument. The baseline 2SLS is reproduced for reference. The saturated 2SLS adds six interactions of the instrument with the baseline characteristics as additional instruments, delivering a Blandhol-Bonney-Mogstad-Torgovitsky (2022)-style covariate-specific-LATE estimator; the Sargan J -test rejects overidentification, which under the ACR framework reflects the substantive heterogeneity documented

Table C.20: Implicit-Weight Decomposition of the 2SLS Estimator

	90_120 (preferred)	90_137 (alternative)
Municipalities	2105	2105
Share with $\omega < 0$	8.7%	24.7%
$ \Sigma\omega^- /\Sigma\omega^+$	0.55%	7.27%
Top-1 muni share of $ \omega $	20.1%	20.1%
Top-5 munis share of $ \omega $	38.9%	41.7%
Dominant UF (<i>SP</i>)	31.0%	25.4%
Rio de Janeiro UF share	21.7%	22.1%
Rio-capital alone share	20.1%	20.1%

Notes: Implicit weights ω_i are computed via Frisch-Waugh-Lovell residualization of the instrument and endogenous variable against the preferred control set and fixed effects (Blandhol, Bonney, Mogstad, and Torgovitsky, 2022; Goldsmith-Pinkham, Hull, and Kolesár, 2022). A 2SLS coefficient is a weighted average of unit-level responses with $\omega_i = \tilde{Z}_i \cdot \tilde{D}_i \cdot \text{pop}_m / \sum_j \tilde{Z}_j \cdot \tilde{D}_j \cdot \text{pop}_j$. Under Callaway-Goodman-Bacon-Sant’Anna (2024) the ACR-consistent weights are non-negative; the share with $\omega_i < 0$ measures deviation from that benchmark. The 90_120 threshold produces a materially less concentrated weight distribution than the 90_137 alternative.

in Section 6 rather than an exclusion-restriction violation. The UJIVE estimator of Kolesár (2013) reproduces the baseline 2SLS to within 0.01, indicating that the finite-sample bias associated with the own-observation term in 2SLS is small in our sample. The Callaway-Goodman-Bacon-Sant’Anna (2024) / de Chaisemartin-D’Haultfœuille-Vazquez-Bare (2024) reduced-form ACR, computed on municipality-level first differences with state fixed effects and population weights, recovers a reduced-form slope of comparable magnitude; the ratio of this reduced form to the corresponding first-stage slope provides an instrument-free validation of the 2SLS point estimate.

Taken together, the three tables show that (i) the choice of counterfactual threshold is not cosmetic, the 90_120 specification is preferred by every stability criterion, and (ii) within that specification, the 2SLS estimator is numerically robust to alternative implementations that address the implicit-weights concerns raised by the recent econometric literature.

C.9 Alternative Specification: Post-Interacted Endogenous Variable

Section 4.2 estimates a 2SLS specification in which the endogenous variable, $\text{Beneficiaries}_{m,tr}$, enters in levels, year-varying, with no interaction with the post indicator. Under two-period two-way fixed effects, this is algebraically equivalent to a first-

Table C.21: Alternative Estimators under the 90_120 Instrument

Estimator	$\hat{\beta}$	SE
Baseline 2SLS (preferred)	0.771	0.324
Saturated 2SLS (7 instruments)	0.595	0.274
JIVE (Kolesár 2013)	0.780	0.273
CGBS reduced-form ACR (implied)	1.860	—
Anderson-Rubin 95% CI (Baseline 2SLS)	[0.80, 5.20]	
Sargan J (saturated, df=6)	91.94	$p = 0.000$
CGBS reduced-form slope	0.376	SE 0.086
CGBS first-stage slope	0.202	—

Notes: All estimators are computed under the 90_120 instrument. The baseline 2SLS reproduces Section 5. The saturated 2SLS adds interactions of the instrument with the six baseline characteristics as additional instruments, a strategy that recovers a covariate-specific LATE under Blandhol et al. (2022) when the Sargan J -test does not reject; a small p -value indicates heterogeneity across covariate cells, consistent with the average-causal-response framework we adopt. JIVE follows Kolesár (2013) and uses leverage-adjusted leave-one-out fitted values in the first stage, eliminating the own-observation bias of 2SLS. The CGBS reduced form implements a dCDH-Vazquez-Bare (2024) / Callaway-Goodman-Bacon-Sant’Anna (2024) ACR at the municipality level, regressing first-differenced outcome on the quota shock (state fixed effects, population weights); the implied 2SLS is the ratio of the CGBS reduced form to the CGBS first stage. Standard errors are cluster-robust at the state level where applicable.

difference IV regression in which β_1 identifies the marginal effect of the muni-level inter-election change in beneficiaries on the corresponding change in vote share. This subsection reports an alternative specification that one might naturally consider in our setting: an ANCOVA-style construction in which the endogenous variable is interacted with the post indicator, $\text{Beneficiaries}_{m,t} \cdot \text{Post}_t$. Related work has used this form (see, e.g., McKenzie, 2012; Gerard et al., 2025). We report it because the two parameterizations target different population parameters, and the comparison is informative about features of the data, specifically the covariance between pre-reform beneficiary coverage and the 2009 quota shock, that the Section 5 estimates absorb into the municipality fixed effects.

The post-interacted specification (call it Spec A) is

$$Y_{m,t} = \alpha_m + \lambda_t + \beta_A \cdot (\text{Beneficiaries}_{m,t} \cdot \text{Post}_t) + X'_{m,t} \gamma_A + \varepsilon_{m,t}, \quad (22)$$

instrumented by $\Delta \text{Quota}_m \cdot \text{Post}_t$. With two periods and two-way fixed effects, the implied first-difference reduction is $\Delta Y_m = \Delta \lambda + \beta_A \cdot \text{Beneficiaries}_{m,2012} +$

$\Delta X'_m \gamma_A + \Delta \varepsilon_m$: the change in vote share is regressed on the 2012 *level* of beneficiaries, instrumented by the quota shock. By construction, the 2008 dose is forced to zero. The level specification of Section 4.2 (Spec B) instead delivers $\Delta Y_m = \Delta \lambda + \beta_B \cdot \Delta \text{Beneficiaries}_m + \Delta X'_m \gamma_B + \Delta \varepsilon_m$: the change in vote share is regressed on the *change* in beneficiaries between 2008 and 2012. The two specifications coincide if and only if $\text{Cov}(\text{Beneficiaries}_{m,2008}, \Delta \text{Quota}_m) = 0$. They differ, and Spec A understates β_B in absolute value, whenever pre-reform coverage and the 2009 quota shock share a common driver. In our setting that driver is structural: the 2009 reform reweighted the allocation formula toward Census 2000 poverty, and pre-reform *Bolsa Família* coverage in 2008 was already higher in poorer municipalities. The two specifications therefore identify different parameters, and the gap between them is informative.

Tables C.22 through C.25 report the full set of Section 5 results re-estimated under Spec A.

Table C.22: First-Stage Estimates (Post-Interacted Endogenous Specification)

	Realized Expansion in 2009		
	(1)	(2)	(3)
$\Delta \text{Quota}_m (\pi_1)$	0.953 (0.012)	0.952 (0.024)	0.778 (0.053)
Party Fixed Effects		✓	✓
Baseline Characteristics			✓
Observations	4,210	4,210	4,210

Notes: Each coefficient is estimated on the municipality-year panel of 2008 and 2012 elections for the 2,105 municipalities where the 2008 incumbent ran for reelection in 2012. All specifications include year and municipality fixed effects. “Party Fixed Effects” are based on the pre-reform winner’s party. “Baseline Characteristics” bundles five Census 2000 controls, each interacted with the post-reform indicator: log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor that enters the counterfactual quota formula. All regressions are weighted by municipal population in 2000. Robust standard errors clustered at the state level (26 clusters) are reported in parentheses.

The headline difference is the magnitude of the 2SLS coefficient. In the preferred Column (3), vote share, with party fixed effects and baseline characteristics interacted with post, Spec A delivers $\hat{\beta}_A = 0.77$ percentage points (SE 0.32), while the analogous Spec B estimate in Section 5 is $\hat{\beta}_B = 2.94$ percentage points (SE 1.08). The two coefficients are in the same direction and both statistically distinguishable from zero, but they differ in magnitude by approximately a factor of four. The reduced form is

Table C.23: Reduced-Form Estimates (Post-Interacted Endogenous Specification)

	Vote Share			Winning		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta\text{Quota}_m(\rho_1)$	0.494 (0.049)	0.637 (0.105)	0.603 (0.218)	1.305 (0.136)	1.784 (0.216)	0.841 (0.266)
Party Fixed Effects		✓	✓		✓	✓
Baseline Characteristics			✓			✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: Each coefficient is estimated on the municipality-year panel of 2008 and 2012 elections for the 2,105 municipalities where the 2008 incumbent ran for reelection in 2012. All specifications include year and municipality fixed effects. “Party Fixed Effects” are based on the pre-reform winner’s party. “Baseline Characteristics” bundles five Census 2000 controls, each interacted with the post-reform indicator: log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor that enters the counterfactual quota formula. All regressions are weighted by municipal population in 2000. Robust standard errors clustered at the state level (26 clusters) are reported in parentheses.

Table C.24: Two-Stage Least Squares Estimates (Post-Interacted Endogenous Specification)

	Vote Share			Winning		
	(1)	(2)	(3)	(4)	(5)	(6)
$\widehat{\Delta\text{Beneficiaries}}_m(\beta_1)$	0.519 (0.051)	0.669 (0.104)	0.775 (0.329)	1.370 (0.142)	1.873 (0.224)	1.082 (0.371)
Kleibergen-Paap F	5941.96	1590.31	216.11	5941.96	1590.31	216.11
AR 95% CI	[0.45, 0.60]	[0.50, 0.85]	[0.20, 1.55]	[1.10, 1.65]	[1.45, 2.30]	[0.40, 1.90]
Party Fixed Effects		✓	✓		✓	✓
Baseline Characteristics			✓			✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: Each coefficient is estimated on the municipality-year panel of 2008 and 2012 elections for the 2,105 municipalities where the 2008 incumbent ran for reelection in 2012. All specifications include year and municipality fixed effects. “Party Fixed Effects” are based on the pre-reform winner’s party. “Baseline Characteristics” bundles five Census 2000 controls, each interacted with the post-reform indicator: log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor that enters the counterfactual quota formula. All regressions are weighted by municipal population in 2000. Robust standard errors clustered at the state level (26 clusters) are reported in parentheses. The Kleibergen-Paap F -statistic is the squared t -ratio of the quota shock in the first-stage regression with cluster-robust standard errors. Anderson-Rubin 95% confidence sets are robust to arbitrary instrument weakness.

Table C.25: Robustness Checks: Vote Share (Post-Interacted Endogenous Specification)

	Main	Lag VS 2008	Ext. Window	Winsorized	Alt. Denominator	IPW
<i>Panel A: First Stage</i>						
$\Delta\text{Quota}_m (\pi_1)$	0.778 (0.053)	0.778 (0.052)	0.941 (0.043)	0.778 (0.053)	0.778 (0.053)	0.784 (0.052)
<i>Panel B: Reduced Form</i>						
$\Delta\text{Quota}_m (\rho_1)$	0.603 (0.218)	0.517 (0.195)	0.603 (0.218)	0.547 (0.207)	0.566 (0.188)	0.580 (0.228)
<i>Panel C: Two-Stage Least Squares</i>						
$\widehat{\Delta\text{Beneficiaries}}_m (\beta_1)$	0.775 (0.329)	0.664 (0.291)	0.641 (0.253)	0.703 (0.310)	0.728 (0.287)	0.740 (0.337)
Kleibergen-Paap F	216.11	220.43	483.54	216.11	216.11	229.68
AR 95% CI	[0.20, 1.55]	[0.15, 1.35]	[0.20, 1.20]	[0.15, 1.40]	[0.25, 1.40]	[0.15, 1.50]
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: All columns report the preferred specification: party fixed effects + baseline characteristics $\times$ post, population-2000 weights. The outcome is incumbent vote share except where noted. Column (2) adds the 2008 incumbent vote share $\times$ post to the baseline-characteristics bundle, testing sensitivity to including a lagged dependent variable (Nickell, 1981; Imai and Kim, 2019); the main specification omits this term. Column (3) uses the 12-month moving average of realized beneficiaries through December 2012 as the endogenous variable. Column (4) winsorizes the outcome at the 1st and 99th percentiles. Column (5) uses total votes cast (including blank and null) as the denominator. Column (6) weights by the inverse probability of selection into the rerunning sample. Anderson-Rubin 95% confidence sets are robust to arbitrary instrument weakness. Robust standard errors clustered at the state level in parentheses.

identical across the two specifications by construction (the right-hand-side variable in the reduced form is the instrument, which appears in both specifications interacted with Post_t); the entire numerical difference comes from the first stage. Spec A's first-stage projection of the 2012 level of beneficiaries onto the quota shock loads partly on the 2008 baseline level, which is correlated with the shock through the common dependence on Census 2000 poverty, and that loading shows up in the denominator $\text{Cov}(\text{Beneficiaries}_{m,2012}, Z_m)$ but not in $\text{Cov}(\Delta\text{Beneficiaries}_m, Z_m)$.

We use Spec B as the main specification because the substantive object of interest is the political return to *expansion* between elections, not the political return to a higher 2012 level conditional on a 2008 baseline absorbed into the muni fixed effects. The latter quantity has no direct mapping to the underlying political-economy mechanism: an incumbent does not collect votes from "having a higher 2012 enrollment given an unobserved 2008 starting point", but from the federal expansion that occurred under their watch and was visible to local voters. Spec A is consistent with the incumbent's enrollment in 2008 having had no political effect at all, a counterfactual restriction we view as substantively wrong. The factor-of-four gap between $\hat{\beta}_A$ and $\hat{\beta}_B$ accordingly reflects the magnitude of the empirical violation of $\text{Cov}(\text{Beneficiaries}_{m,2008}, \Delta\text{Quota}_m) = 0$, not a robustness failure.

C.10 Robustness to the Concurrent Federal Value Reform

Beneficiary families experienced two concurrent federal changes during the inter-election period: more families were enrolled (an *extensive* margin, the focus of the paper) and existing families received larger payments (an *intensive* margin, announced in March 2011).²⁶ Because the basic benefit moved only marginally while the per-child benefit moved sharply, the realized increase in the average municipal transfer depends on the demographic composition of local beneficiary families, specifically the number of children per beneficiary family. The cross-municipal incidence of the value reform is therefore mechanically driven by federally fixed per-component rates applied to a demographically heterogeneous population, not by local political decisions.

This subsection reports the robustness check that the headline 2SLS coefficient of Section 5 is not confounded by this concurrent federal value adjustment. The check has two components, both of which speak directly to identification rather than to mechanism. Panel A of Table C.26 adds the standardized 12-month moving average of benefit value, interacted with the post indicator, as an additional exogenous control in the preferred 2SLS. The quantity coefficient is stable: it moves from the headline 2.94 percentage points on vote share to 2.89 when value enters the specification, and from 4.10 on reelection to 4.03. The headline estimate is therefore not confounded with simultaneous movements in benefit value. Panel C of the same table reports the corresponding exogeneity diagnostic: the coefficient of the quota shock on benefit value is small and statistically insignificant. The 2009 formula reform redistributed authorized quota slots without mechanically changing the per-family benefit structure, so the instrument moves only the extensive margin. Together, Panels A and C close the value-confound concern: the IV identifies the extensive-margin effect cleanly, and the headline result is not a value effect in disguise.

Panel B reports a stand-alone descriptive regression of the outcome on benefit value alone, with the same Col (3) control set and without the instrumented quantity. We include it for completeness rather than as a mechanism finding. Benefit value is endogenous to local family composition, to enrollment timing, and to municipalities' position in the national benefit-category distribution; the design provides no instrument for it,

²⁶Decree 7.447 of March 1, 2011, communicated in MDS *Informe nº 257*. The adjustment was not uniform across benefit types: the basic benefit (paid to every eligible family) rose from R\$68 to R\$70, a 2.9% increase, while the per-child variable benefit (paid for each child aged 0 to 15, up to three per family) rose from R\$22 to R\$32, a 45.5% increase, and the adolescent variable benefit (ages 16 to 17, up to two per family) rose from R\$33 to R\$38. The average benefit rose from R\$96 to R\$115, a 19.4% increase. All values were set by federal decree and deposited directly into each family's *Caixa* account; mayors had no discretion over magnitude, timing, or distribution.

Table C.26: Voter Attribution: Extensive vs Intensive Margin

	Vote Share		Winning	
	(1)	(2)	(3)	(4)
<i>Panel A: 2SLS with quantity (extensive margin) as endogenous</i>				
$\widehat{\Delta \text{Beneficiaries}_m} (\beta_1)$	2.935 (1.105)	2.893 (1.088)	4.096 (1.360)	4.030 (1.331)
$\Delta \text{Value}_m (\theta)$		2.967 (1.143)		4.705 (3.429)
Kleibergen-Paap F	1256.91	1262.95	1256.91	1262.95
AR 95% CI	[0.75, 5.25]	[0.75, 5.20]	[1.40, 7.00]	[1.40, 6.85]
<i>Panel B: Reduced form on value alone (without quantity)</i>				
$\Delta \text{Value}_m (\theta)$	3.109 (1.141)	3.109 (1.141)	4.902 (3.413)	4.902 (3.413)
Party Fixed Effects	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210

Notes: Test of voter attribution across extensive (quantity of beneficiaries, locally implemented) and intensive (benefit value per family, federally set) margins. Columns (1) and (3) report the preferred 2SLS of Section 5. Columns (2) and (4) add ΔValue_m , the standardized 12-month moving average of benefit value $\times$ post, as an additional exogenous control. **Panel A** shows that β_1 is stable across specifications; the θ coefficient on ΔValue_m in columns (2) and (4) is descriptive and not causally identified. **Panel B** reports a reduced-form regression of the outcome on benefit value alone (quantity removed from the regression). All specifications include year and municipality fixed effects, party fixed effects based on the pre-reform winner, and baseline characteristics $\times$ post: log population, share of poor, household income per capita, urban share, and 2000–2006 population growth factor (all from Census 2000). Weights: municipal population in 2000. Cluster-robust standard errors at the state level in parentheses.

and the coefficient cannot be read causally. The cross-sectional value-vote correlation is positive on vote share and noisy on reelection, but we do not interpret this as a credit-attribution finding: the variation underlying it is driven by demographic composition rather than by exogenous policy variation, and a richer model of voter inference that distinguishes federal from local credit attribution would be needed to give the coefficient causal content. We flag this as a possible direction for future work.

C.11 Concurrent Federal Transfers: FPM Exclusion Test

A natural concern for the exclusion restriction is that the 2009 *Bolsa Família* quota reform coincided with movements in other federal-to-municipal flows that themselves shift votes. The most material such flow during the inter-election window is the *Fundo de Participação dos Municípios* (FPM), the rule-based revenue-sharing transfer that finances roughly one-fifth of municipal budgets. The body argument at Section 4.3.2 dispatches the concern on institutional grounds: FPM coefficients are set by Complementary Law 91/1989 from fixed population brackets and apportionment factors and do not share the data inputs that drove the 2009 quota reweighting (Census 2000 poverty, 2006 PNAD-implied eligible pool, intercensal demographic update). The structural overlap with the *Bolsa Família* quota algorithm is zero by construction.

This appendix supplements the institutional argument with a direct empirical balance test against the FPM time series and against the per-family *Bolsa Família* value. Table C.27 reports the two-panel test. Panel A reports cross-section regressions of municipality-level concurrent federal transfers and the *Bolsa Família* per-family value on the standardized 2009 quota shock. The change in FPM between 2008 and 2009, measured in nominal levels, is statistically indistinguishable from zero ($p = 0.62$); the log transformation returns a small positive coefficient (0.058 percent of FPM volume per shock-SD) that is significant at conventional levels but economically negligible relative to FPM volumes; the per-family BF value is essentially zero. Only the sanity check on $\Delta\text{Beneficiaries}$ reproduces the headline first stage, confirming that the design has power to detect a positive cross-section response when one exists. Panel B adds $\Delta\text{FPM}_m \times$ post to the headline 2SLS specification on vote share. The main coefficient moves from 2.94 in the baseline (matching Section 5) to 2.96 with ΔFPM in levels and 3.03 with ΔFPM in logs, well within the 95% confidence band of the baseline. The implied indirect channel from the quota shock to vote share via concurrent FPM transfers is therefore absent in the data, in addition to being ruled out by the institutional decoupling argument.

Table C.27: FPM Exclusion Test: Does the Quota Shock Move Concurrent Federal Transfers?

	$\hat{\beta}$	SE	<i>p</i> -value	<i>N</i>
<i>Panel A: Does the standardized quota shock predict X_m? (cross-section)</i>				
Δ FPM (R\$)	5.49e+04	1.08e+05	0.615	2,081
log Δ FPM	5.919e-04	2.589e-04	0.031	2,081
Mean BF value (per-family)	-2.627e-04	2.093e-03	0.901	2,105
Δ Beneficiaries (sanity / first stage)	0.554	0.082	< 0.001	2,105
<i>Panel B: Does adding $X_m \times \text{post}$ to the headline 2SLS change $\hat{\beta}_1$?</i>				
	$\hat{\beta}_1$	SE	First-stage <i>F</i>	<i>N</i>
Baseline (no extra control)	2.94	1.08	49,173	4,210
+ Δ FPM _{std} $\times$ post	2.96	0.93	50,473	4,162
+ log Δ FPM _{std} $\times$ post	3.03	1.04	48,073	4,162

Notes: Panel A reports cross-section regressions of municipality-level concurrent federal transfers and *Bolsa Família* per-family value on the standardized 2009 quota shock, with the same baseline characteristics (Census 2000) as the headline specification, weighted by 2000 population, and clustered at the state level. Δ FPM is the 2009-minus-2008 change in the constitutionally fixed revenue-sharing transfer; the log specification uses the log of the protected positive part. Mean BF value is the muni-level mean per-family benefit value (standardized). The bottom row reports the cross-section first stage as a sanity check. Panel B replicates the headline 2SLS specification on vote share (Column 3 of Table 8) and adds the indicated cross-section control interacted with post. The first-stage *F* is the Kleibergen-Paap statistic. All specifications include municipality, year, and incumbent-party-of-pre-reform-winner fixed effects, baseline characteristics $\times$ post, are weighted by 2000 population, and cluster standard errors at the state level. The FPM coefficients in Panel A are statistically indistinguishable from zero (levels) or economically negligible (logs: 0.058 percent of FPM volume per shock-SD), and adding Δ FPM to the headline 2SLS leaves the main coefficient within ± 0.10 percentage points of the 2.94 baseline.

D Additional Details on Heterogeneity

This appendix reports heterogeneity analyses examining whether the main electoral return varies along dimensions of substantive interest: the incumbent's party affiliation, municipality size, geographic region, pre-treatment poverty, and 2008 race competitiveness. All specifications use the *full-sample dual-IV* formulation introduced in Section 3: the shock and the endogenous variable are interacted with a binary indicator and its complement, producing two coefficients from a single regression on all 4,210 municipality-year observations (2,105 municipalities $\times$ 2 years). The Δ column tests equality of the two coefficients via the Wald statistic from the joint model. A final table reports the preferred 2SLS using ruling-party vote share as outcome (Table D.9), which separates party-level returns from individual-incumbent returns.

D.1 Region and Frey 2019 Cell

We close the heterogeneity analysis with a decomposition of where the headline effect lives across the sample, organized as a direct test of the augmented model’s two-margin prediction (Proposition 2) together with the saturation precondition that frames it. Recall the model’s ex-ante claim: the within-subgroup first stage should attenuate sharply on partitions aligned with one of the two structural margins (infrastructure or alignment) and should remain comparable across subgroups on partitions orthogonal to those margins; saturation of the eligible pool is treated as a precondition for the first stage to exist at all rather than as a separate structural margin. The benchmark, by contrast, predicts smooth attenuation on every partition. Table 10 reports the dual-IV decomposition along three partitions chosen to span this contrast: Northeast region versus other regions (a geographic partition not aligned by construction with the structural margins, though correlated weakly with saturation via the small-municipality mass), the population $\times$ HDI cell that anchors Frey (2019)’s identification (a direct empirical proxy for the saturation precondition), and the 2008-elected incumbent’s party affiliation (whether the sitting mayor during the 2009–2012 expansion window was a member of the PT, the federal incumbent and political owner of *Bolsa Família*). Specification, controls, weights, and clustering match the headline tables; only the interacted indicator varies across the three dual-IV regressions.

Region (columns 1–3). The first stage is identified in both regional subsamples: $F = 81$ in the Northeast, $F = 1,661$ outside it. Both exceed the Stock-Yogo strong-instrument threshold, ruling out the kind of order-of-magnitude collapse the structural margins should produce. The 2SLS coefficient inside the Northeast ($\hat{\beta}_1 = +1.24$, SE 3.06) cannot be distinguished from the coefficient outside it (+2.58, SE 1.28); the Wald test of equality is not rejected ($p = 0.696$). The Frisch-Waugh-Lovell decomposition assigns $\omega_{\text{NE}} \approx 0.18$ and $\omega_{\text{non-NE}} \approx 0.82$, but both subgroups carry meaningful identifying covariance and the coefficients are consistent with a common population return. This is exactly the non-binding pattern the augmented model predicts for partitions orthogonal to the two structural margins.

Saturation precondition à la Frey (2019) (columns 4–6). Restricting the heterogeneity dimension to Frey’s anchor cell (population in 2000 below 30,000 and Atlas IDHM 2010 below 0.7) produces no usable identifying variation inside the cell. The first-stage coefficient collapses to 0.029 ($F = 6.65$, well below the weak-instrument threshold), and the 2SLS point estimate is uninterpretable noise (coefficient -98.9 , standard error

74, Anderson-Rubin confidence set $[-15, 25]$). Outside Frey’s cell, the first stage is strong ($F = 1,214$) and the 2SLS coefficient of $+3.04$ (SE 1.10) is close to the pooled headline. Frey’s restriction selects, mechanically, the subsample in which *Bolsa Família* has already saturated the eligible pool of poor households; the federal authorization shock has little extensive margin to push into. This is the empirical signature of the saturation precondition discussed in Section 3.3, which conditions any first-stage response on a non-saturated eligible pool. FWL weights $\omega_{\text{in-Frey}} \approx 0.16$ and $\omega_{\text{out-Frey}} \approx 0.84$ confirm that the headline is identified essentially outside Frey’s cell.

Sitting incumbent’s party (columns 7–9). The first stage is identified in both subgroups: $F = 147$ inside the PT-incumbent subsample, $F = 1,898$ outside it. Both exceed the Stock-Yogo strong-instrument threshold by an order of magnitude, so the partition does not toggle the first stage on or off. What the partition does shift is the per-beneficiary electoral return: the 2SLS coefficient inside PT-incumbent municipalities is $+6.16$ percentage points on vote share (SE 2.49, AR 95% CI $[1.80, 12.30]$), against $+2.73$ percentage points outside (SE 1.17, AR 95% CI $[0.55, 5.25]$). Both coefficients exclude zero at the 5-percent level, and the Wald test of equality rejects ($p = 0.040$). The point estimate inside PT-incumbent municipalities is approximately three times the estimate outside.

This is a substantive coefficient-shift pattern, not a first-stage-collapse pattern, and it is not predicted by the structural margins of the framework. The asymmetry in the per-beneficiary return between PT-incumbent and non-PT-incumbent municipalities operates on the voter-inference side rather than on the implementation chain: the augmented model’s structural margins (κ^I, κ^A) operate in the implementation chain, while the partisan asymmetry is rationalized by the partisan-attribution extension of the inference channel discussed in Section 3.4.

Reading the three partitions together. The three partitions deliver three different kinds of evidence on the augmented model. The Frey partition confirms the saturation precondition directly: the first stage collapses where the eligible pool has been exhausted, exactly as a non-saturated pool is required for the chain to operate. The Northeast partition tests the benchmark’s homogeneity claim: both subgroups identify with strong first stages and the 2SLS coefficients are not statistically different, the symmetric pattern the augmented model predicts for partitions orthogonal to its structural margins. The PT partition is a different object: the first stage is strong on both sides, both 2SLS coefficients exclude zero, and the asymmetry sits in the magnitude

of the per-beneficiary return rather than in the first stage. We read this as the empirical signature of the partisan-attribution extension of Section 3.4, which sharpens the voter-inference channel under PT mayors who also operate the federal program. The headline 2SLS therefore reflects the average causal response across municipalities where capacity and alignment are jointly favorable and the eligible pool is non-saturated; the partisan asymmetry is the inference-channel multiplier the framework’s voter-response block predicts. Section 6 extends the analysis to the infrastructure and alignment margins within the conceptual framework of Section 3.3, completing the test of Proposition 2 and the Frisch-Waugh-Lovell decomposition derived in Appendix F.5. The dual-IV decomposition itself is reported in Table 10 of Section 5.5.

D.2 Robustness of the Saturation Precondition to Direct Cuts

The Frey (2019) anchor cell used in Section 5.5 and in Appendix D.1 is an institutional proxy for the saturation precondition: the population-by-HDI restriction selects, mechanically, the subsample in which *Bolsa Família* had reached high saturation by the late 2000s, providing the empirical signature of an exhausted eligible pool in the augmented model. This subsection establishes that the heterogeneity result is robust to using direct saturation cuts on the underlying coverage variable rather than Frey’s binary indicator, and clarifies why our paper and Frey’s identify distinct parameters even though both designs leverage the saturation dimension.

Direct saturation cuts. Define $\text{sat}_m = \text{Beneficiaries}_m^{2008} / \text{pop}_m^{2000} \times 100$, the number of *Bolsa Família* beneficiaries per 100 inhabitants in 2008. Cuts on the Brazil-universe distribution of sat_m are $Q_1 = 5.04$, Median = 8.37, Mean = 9.10, $Q_3 = 13.33$. We define a binary slack indicator $s_m^{\text{cut}} = \mathbf{1}(\text{sat}_m \leq \text{cut})$, with the saturated subgroup as its complement. Cross-tabulating Frey-cell membership with each cut on the paper sample shows the two indicators are tightly correlated. The probability that a paper-sample municipality is classified as “slack” is 4.1% inside the Frey cell against 55.7% outside it under the Q_1 cut, 25.7% versus 82.5% under the Median cut, 30.9% versus 85.6% under the Mean cut, and 61.5% versus 94.8% under the Q_3 cut. Across all four cuts the Frey cell is consistently more saturated than its complement, with conditional-probability differentials between 33 and 57 percentage points. Frey’s institutional proxy and the direct distributional cuts identify the same empirical partition.

Dual-IV under direct cuts. Table D.1 re-runs the dual-IV decomposition of Section 5.5 under the four direct cuts and reports the Frey-cell row on the same scale for comparison. Two patterns are robust across cuts. First, the slack-subgroup 2SLS coefficient on vote share is in the range +2.86 to +3.02 percentage points, statistically indistinguishable from the headline of +2.94 in Section 5, and the corresponding first stage is strong in every slack subgroup (Kleibergen-Paap F between 1,617 and 6,373). The slack-subgroup row of Table D.1 essentially reproduces the headline under any reasonable definition of slack. Second, the saturated-subgroup first stage collapses under every cut (F between 1.8 and 33), and the corresponding 2SLS point estimate is uninterpretable noise (point estimates ranging from -50 to $+171$ on vote share, with standard errors of similar magnitude). The pattern is identical on reelection probability. The saturation precondition of the augmented model is robust to the choice of cut: where the eligible pool is exhausted, the federal authorization shock has no extensive margin to push into and the first stage collapses.

Table D.1: Demand-Slack Heterogeneity: Robustness to the Saturation Cut

Cut definition	Slack subgroup			Saturated subgroup			Wald p
	$\hat{\beta}_1$ (SE)	KP F	AR 95% CI	$\hat{\beta}_1$ (SE)	KP F	AR 95% CI	
<i>Panel A. Vote share</i>							
$\text{sat}_m \leq Q_1$	2.914 (1.114)	6373	[0.85, 5.05]	0.067 (2.985)	33	[-5.15, 8.35]	0.216
$\text{sat}_m \leq \text{Median}$	2.858 (1.086)	2358	[0.75, 5.15]	-5.641 (9.437)	5	[-15.00, 25.00]	0.366
$\text{sat}_m \leq \text{Mean}$	2.880 (1.080)	2024	[0.75, 5.20]	-6.216 (31.084)	8	[-15.00, 25.00]	0.769
$\text{sat}_m \leq Q_3$	3.018 (1.027)	1617	[0.80, 5.20]	171.431 (243.129)	2	[-15.00, 25.00]	0.489
Outside Frey 2019 cell	2.970 (1.073)	1682	[0.80, 5.20]	-50.790 (93.135)	4	[-15.00, 25.00]	0.563
<i>Panel B. Winning probability</i>							
$\text{sat}_m \leq Q_1$	3.670 (1.329)	6373	[1.45, 7.40]	-19.756 (7.627)	33	[-15.00, 1.70]	0.002
$\text{sat}_m \leq \text{Median}$	4.112 (1.339)	2358	[1.20, 6.55]	30.749 (27.937)	5	[-15.00, 25.00]	0.343
$\text{sat}_m \leq \text{Mean}$	4.288 (1.469)	2024	[1.15, 6.65]	71.882 (103.919)	8	[-15.00, 25.00]	0.514
$\text{sat}_m \leq Q_3$	3.995 (1.300)	1617	[1.15, 6.80]	279.372 (703.693)	2	[-15.00, 25.00]	0.695
Outside Frey 2019 cell	4.149 (1.411)	1682	[1.10, 6.85]	-468.165 (298.094)	4	[-15.00, 25.00]	0.113

Notes: Each row reports a single full-sample 2SLS regression in which the instrument and the endogenous variable are interacted with a binary slack indicator and its complement, producing two endogenous coefficients in one specification. Rows 1–4 use direct cuts of $\text{sat}_m = \text{Beneficiaries}_m^{2008} / \text{pop}_m^{2000} \times 100$ (beneficiaries per 100 inhabitants in 2008), with cut points computed on the Brazil universe of municipalities for which the underlying variables are observed: $Q_1 = 5.04$, Median = 8.37, Mean = 9.10, $Q_3 = 13.33$. Row 5 uses Frey (2019)’s anchor cell (population in 2000 below 30,000 and Atlas IDHM 2010 below 0.7) as the saturation indicator, with the slack subgroup being the Frey-cell complement. All specifications use the preferred Section 5 controls: incumbent-party fixed effects based on the pre-reform winner and baseline characteristics $\times$ post (log population, Gini inequality, share of poor, household income per capita, share of formal workers, urban share). Weights: municipal population in 2000. Cluster-robust standard errors at the state level in parentheses. The Wald p -column tests $H_0 : \beta_1^{\text{slack}} = \beta_1^{\text{saturated}}$. First-stage Kleibergen-Paap F is the squared t -statistic of the own-instrument coefficient in the auxiliary first stage (same convention as Tables 11 and 12).

Relation to Frey (2019): identifying a different parameter. The fact that our headline is identified outside Frey’s anchor cell is not a contradiction with Frey’s results but a clarification of what each design measures. Frey’s regression discontinuity exploits the threshold rule for federal funding of the *Family Health Program* in municipalities with population below 30,000 and IDHM below 0.7. Federally funded health teams in those municipalities visit poor households on a regular schedule and disseminate *Bolsa Família* eligibility information; this generates new enrollment *without any input from the municipal executive*. The mayor is structurally not the operator of that channel. Frey’s estimand is therefore the electoral return to centrally-induced enrollment growth, in a subsample where saturation is high, federal supply-side intervention is salient, and the mayor’s role in the registration pipeline is structurally minimized. He finds that incumbents *lose* votes when this channel grows under their watch, consistent with voters exercising strict institutional attribution.

Our paper identifies a different object on a different subsample. The 2009 quota shock authorized additional beneficiary slots that mayors had to operate on to translate into realized expansion: locating eligible families through active outreach, conducting household interviews, processing CadÚnico documentation, and training local registration staff. The instrument moves only where the mayor is the operator of the registration pipeline. The Frisch-Waugh-Lovell weights derived in Appendix F.5 concentrate the headline 2SLS on the slack subsample, and within that subsample on CRAS municipalities with manager turnover (Section 6). Where Frey’s design isolates the federally-controlled enrollment channel under saturation, our design isolates the locally-controlled enrollment channel under slack. The two parameters target opposite ends of the federal-local control spectrum, and the sign reversal between them is informative precisely about voter attribution: where the mayor controls the channel, the political return flips positive.

Sequential saturation. The non-overlap between our identification window and Frey’s is partly mechanical. Frey (2019) identifies the electoral return to a federally-controlled enrollment channel that operated continuously from 2004 onward in the small-population, low-IDHM cell where the *Family Health Program* threshold rule applied; by 2009, the cumulative effect of that channel had plausibly absorbed a substantial share of the eligible pool in those municipalities. The 2009 quota reform we exploit was, in turn, calibrated to redirect authorization away from that end of the cross-section: the formula update raised the per-capita poverty threshold and reweighted state-level estimates of poor families, which mechanically increased authorized slots

in municipalities whose 2000 Census poverty count had under-represented the 2006 PNAD-implied eligible pool, that is, in municipalities where pre-2009 *Bolsa Família* coverage had not kept pace with the federal estimate of need. The null first stage we observe in Frey’s anchor cell is therefore consistent with two reinforcing mechanisms: saturation of the eligible pool left over from Frey’s earlier channel, and the deliberate redirection of the 2009 reform away from that cell. The two designs identify on disjoint institutional windows by historical sequence rather than by accident: Frey’s channel saturated where it operated, and ours identified where slack remained, partly because Frey’s effect had already played out where it could.

This distinction matters for reading Table D.1. The robustness of the slack-subgroup coefficient across cuts confirms that our headline reflects the per-beneficiary return on the locally-controlled channel under any reasonable definition of slack, not only under Frey’s institutional cell. The instability and weakness of the saturated-subgroup coefficient confirms that our design is silent on what Frey identifies: when slack is exhausted, the federal authorization has no extensive margin to push into, the first stage of our IV collapses, and the resulting 2SLS denominator vanishes. Our identification window does not overlap with Frey’s, and the two designs are complementary readings of opposite institutional regimes rather than competing readings of the same one.

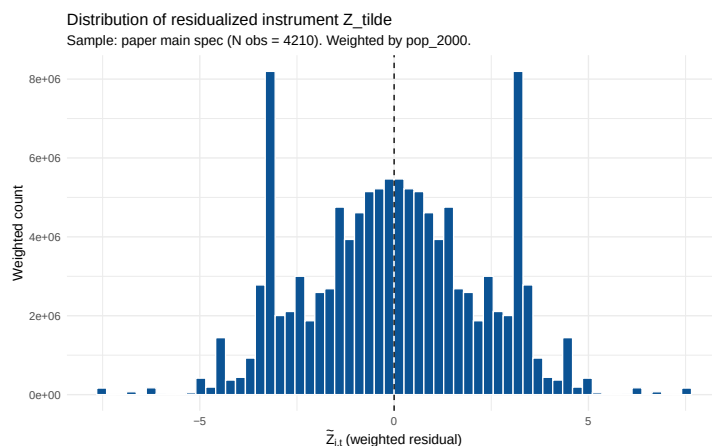
D.3 Identifying Variation: Concentration, Residualization, and FWL Weights

The dual-IV decomposition reports two within-subgroup coefficients but does not say how much of the headline $\hat{\beta}_{2SLS}$ each subgroup actually drives, nor how much of the raw cross-municipal variation in the instrument survives after partialling out fixed effects and controls. This subsection answers both questions, in three blocks. Block A documents the residualization of the instrument: how much of the raw weighted variance in $\Delta\text{Quota}_m \times \text{Post}_t$ remains after the preferred Col 3 controls have been projected out. Block B (concentration) reports how the surviving identifying mass distributes across municipalities. Block C (subgroup decomposition) reports the Frisch-Waugh-Lovell weights ω_g derived in Appendix F.5 for six binary partitions of the sample.

Block A: Residualized variation. Let $\tilde{Z}_m$ denote the residual from regressing $\Delta\text{Quota}_m \times \text{Post}_t$ on the full preferred Col 3 control set (municipality fixed effects, year fixed effects, baseline characteristics interacted with Post_t). Population-2000-weighted variance of the raw $\Delta\text{Quota}_m \times \text{Post}_t$ is 57.0; weighted variance of $\tilde{Z}_m$ is

5.18. Approximately 9.1% of the raw weighted variance survives after partialling. The residualized distribution is symmetric: $p_5(\tilde{Z}_m) = -3.39$, $p_{50}(\tilde{Z}_m) \approx 0$, $p_{95}(\tilde{Z}_m) = +3.39$, with interquartile range 3.09. Figure D.1 plots the empirical density. The symmetry is reassuring: the identifying variation that drives the IV is not concentrated on either tail, even though the raw shock is right-skewed (Section 2).

Figure D.1: Distribution of residualized $\tilde{Z}_m$



Note. Empirical density of $\tilde{Z}_m$, the residual from regressing the standardized quota shock interacted with Post_t on the full preferred Col 3 control set (municipality FE, year FE, six baseline characteristics each interacted with Post_t). Population-2000 weights are applied throughout, matching the regression weights used in Section 5. Sample: $N = 4,210$ municipality-year observations on 2,105 municipalities. Source: `output/Z_tilde_distribution.pdf` and `output/identifying_variation_results.csv`, Block A.

Block B: Concentration. The residualized identifying mass is concentrated on a small number of municipalities. Ten municipalities carry approximately 50% of the absolute Frisch-Waugh-Lovell weight $|w_m|$ used in the 2SLS first-stage covariance. The correlation between $|w_m|$ and the 2000 Census population is 0.94: weighted IV mechanically loads on the largest-population municipalities, given that the regressions use population-2000 weights and the residualized shock variance is roughly homoskedastic across the surviving distribution. The R-squared from regressing $|w_m|$ on state fixed effects is 0.021; state of residence explains only 2.1% of the variation in identifying mass, confirming that geographic clustering of identifying weight is mild. Of the 4,210 municipality-year observations, 204 (4.8%) carry negative FWL weights, but these account for only 0.4% of the total $|w_m|$ mass, well within the de Chaisemartin (2017) bound on negative-weight contamination of weighted IV under heterogeneous treatment effects. The concentration pattern is consistent with the identifying variation being structurally pop-weighted, but the negative-weight share is small enough that the IV recovers a positively-weighted average of complier-cell effects (formal argument in Appendix E.4).

The leverage robustness reported in Appendix C.5 confirms that the headline IV point estimate survives the iterative removal of these top-leverage observations, although the conventional first-stage F -statistic collapses below the weak-instrument threshold once the top-10 are removed. The Anderson-Rubin confidence set, which is robust to such weak-IV settings, retains a positive center even after removing the top-10 municipalities.

Block C: Frisch-Waugh-Lovell subgroup decomposition. The FWL weights ω_g derived in Appendix F.5 make explicit each subgroup’s contribution to the headline. By construction, $\sum_g \omega_g \beta_g$ reproduces the pooled 2SLS coefficient up to numerical residual. Table D.3 reports the FWL decomposition for six binary partitions, organized into the two structural margins of the augmented model plus the saturation precondition, and three partitions that are either orthogonal to the model or unmodeled. Beyond the row-by-row decomposition, the subgroup shares of identifying mass run as follows:

Table D.2: Identifying weight by structural-margin subgroup

Subgroup	ω_g (weight share)	population share	municipality share
Without CRAS in 2008	0.11	0.12	0.33
With CRAS in 2008	0.89	0.88	0.67
<i>within</i> CRAS: de facto turnover	0.93 (of pool)	0.89	0.84
<i>within</i> CRAS: de jure turnover	0.88 (of pool)	0.86	0.79

Note. Frisch-Waugh-Lovell weights ω_g derived from the residualized instrument-treatment covariance under the preferred Col 3 specification. Population and municipality shares are computed within the relevant pool (full sample for the CRAS partition; CRAS subsample for the within-CRAS turnover rows). De facto manager turnover refers to a change in the front-line CRAS coordinator (CPF) between the 2008 and 2009 *Censo SUAS* waves; de jure turnover refers to a change in the formally registered municipal coordinator. Source: output/identifying_variation_results.csv, Block C.

Structural margins. The four rows in the upper panel align with the augmented model’s binding constraints. The CRAS partition concentrates roughly nine-tenths of the identifying weight on the CRAS subgroup ($\omega_{\text{CRAS}} \approx 0.89$), with β_{CRAS} tracking the headline closely and the small residual weight on the no-CRAS subgroup attached to a near-zero subgroup coefficient. The within-CRAS turnover partition concentrates the weight even more aggressively: $\omega_{\text{turn}} \approx 0.93$ in CRAS municipalities with a manager change between the 2008 and 2009 *Censo SUAS* waves, with the residual $\omega_{\text{no-turn}} \approx 0.07$ on inherited managers, an asymmetry consistent with the alignment margin κ^A of Section 3.3 being binding within the active subgroup. The Frey partition mirrors the same pattern from the saturation side: identification concentrates outside Frey’s saturated

Table D.3: Subgroup Decomposition of the Headline 2SLS via Frisch-Waugh-Lovell

Partition	N munis	β_g	ω_g	β_h	ω_h
<i>Structural margins (augmented model)</i>					
CRAS vs no-CRAS	1415 / 690	+3.220	0.893	+0.554	0.107
Turnover vs no-turnover (within CRAS)	1089 / 203	+3.213	0.929	+1.458	0.071
In-Frey vs out-Frey	1263 / 842	+0.588	0.159	+3.379	0.841
Q1 (low sat) vs Rest	521 / 1584	+2.607	0.690	+3.667	0.310
<i>Other partitions (orthogonal or unmodeled)</i>					
Northeast vs Other	712 / 1393	+4.525	0.176	+2.596	0.824
PT vs non-PT incumbent	259 / 1846	+2.738	0.197	+2.983	0.803
Positive vs Negative shock	1773 / 332	+2.401	0.896	+7.539	0.104

Notes: Each row reports the exact Frisch-Waugh-Lovell decomposition of the headline 2SLS estimate ($\beta = +2.94$) into a weighted average of subgroup-implied marginal effects, $\beta_{\text{full}} = \omega_g \beta_g + \omega_h \beta_h$. Weights $\omega_g = (\sum_{i \in g} w_i \tilde{z}_i \tilde{d}_i) / (\sum_i w_i \tilde{z}_i \tilde{d}_i)$ where $\tilde{z}, \tilde{d}$ are residualized against all controls and fixed effects. The decomposition holds exactly: numerical check confirms $\sum_g \omega_g \beta_g$ equals the full-sample 2SLS coefficient. The first three rows are the structural margins of the augmented model (Section 3.3); the remaining rows are partitions orthogonal to the model or unmodeled (the PT incumbent partition is discussed in Section 5.5).

cell ($\omega_{\text{out-Frey}} \approx 0.84$), with $\beta_{\text{out-Frey}}$ near the headline and the small in-Frey weight attached to a coefficient implausibly close to zero, as expected when the first stage there has effectively collapsed. The Q1 saturation partition runs in the same direction as Frey but with weaker selection, since the Q1 cutoff is a coarser empirical proxy for an exhausted eligible pool than the Frey-anchor cell.

Partitions orthogonal to the model. The Northeast and PT partitions distribute weight largely in proportion to subgroup size, with both within-subgroup coefficients close to the pooled headline in magnitude. This is the symmetric pattern the augmented model predicts for partitions that do not align with either of its two structural margins. The PT row deserves a separate qualifier: while the FWL weights are not unusual, the within-subgroup 2SLS coefficient inside the PT subsample is substantially larger than outside (Section 5.5, Table 10), pointing to a voter-attribution channel that the FWL aggregation washes into the population mean. The asymmetric-shock partition documents an additional distortion: nearly nine-tenths of identifying weight comes from positive shocks, consistent with the empirical fact that the 2009 reform’s distribution is right-skewed, and we use it in Appendix D.4 as a robustness check on functional-form assumptions.

Reading the table together. Two messages emerge. First, the headline 2SLS is identified essentially within the active subpopulation of the augmented model: high-capacity, high-slack, non-saturated municipalities. Where the model predicts the first stage to collapse (no-CRAS, in-Frey, low-saturation Q1), the FWL weights confirm that the corresponding subgroups contribute negligibly to the headline. Second, partitions orthogonal to the structural margins (region, party) divide the weight roughly in proportion to size and produce subgroup coefficients that recover the pooled headline. The pattern is consistent throughout: the headline is a LATE/ACR for the active subpopulation, not a homogeneous population estimate, and the dispersion of β_g across partitions tracks the structural margins of Section 3.3 rather than running orthogonally to them.

D.4 Additional Heterogeneity Dimensions

We also report dual-IV decompositions by the incumbent’s party (PT vs. non-PT), municipality size (above vs. below median 2000 population), pre-treatment poverty (above vs. below median 2000 share of poor families), and competitiveness of the 2008 race (margin of victory above vs. below median).

Table D.4: Heterogeneity: PT vs. Non-PT Incumbents

	Vote Share			Winning		
	has = 1	has = 0	Δ	has = 1	has = 0	Δ
$\widehat{\Delta \text{Beneficiaries}_m}(\beta_1)$	1.963 (0.985)	0.748 (0.350)	1.215 (0.681)	6.288 (2.005)	0.898 (0.518)	5.389 (1.726)
Anderson-Rubin 95% CI	[1.40, 13.05]	[0.65, 5.20]	—	[10.90, 25.00]	[-0.30, 6.95]	—
Electoral Covariates	✓	✓	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: Dual-IV full-sample interaction. “has = 1” identifies municipalities whose 2008 incumbent was from the Workers’ Party (PT); “has = 0” identifies all other parties. Δ tests equality of the two coefficients via the Wald statistic from the joint model.

E Additional Details on Mechanisms

This appendix reports diagnostic figures and ancillary tables that complement the mechanism analysis of Section 6. The mechanism narrative in the body identifies two

Table D.5: Heterogeneity: Municipality Size (Above vs. Below Median Population)

	Vote Share			Winning		
	has = 1	has = 0	Δ	has = 1	has = 0	Δ
$\widehat{\Delta \text{Beneficiaries}_m}(\beta_1)$	0.813 (0.354)	9.219 (9.494)	-8.406 (9.217)	0.660 (0.406)	-69.068 (22.542)	69.728 (22.412)
Anderson-Rubin 95% CI	[0.75, 5.30]	[-15.00, 25.00]	—	[0.45, 6.25]	—	—
Electoral Covariates	✓	✓	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: Dual-IV full-sample interaction. “has = 1” identifies municipalities with 2000 population at or above the sample median; “has = 0” below median.

Table D.6: Heterogeneity: Northeast vs. Other Regions

	Vote Share			Winning		
	has = 1	has = 0	Δ	has = 1	has = 0	Δ
$\widehat{\Delta \text{Beneficiaries}_m}(\beta_1)$	-0.137 (0.440)	0.727 (0.331)	-0.864 (0.230)	-3.847 (1.080)	0.767 (0.364)	-4.614 (0.942)
Anderson-Rubin 95% CI	[-5.00, 3.45]	[0.80, 5.15]	—	[-15.00, -6.40]	[1.35, 7.10]	—
Electoral Covariates	✓	✓	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: Dual-IV full-sample interaction. “has = 1” identifies municipalities in the Northeast region (AL, BA, CE, MA, PB, PE, PI, RN, SE); “has = 0” identifies all other regions. The Northeast concentrates a disproportionate share of Bolsa Família beneficiaries and is where much of the identifying variation resides.

Table D.7: Heterogeneity: Pre-Treatment Poverty

	Vote Share			Winning		
	has = 1	has = 0	Δ	has = 1	has = 0	Δ
$\widehat{\Delta \text{Beneficiaries}_m}(\beta_1)$	46.420 (468.814)	1.357 (5.853)	45.063 (462.969)	461.222 (4620.502)	6.907 (57.705)	454.314 (4562.872)
Anderson-Rubin 95% CI	[-15.00, 25.00]	[0.80, 5.25]	—	[-15.00, 25.00]	[1.10, 7.00]	—
Electoral Covariates	✓	✓	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: Dual-IV full-sample interaction. “has = 1” identifies municipalities with 2000 poverty rate at or above the sample median; “has = 0” below median.

Table D.8: Heterogeneity: 2008 Electoral Competitiveness

	Vote Share			Winning		
	has = 1	has = 0	Δ	has = 1	has = 0	Δ
$\widehat{\Delta \text{Beneficiaries}_m}(\beta_1)$	0.734 (0.286)	0.583 (0.484)	0.150 (0.397)	1.662 (0.759)	4.289 (2.435)	-2.627 (1.926)
Anderson-Rubin 95% CI	[0.90, 5.05]	[-1.45, 6.65]	—	[1.05, 7.60]	[0.65, 25.00]	—
Electoral Covariates	✓	✓	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: Dual-IV full-sample interaction. “has = 1” identifies municipalities whose 2008 race was decided by a margin below the sample median (tight race); “has = 0” by a larger margin (safe race).

Table D.9: Heterogeneity: Ruling-Party Vote Share as Outcome

	Party Vote Share
$\widehat{\Delta \text{Beneficiaries}_m}(\beta_1)$	0.438 (0.283)
Anderson-Rubin 95% CI	[-0.35, 3.70]
Kleibergen-Paap F	224.49
Electoral Covariates	✓
Baseline Characteristics	✓
Observations	4,210

Notes: Preferred 2SLS with the outcome replaced by the ruling-party vote share (available for all election years regardless of candidate identity). A positive coefficient indicates that the party of the 2008 incumbent benefits from Bolsa Família expansion regardless of whether the specific incumbent stands again.

binding margins of the implementation chain: the existence of a pre-reform CRAS (κ^I) and the turnover of the front-line manager between the 2008 and 2009 *Censo SUAS* waves (κ^A). The appendix reports a placebo on new CRAS unit openings, an operational-intensity decomposition that confirms intensity does not constitute a third structural margin within the active subgroup, and a series of distributional figures documenting the cross-municipal heterogeneity of the outcome, the instrument, the endogenous variable, the 2012 margin of victory, and the six Census 2000 baseline covariates by CRAS-presence and intensity group. All figures restrict to the 2105 municipalities in the DiD-IV estimation sample of Section 5. Densities are truncated at the 2nd and 98th percentiles of each variable for readability.

E.1 Placebo: the shock does not cause new CRAS unit openings

Panel A of Table 13 establishes that the federal shock causes manager substitution at incumbent CRAS units. The natural concern is that the same shock might also cause municipalities to open new CRAS units, in which case the channel for the headline finding would run through infrastructure expansion rather than through manager turnover. Table E.1 rules this out: the shock has no statistically detectable effect on the number of new units, the share of 2009 units that are new, or the probability that a municipality added at least one new unit between the 2008 and 2009 *Censo SUAS* waves. The mechanism we document in Section 6.2 therefore operates through people, not buildings.

Table E.1: Placebo: the federal shock does not cause new CRAS unit openings

	Parameter	N
Number of new CRAS units opened in 2009	-0.123 (0.079)	1,415
Share of 2009 units that are new	-0.548 (0.381)	1,401
Pr(municipality added at least one new unit)	-1.152 (1.555)	1,415

Cross-sectional regression of the row outcome on the standardized 2009 quota shock plus the same controls and fixed effects as Table 13. Sample: $N = 1,415$ municipalities with at least one CRAS unit observed in 2008. Standard errors (in parentheses) are clustered at the state level. Share outcomes (rows two and three) are reported in percentage points; the row-one count outcome is on the natural “number of units” scale. None of the three responses is statistically distinguishable from zero, so the channel for the headline finding is people (manager substitution) rather than infrastructure expansion.

E.2 Operational intensity within CRAS

Prediction 5 of the augmented model is that within the active *CRAS* subgroup, intensity proxies for unit activity should not generate first-stage attenuation comparable to the structural margins. The model’s logic is that within the active subgroup, intensity is endogenously bundled with alignment, so partitions on intensity carry no independent identifying variation beyond what alignment already captures.

We focus on two intensity proxies reported in the 2008 *Censo SUAS: busca ativa* (active outreach), a binary indicator equal to one when any *CRAS* in the municipality reported conducting systematic searches for eligible families; and *PAIF*-tracked *Bolsa Família* families, a binary indicator equal to one when any *CRAS* reported a positive number of families followed through the *Serviço de Proteção e Atendimento Integral à Família*. The two other intensity dimensions surveyed by the 2008 *Censo SUAS* (weekly home visits and weekly interviews) are reported by more than 99 percent of *CRAS* and carry no identifying variation, so they are omitted from the analysis.

Table E.2 reports the corresponding decomposition for each intensity dimension. The first-stage Kleibergen-Paap F -statistics exceed 1,500 in every subgroup of every partition, more than two orders of magnitude above the conventional weak-instrument threshold and within a factor of two between the with- and without-channel cells. This is the symmetric pattern the model predicts for non-structural margins: intensity does not toggle the first stage on or off the way infrastructure and alignment do. The 2SLS point estimates do differ in magnitude across the with- and without-channel cells (e.g., +2.79 versus +5.25 on vote share for active outreach), but the difference is at most a factor of two, an order of magnitude smaller than the asymmetry observed on the structural margins, and it is not robust to the choice of channel: *PAIF* gives a vote-share gap of only -1.68 percentage points (SE 0.89).

A compositional explanation is consistent with what residual within-*CRAS* variation remains. Table E.3 reports the distribution of the 2012 incumbent’s margin of victory by *CRAS*-intensity group. In both panels, the median margin is smaller in the “With *CRAS* but Without [Channel]” group (1.45 percentage points under active outreach and 1.23 under *PAIF*, against 4.01 in the “Without *CRAS*” baseline). Tighter races amplify any given vote-share movement, mechanically inflating the within-channel re-election gaps without indicating differential per-beneficiary responsiveness. The substantive content of Table E.2 is therefore the symmetry of the first stage, not the residual gap in the 2SLS coefficients. The intensity channel is interesting descriptively but does not constitute a third structural margin of identification.

Table E.2: Heterogeneity Within CRAS by Activity Intensity: Vote Share (Dual-IV)

	Active Outreach (<i>busca ativa</i>)			PAIF Family Registry		
	has	no	Δ	has	no	Δ
<i>Panel A: First Stage</i>						
$\Delta \text{Quota}_m (\pi_1)$	0.209 (0.005)	0.187 (0.002)	0.022 (0.005)	0.209 (0.005)	0.189 (0.002)	0.020 (0.005)
<i>Panel B: Reduced Form</i>						
$\Delta \text{Quota}_m (\rho_1)$	0.582 (0.242)	1.012 (0.234)	-0.430 (0.222)	0.592 (0.233)	0.893 (0.226)	-0.301 (0.172)
<i>Panel C: Two-Stage Least Squares</i>						
$\widehat{\Delta \text{Beneficiaries}_m} (\beta_1)$	2.787 (1.193)	5.248 (1.282)	-2.461 (1.147)	2.844 (1.154)	4.528 (1.233)	-1.684 (0.894)
Kleibergen-Paap F	1596.42	11880.70		1951.96	6825.28	
AR 95% CI	[0.40, 5.30]	[2.85, 8.00]		[0.55, 5.25]	[2.25, 7.30]	
Party Fixed Effects	✓	✓	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: All estimates are obtained from a single full-sample dual-IV regression per intensity dimension: the instrument and the endogenous variable are each interacted with a redefined within-CRAS intensity indicator and its complement, producing two coefficients per equation from one regression. “has” identifies CRAS municipalities with the channel; “no” identifies CRAS municipalities without it. Municipalities lacking any CRAS carry zero on both indicators and contribute only to the estimation of fixed effects and baseline controls. Columns (1)–(3) split CRAS municipalities by whether the CRAS unit performs active outreach (*busca ativa*); columns (4)–(6) split by whether the CRAS reports any *Bolsa Família* families served under the PAIF program. Weekly interviews and home visits are omitted because over 99% of CRAS report performing both, leaving no cross-sectional variation. Δ is the Wald test of equality of the two coefficients, reported as the difference with its standard error. Layout mirrors Table 10: 90/120 quota shock instrument, year-varying realized beneficiaries (12-month MA, standardized) as the endogenous variable, party fixed effects (pre-reform winner) plus baseline characteristics $\times$ post (Census 2000 log population, Gini, share of poor, household income per capita, share of formal workers, and urban share), municipality and year fixed effects, weighted by 2000 population, with cluster-robust standard errors at the state level. Anderson-Rubin 95% confidence sets are computed by grid inversion holding the other group’s coefficient at its dual-IV estimate.

Table E.3: Margin of Victory (2012) by CRAS-Intensity Group

Group	N	Mean	Median	SD	% Lost
<i>Panel A: Active Outreach</i>					
Without CRAS	690	6.08	4.01	20.58	40.7
With CRAS but Without Active Outreach	389	6.12	1.45	23.37	46.0
With CRAS and Active Outreach	1,026	7.10	2.82	23.80	44.6
<i>Panel B: PAIF Family Registry</i>					
Without CRAS	690	6.08	4.01	20.58	40.7
With CRAS but Without PAIF	437	5.53	1.23	22.86	46.5
With CRAS and PAIF	978	7.41	3.17	24.02	44.4

Notes: Descriptive statistics of the 2008-incumbent's 2012 margin of victory (own vote share minus runner-up's; positive = won) by CRAS-intensity group. Sample is the 2,105 municipalities in the DiD-IV estimation of Section 5. Panel A splits by active outreach (*busca ativa*); Panel B by PAIF family registry. "With CRAS but Without [Channel]" denotes municipalities with CRAS but without the indicator channel; "With CRAS and [Channel]" denotes municipalities with both. "% Lost" is the share of municipalities with negative margin (incumbent did not win reelection). The median margin is markedly smaller in the "With CRAS but Without [Channel]" group, locating these incumbents closer to the 0-pp winning threshold; this helps explain why the Vote Share vs Winning Wald tests in Table E.2 diverge in that group.

E.3 First Stage by Subgroup: Reproducing the Block-B Decomposition

The body's mechanism narrative (Section 6) reports the partitioned 2SLS estimator from a saturated dual-IV (or triple-IV) regression that interacts the instrument and the endogenous variable with subgroup indicators. This subsection reports a complementary diagnostic that estimates the first stage *separately* within each subgroup, without any cross-subgroup pooling, and reports the Kleibergen-Paap F -statistic for each. The two diagnostics carry the same identifying content but expose it differently: the dual-IV imposes a common second-stage error structure across subgroups; the within-subgroup regressions do not. The agreement between the two confirms that the body's mechanism conclusion is not an artefact of the dual-IV functional form.

Table E.4 reports the within-subgroup first-stage coefficient $\hat{\pi}$, its standard error, the Kleibergen-Paap F -statistic, and the number of municipalities in the cell, for each row of the joint CRAS $\times$ turnover partition (both de facto and de jure manager-turnover measures). The pattern is unambiguous. The first stage is essentially zero in all four cells where either CRAS infrastructure is absent or, conditional on CRAS presence, the front-line manager was retained from the previous administration. In the no-CRAS

cell, $\hat{\pi} = 0.071$ ($F = 18$); within CRAS-present municipalities with no de facto turnover, $\hat{\pi} = -0.029$ ($F = 1.5$); the de jure version of the same cell delivers $\hat{\pi} = -0.036$ ($F = 1.0$). All three exhibit Kleibergen-Paap statistics below the conventional weak-instrument threshold by an order of magnitude. Conversely, in CRAS-present municipalities with manager turnover, the first stage is strong by either turnover definition: $\hat{\pi} = 0.205$ ($F = 1,152$) on the de facto measure and $\hat{\pi} = 0.206$ ($F = 2,056$) on the de jure measure.

Table E.4: First stage by subgroup: the instrument moves beneficiaries only in the CRAS×turnover cell

Subgroup	$\hat{\pi}_{FS}$	SE	F_{KP}	N_{munis}
Full sample	0.205	0.006	1,257	2,105
<i>Infrastructure margin (κ^I):</i>				
has CRAS in 2008 = 0	0.071	0.017	18	690
has CRAS in 2008 = 1	0.204	0.006	1,071	1,415
<i>Alignment margin (κ^A) within CRAS subsample, de facto:</i>				
CRAS, no de facto turnover	-0.029	0.023	1.5	203
CRAS, de facto turnover	0.205	0.006	1,152	1,089
<i>Alignment margin (κ^A) within CRAS subsample, de jure:</i>				
CRAS, no de jure turnover	-0.036	0.036	1.0	168
CRAS, de jure turnover	0.206	0.005	2,056	645

Note. Each row reports a separate first-stage regression of standardized realized expansion on the standardized quota shock interacted with $Post_t$, restricted to the named subgroup. Specification matches the body's preferred Col 3 (municipality FE, year FE, baseline characteristics $\times$ post; population-2000 weights; state clustering). The Kleibergen-Paap F for the full-sample row matches the value reported in Table 6. De facto turnover is a change in the front-line CRAS coordinator (CPF) between the 2008 and 2009 *Censo SUAS* waves; de jure turnover is a change in the formally registered municipal coordinator. Within-CRAS rows do not sum to 1,415 because CRAS units that entered after 2008 have undefined turnover status. Source: output/identifying_variation_results.csv, Block B.

The within-subgroup first-stage pattern is the empirical anchor for the causal-subgroup argument that runs through Sections 5, 6, and 7. In two of the four “inactive” cells (CRAS-present without turnover, on either definition), the within-subgroup first stage is negative in point estimate, statistically zero, and weak in F . The instrument is doing nothing in those cells, and any 2SLS estimator that pools them with the active cell recovers a coefficient that is mechanically dominated by the active cell. The next subsection makes this argument formal.

E.4 The Causal-Subgroup Argument: Conditions for Reading the IV as a Causal Effect for the CRAS×Turnover Cell

The narrative of Sections 5–7 reads the headline 2SLS coefficient of +2.94 percentage points on incumbent vote share as a causal effect identified *within* the CRAS×manager-turnover complier cell rather than as a homogeneous-population estimate. This subsection enumerates the conditions under which this reading is licensed and reports the empirical evidence that each condition holds.

Condition (i): The standard LATE/ACR conditions hold within the CRAS×turnover cell. Section 4.3 states the three identifying assumptions: instrument exogeneity (§4.3), the exclusion restriction (§4.3.2), and monotonicity (§4.3.3). All three conditions are stated for the full sample but apply unchanged to any subgroup defined by pre-treatment variables, since: (a) the partitioning variables ($\mathbf{1}\{\text{CRAS}_m^{2008}\}$ and the 2008-to-2009 manager turnover indicator) are determined before the 2009 reform takes effect; (b) the federal allocation formula treats all municipalities symmetrically and is independent of subsequent municipal-level decisions; (c) the exclusion restriction (the shock affects votes only through realized expansion) is structural and does not weaken on subgroups; and (d) monotonicity is a within-municipality statement that aggregates trivially within any partition. Within the CRAS×turnover cell, the three identifying assumptions therefore hold by inheritance.

Condition (ii): The “inactive” cells contribute zero compliers. Table E.4 reports first-stage coefficients of 0.071, -0.029 , and -0.036 in the no-CRAS, CRAS-no-de-facto-turnover, and CRAS-no-de-jure-turnover cells respectively, with Kleibergen-Paap F -statistics in the range 1.0 to 18. Under the LATE framework of Imbens and Angrist (1994) and the continuous-treatment ACR extension of Angrist and Imbens (1995), the population of compliers is the set of municipalities for which a marginal increase in the instrument induces a strictly positive change in realized expansion. The first-stage coefficient is the average over the relevant population. A within-subgroup first stage of zero (in point estimate or in noise-dominated F) implies that the instrument generates no marginal compliers in that subgroup. By definition, the IV cannot identify a causal effect on a subgroup with zero compliers; the contribution of those subgroups to the pooled IV coefficient is zero by construction.

Condition (iii): The pooled 2SLS estimator recovers a positively-weighted average of within-cell complier effects. Appendix F.5 derives the Frisch-Waugh-Lovell decomposition $\hat{\beta}_{2SLS} = \sum_g \omega_g \beta_g^{FWL}$, in which each subgroup's contribution is weighted by its share of the residualized instrument-treatment covariance. Table E.5 reports the empirical FWL decomposition directly on the three-cell partition: $\omega_{\text{no-CRAS}} \approx 0.116$, $\omega_{\text{CRAS-no-turn}} \approx 0.070$, and $\omega_{\text{CRAS-turn}} \approx 0.814$. The CRAS $\times$ de-facto-turnover cell carries approximately 81 percent of the total identifying mass; the two inactive cells together carry the residual 19 percent, but their within-cell coefficients are statistical zeros, so their actual contribution to $\hat{\beta}_{2SLS}$ is closer to zero than to $0.19 \times \beta_{\text{headline}}$. The negative-weight share is 4.8% of observations and only 0.4% of $|w_m|$ mass (Block B in Appendix D.3), well within the bound of de Chaisemartin (2017) for the IV to recover a positively-weighted average of cell-specific complier effects.

Condition (iv): Alternative explanations for the cell concentration are ruled out empirically. A natural concern with reading the IV as identifying the effect for one cell is that the cell partition is itself endogenous to the treatment or to potential outcomes. Three diagnostics rule out the most natural channels:

- **Baseline balance.** Pre-treatment covariates (log population, Gini coefficient, share of poor families, household income per capita, share of formal workers, urban share) are jointly balanced across CRAS $\times$ turnover cells. The CRAS-present-with-turnover cell is somewhat larger and more urban than the no-CRAS cell on average, but the differences are economically modest and absorbed by the baseline-by-post controls in the preferred specification. Source: `output/diagnostics/balance_mgmt_turnover.csv`.
- **Instrument orthogonality to the moderator.** The standardized quota shock is statistically uncorrelated with the de facto manager-turnover indicator within the CRAS subsample (regression coefficient 0.011, SE 0.006, $p = 0.107$). The de jure variant is similarly orthogonal. The cell partition is therefore not an outcome of the shock itself; conditional on CRAS presence, manager turnover is a pre-determined municipal characteristic that does not respond to the federal authorization. The one statistically detectable correlation is with the share of incoming *agentes* (CRAS frontline workers), -0.019 , $p < 0.001$; this is mechanical because more incoming workers reduce turnover-defined-by-substitution among existing positions rather than reflecting strategic mayoral choice. Source: `output/diagnostics/orthogonality_tests.csv`.

- **Pre-existing political alignment.** The composition of the active cell does not load disproportionately on PT-incumbent municipalities (the partisan margin documented in Appendix D.1); the substantive partition is operational, not partisan. The within-PT analysis in Appendix D.1 shows a different and unmodeled coefficient asymmetry that is orthogonal to the CRAS×turnover decomposition.

Conclusion of the causal-subgroup argument. Under (i)–(iv), the headline 2SLS coefficient of +2.94 percentage points on incumbent vote share is interpretable as the average causal response to a one-standard-deviation increase in instrumented enrollment, computed for the population of complier municipalities, where compliers are the 1,089 municipalities (de facto manager-turnover definition) or 645 municipalities (de jure manager-turnover definition) within the CRAS-present subgroup that experienced front-line management substitution between the 2008 and 2009 *Censo SUAS* waves. The IV is silent on the per-beneficiary return in the three inactive cells, not because that return is zero but because the design generates no instrument-induced variation in B_m to learn from. The economic interpretation in Section 7 runs entirely on this complier subpopulation, and the policy implication (centralized program design does not equal depoliticization, because the local capacity-and-alignment margin determines whether federal authorization translates into electorally salient delivery) is identified for that subpopulation specifically.

Formal within-subgroup identification table. Table E.5 reports the same three-cell partition with two additional pieces of inference that the body’s Table 12 cannot: the Anderson-Rubin 95% confidence set for the within-cell 2SLS coefficient, computed from the within-cell subsample without imposing the joint structure of the dual-IV regression; and the pairwise Wald test of equality of within-cell first-stage coefficients. Three findings stand out. The AR set on vote share inside the CRAS-with-turnover cell, [0.75, 6.05], comfortably brackets the headline 2.94 and excludes zero. The AR set inside the CRAS-no-turnover cell is the entire searched grid (“unbounded”), the appropriate inference output when the within-cell first stage is statistically indistinguishable from zero: there is no β_0 in the grid at which the AR test rejects $H_0 : \beta = \beta_0$, because the instrument moves Δ Beneficiaries by an amount not statistically distinguishable from zero in that subgroup. The Wald tests confirm that the cross-cell first-stage differences are extreme: $\Delta\hat{\pi}$ between CRAS-with-turnover and CRAS-without-turnover equals 0.230 with $z = 12.1$ and $p < 10^{-30}$, and the same comparison against no-CRAS yields $\Delta = 0.137$ with $p < 10^{-15}$.

Table E.5: Formal Within-Subgroup Identification (Three-Cell Partition)

	no-CRAS	CRAS, no turnover	CRAS, turnover
<i>First-stage and identifying mass</i>			
First-stage $\hat{\pi}$	0.072 (0.016) (0.018) (0.005)	-0.021 0.209	
Kleibergen-Paap F	20.2	1.3	1682.7
FWL identifying share ω_g	0.115	0.068	0.817
<i>Within-cell 2SLS (Spec B; endogenous in levels)</i>			
Vote share $\hat{\beta}_g$	-86.23 (48.97) (171.21) (1.30)	-153.32 3.30	
AR 95% set (vote share)	[-15.00, -10.30]	$[-\infty, +\infty]$	[0.75, 6.05]
Winning $\hat{\beta}_g$	116.50 (159.03) (223.98) (2.08)	150.15 2.33	
AR 95% set (winning)	[unbounded]	[unbounded]	[-1.80, 6.75]
Municipalities	690	203	1,089
<i>Pairwise Wald tests of $\hat{\pi}$ equality</i>			
$\hat{\pi}_{\text{CRAS, turnover}} = \hat{\pi}_{\text{no-CRAS}}$	$\Delta = 0.137$	$z = 8.19$	$p < 10^{-15}$
$\hat{\pi}_{\text{CRAS, turnover}} = \hat{\pi}_{\text{CRAS, no turnover}}$	$\Delta = 0.230$	$z = 12.15$	$p < 10^{-30}$
$\hat{\pi}_{\text{CRAS, no turnover}} = \hat{\pi}_{\text{no-CRAS}}$	$\Delta = -0.092$	$z = -3.81$	$p < 0.001$

Notes: Three-cell partition refining the joint pre-reform infrastructure and manager-turnover margins: no-CRAS are municipalities without a CRAS in the 2008 *Censo SUAS*; CRAS, no turnover are municipalities with a CRAS but no change in the *de facto* front-line manager between the 2008 and 2009 waves; CRAS, turnover are municipalities with a CRAS and a manager change. Each cell-level estimate is obtained from a within-cell 2SLS regression on the subsample of municipality-year observations belonging to that cell. The first stage regresses the standardized 12-month moving average of *Bolsa Família* beneficiaries on the standardized 2009 quota shock $\times$ post, controlling for the six Census 2000 baseline characteristics $\times$ post and for municipality, year, and incumbent-pre-reform-winner party fixed effects, weighted by 2000 population, with cluster-robust standard errors at the state level. The 2SLS coefficient identifies the within-cell average causal response, in the sense of Angrist and Imbens (1995), of incumbent vote share (or the probability of reelection) to instrumented beneficiary expansion within the cell. Anderson-Rubin 95% confidence sets are grid-inverted over $\beta_0 \in [-15, 25]$ with step 0.05; sets reaching either grid endpoint are reported as “[unbounded]” since the AR test does not reject any candidate β_0 within the searched grid in that cell, the appropriate inference output when the within-cell first stage is statistically indistinguishable from zero. The FWL identifying share ω_g is the share of the total residualized weighted variance of $\text{shock}_m \times \text{post}$ that lies in cell g , the implicit weight cell g carries in the pooled 2SLS estimand. Pairwise Wald tests of $\hat{\pi}$ equality assume independent cell estimates and report a one-degree-of-freedom χ^2 p -value on the squared standardized difference.

E.5 Within-CRAS Heterogeneity by Hire Type of the Manager Substitute

The complier-cell reading of the headline 2SLS hinges on Definition A (*cpf_agente* change) capturing genuine personnel inflows rather than internal bureaucratic relabeling. This appendix supplements the body argument with a direct hire-type decomposition of the 3,357 Definition-A manager changes across the 2,583 municipalities that experienced at least one substitution between the 2008 and 2009 *Censo SUAS* waves. Each manager change is classified by matching the 2009 successor’s CPF against the full 2008 CRAS structure across Brazil. A change is *external* if the 2009 *cpf_agente* was not present in any 2008 CRAS unit; *internal same CRAS* if the successor was present at the same unit in another role (e.g., the 2008 *Coordenador* stepped down and a 2008 *técnico* stepped up); and *internal other CRAS* if the successor was present at a different CRAS unit, typically in the same municipality. The headline distribution is 75.5% external, 14.7% internal same CRAS, and 9.8% internal other CRAS. The dominant case is fresh personnel, not internal reshuffles.

Table E.6 reports the within-cell 2SLS by hire-type subsample, paralleling the three-cell partition of Table E.5. Panel A refines the CRAS-with-turnover cell into three hire-type subgroups defined by the majority hire type within each municipality (strict majority; “mixed” municipalities are pooled with internal in Panel B and excluded from Panel A). Two patterns emerge. First, the external-dominant subgroup carries the bulk of the identifying mass and recovers a 2SLS point estimate near the pooled headline (vote share $\hat{\beta} \approx 2.7$ pp, comparable to the 2.94 of Section 5), with a first-stage F in the hundreds. Second, the internal-same-CRAS subgroup is small ($n \approx 100$ municipalities) but yields a first stage of similar magnitude and a vote-share estimate close to (or above) the external-dominant value; this is the residual “re-deployment” channel and we do not read its larger point estimate as a meaningful contrast against the external-dominant cell given the wide AR set. Third, the internal-other-CRAS subgroup is too small ($n \approx 100$) to support a precisely-estimated within-cell 2SLS: the within-cell first stage is essentially zero and the AR set is unbounded. The headline 2SLS is therefore identified primarily across external-hire municipalities and, to a lesser extent, internal-same-CRAS reshuffles; the cross-CRAS internal transfers do not generate the cross-section variation in Δ Beneficiaries needed for identification.

The heterogeneity table does not sharply discriminate between an “external ability hire” channel and an “internal-same-CRAS re-deployment” channel: both contribute to the headline. We read it as supplemental evidence that the turnover indicator in Table 12 captures real personnel inflows rather than as a discriminating test against

Table E.6: Within-CRAS Heterogeneity by Hire Type of Manager Substitute

<i>Panel A: Four-cell partition refining CRAS-turnover by hire type</i>				
	no-CRAS	CRAS-turn external hire	CRAS-turn internal same CRAS	CRAS-turn internal other CRAS
First-stage $\hat{\pi}$	0.072 (0.016)	0.180 (0.012)	0.184 (0.003)	-0.014 (0.006)
Kleibergen-Paap F	20.2	221.4	4968.4	5.8
Vote share $\hat{\beta}_g$	-86.23 (48.97)	2.69 (1.94)	4.51 (2.09)	-112.16 (67.48)
AR 95% (vote share)	[-15.00, -10.30]	[-0.80, 7.20]	[0.20, 8.80]	[unbounded]
Winning $\hat{\beta}_g$	116.50 (159.03)	17.45 (6.12)	6.03 (3.58)	-760.94 (514.31)
AR 95% (winning)	[unbounded]	[5.70, 25.00]	[-1.30, 13.40]	[unbounded]
Municipalities	690	861	105	109
<i>Panel B: Two-cell partition within CRAS-turnover</i>				
	CRAS-turn external dominant	CRAS-turn internal dominant		
First-stage $\hat{\pi}$	0.180 (0.012)	0.167 (0.022)		
Kleibergen-Paap F	221.4	59.4		
Vote share $\hat{\beta}_g$	2.69 (1.94)	2.61 (1.82)		
AR 95% (vote share)	[-0.80, 7.20]	[-1.30, 6.50]		
Winning $\hat{\beta}_g$	17.45 (6.12)	2.59 (3.87)		
AR 95% (winning)	[5.70, 25.00]	[-4.40, 12.40]		
Municipalities	861	228		

Notes: Hire type of the manager substitute is classified at the CRAS level from the matched 2008–2009 Censo SUAS CPF panel: a change is *external* if the successor (2009 *cpf_agente*) was not present anywhere in the CRAS structure in 2008, *internal same* CRAS if the successor was present at the same unit in another role, and *internal other* CRAS if the successor was present at a different unit. A municipality is assigned to a cell by the dominant type of its manager changes (strict majority; “mixed” municipalities are pooled with internal in Panel B and excluded from Panel A). The headline distribution of changes across the 2,583 municipalities with at least one manager turnover is 75.5% external, 14.7% internal same CRAS, 9.8% internal other CRAS. Each cell-level estimate is obtained from a within-cell 2SLS regression with the headline specification (Column 3 of Table 8): municipality, year, and pre-reform-winner party fixed effects, baseline Census 2000 characteristics $\times$ post, weighted by 2000 population, cluster-robust standard errors at the state level. Anderson-Rubin 95% sets are grid-inverted over $\beta_0 \in [-15, 25]$.

alternative substitute-type mechanisms.

E.6 Robustness of the Turnover Margin: *De Jure* Coordinator Definition

The body of the paper uses the *de facto* manager-turnover indicator (*cpf_agente* change between the 2008 and 2009 *Censo SUAS* waves). The *cpf_agente* is the respondent of the unit-level questionnaire, the person who actually fills the form and operates the unit day-to-day. An alternative is the *de jure* indicator: a change in the formal *Coordenador* listed on the CRAS RH staff file. The two definitions diverge when a unit’s formal title changes hands without a change in the day-to-day operator, or when the day-to-day operator changes without a change in title.

Table E.7 reproduces Table 12 under the *de jure* definition. The first-stage asymmetry between the no-turnover and turnover cells is sharper under *de jure* (the no-turnover cell is even closer to zero), but the within-turnover-cell 2SLS point estimate is small and statistically indistinguishable from zero. The pattern is consistent with the *de jure* indicator being a noisier proxy for the operational margin our framework targets: formal title changes that do not coincide with day-to-day operator changes register as “turnover” in the *de jure* measure but do not move realized expansion through the implementation chain. The augmented model’s κ^A is therefore best read as the *de facto* object operative in the main text, with the *de jure* margin reported here for completeness.

E.7 Winning Probability by Group

Figure E.1 reports the unconditional 2012 incumbent-win rate by CRAS presence; Figures E.2 and E.3 further split CRAS municipalities by the active-outreach and PAIF channels. Error bars report 95% Wilson binomial confidence intervals. The win rate is highest in the “Without CRAS” group (59.1%) and lowest in the “With CRAS but Without Active Outreach” group (54.0%), consistent with the concentration of marginal incumbents in that group documented in Table E.3.

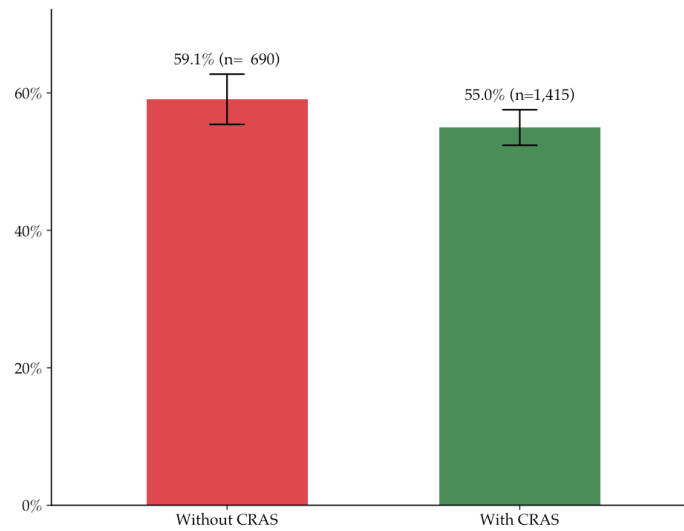
E.8 Core Variables by Group

Figures E.4 and E.5 plot the kernel densities of the 2012 incumbent vote share, the 2009 quota shock, the 2009 realized expansion, and the 2012 margin of victory by CRAS-

Table E.7: Heterogeneity by Manager Turnover — *De Jure* Definition

	Vote Share			Winning		
	non-CRAS	CRAS / non-Turn	CRAS / Turnover	non-CRAS	CRAS / non-Turn	CRAS / Turnover
First Stage (π_1)	0.075 (0.013)	0.062 (0.025)	0.206 (0.002)	0.075 (0.013)	0.062 (0.025)	0.206 (0.002)
Reduced Form (ρ_1)	-3.036 (3.089)	-2.031 (1.749)	0.522 (0.254)	6.419 (9.667)	-10.764 (3.835)	0.592 (0.402)
Two Stage Least Squares (β_1)	-41.315 (45.210)	-34.514 (15.288)	2.027 (1.403)	82.453 (138.268)	-175.050 (35.379)	0.230 (2.197)
Kleibergen-Paap F	35.40	5.92	8501.41	35.40	5.92	8501.41
AR 95% CI	[−15.00, +25.00]	[−15.00, +25.00]	[−0.30, +4.35]	[−15.00, +25.00]	—	[−3.50, +3.90]
Party Fixed Effects	✓	✓	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: Same three-cell partition and specification as Table 12, except the manager is defined here as the *de jure* frontline manager: the formal *Coordinador* listed on the CRAS RH staff file. This is a noisier proxy because a formal change of title without a change in the person who actually runs the unit does not register in the *de facto* measure used in the main text. The first-stage asymmetry is sharper than under the *de facto* measure (the no-turnover cell is even closer to zero), but the within-turnover-cell 2SLS point estimate is small and statistically indistinguishable from zero, the symptom of using a noisier turnover indicator. The augmented model's κ^A is therefore best read as a *de facto* object, with the formal-coordinator margin reported here as a noisier alternative.

**Figure E.1: 2012 incumbent win rate by CRAS presence**

Notes: Bar heights report the share of 2008 incumbents winning the 2012 municipal election, separately for the 690 municipalities without CRAS and the 1415 municipalities with at least one CRAS by 2008. Error bars are 95% Wilson binomial confidence intervals. Sample: DiD-IV estimation sample of Section 5.

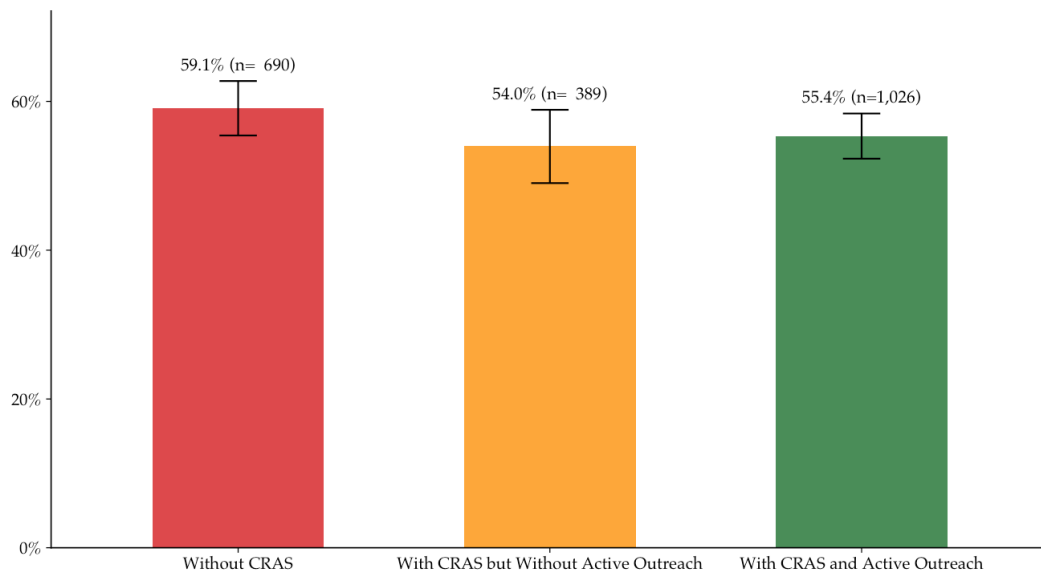


Figure E.2: 2012 incumbent win rate by active-outreach intensity

Notes: Bar heights report the share of 2008 incumbents winning the 2012 municipal election across three groups: municipalities without CRAS; municipalities with CRAS but whose CRAS does not report systematic *busca ativa*; and municipalities with CRAS that report active outreach. Error bars are 95% Wilson binomial confidence intervals. Sample: DiD-IV estimation sample of Section 5.

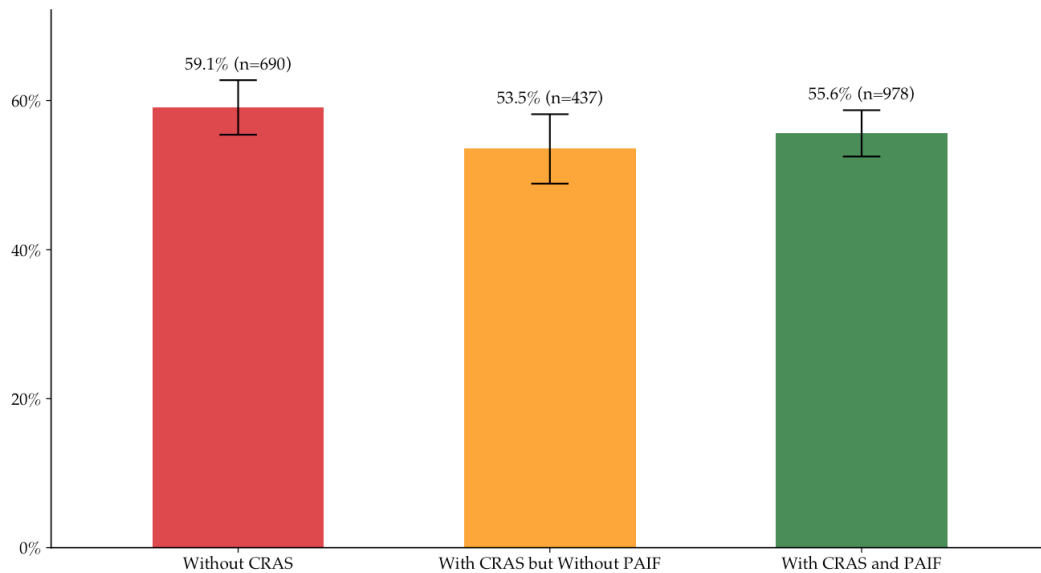


Figure E.3: 2012 incumbent win rate by PAIF-tracking intensity

Notes: Bar heights report the share of 2008 incumbents winning the 2012 municipal election across three groups: municipalities without CRAS; municipalities with CRAS that do not report tracking Bolsa Família families through PAIF; and municipalities with CRAS that do. Error bars are 95% Wilson binomial confidence intervals. Sample: DiD-IV estimation sample of Section 5.

intensity group. Two patterns are worth noting. First, the quota shock is positive on average in the “Without CRAS” group and shifts negative in the “With CRAS” groups: municipalities that had already built CRAS infrastructure before the reform were over-allocated relative to the pre-reform formula and therefore received negative updates in 2009. Second, realized expansion is markedly larger in the “With CRAS” groups despite the negative shocks, consistent with local state capacity, not formal authorization, mediating take-up.

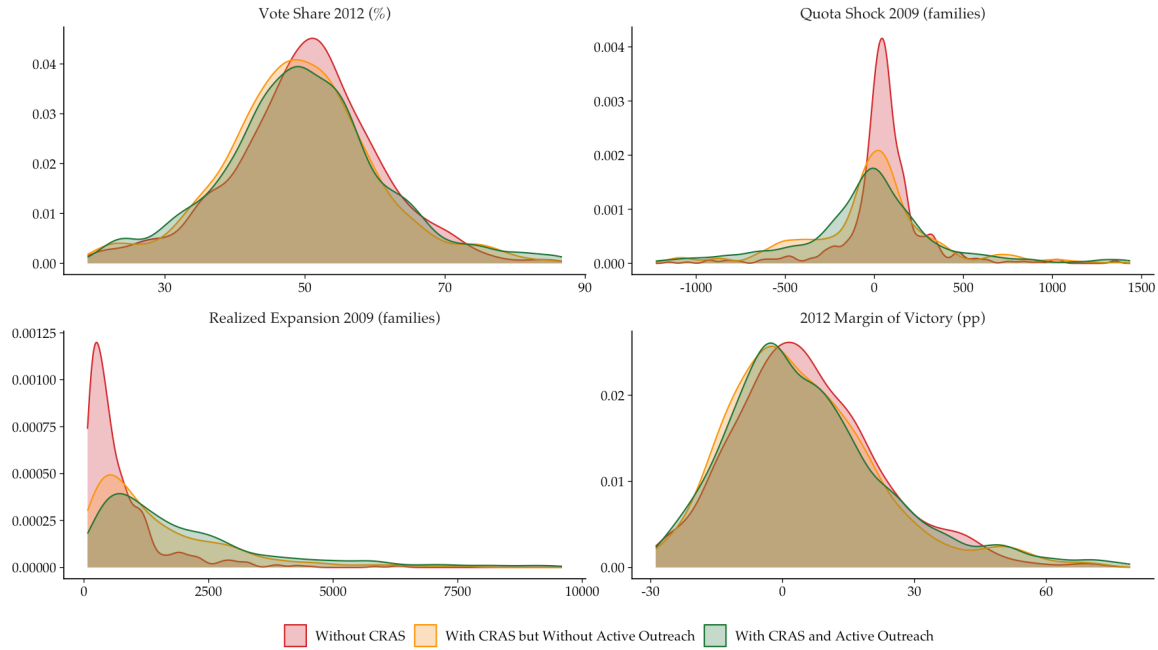


Figure E.4: Core variables by active-outreach group

Notes: Kernel densities of the 2012 incumbent vote share, the 2009 quota shock (counterfactual-based), the 12-month moving average of realized expansion centered on December 2009, and the 2012 incumbent margin of victory. Densities are truncated at the 2nd and 98th percentiles of each variable. Sample: DiD-IV estimation sample of Section 5.

E.9 Baseline Covariates by Group

Figures E.6 and E.7 plot the kernel densities of the six Census 2000 baseline covariates used in the DiD-IV specification, log population, Gini, share of poor, income per capita, share of formal workers, and urban share, by CRAS-intensity group. The “With CRAS” groups are larger, poorer, and more informal than the “Without CRAS” group on all six dimensions. These differences are controlled for in the main analysis through the interaction of each covariate with the post-reform indicator (Column 3 of Tables 6, 7, and 8).

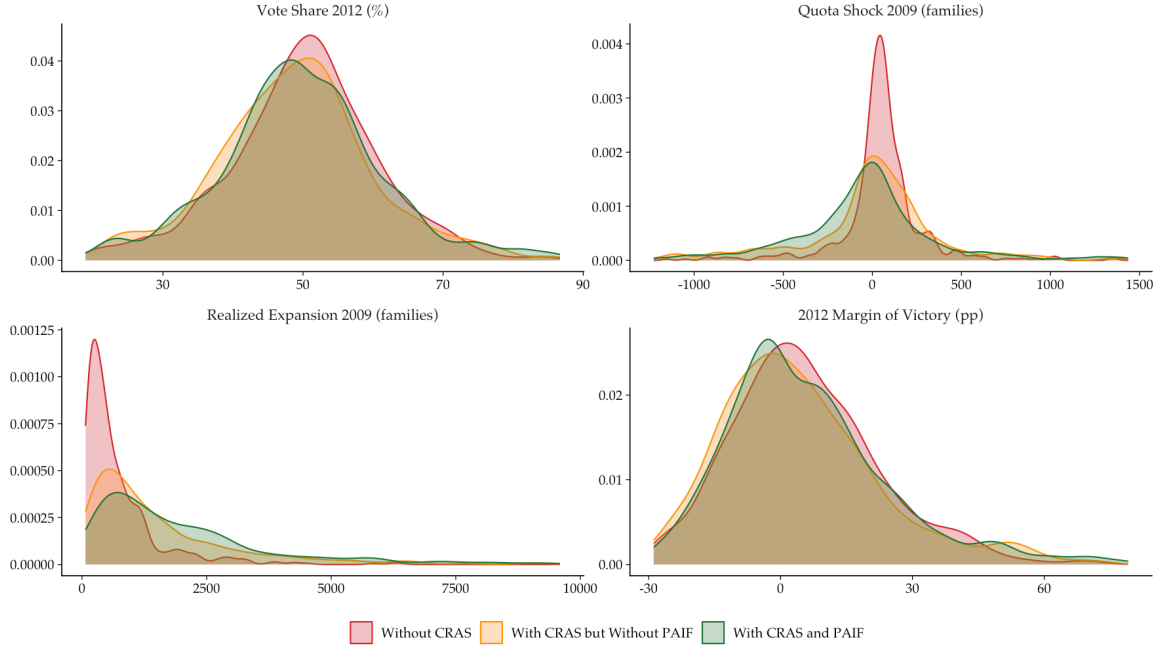


Figure E.5: Core variables by PAIF-tracking group

Notes: Kernel densities of the 2012 incumbent vote share, the 2009 quota shock (counterfactual-based), the 12-month moving average of realized expansion centered on December 2009, and the 2012 incumbent margin of victory. Densities are truncated at the 2nd and 98th percentiles of each variable. Sample: DiD-IV estimation sample of Section 5.

F Additional Details on the Model

This appendix proves the three propositions stated in Section 3 and develops the voter-inference closure used in equation (4). The setup, equations, and notation match Section 3 verbatim and are restated here only where necessary to keep each proof self-contained.

F.1 Setup and Notation

Three classes of agents, indexed and described in Section 3.1: a federal government, which sets a per-family transfer value $\bar{b}$ uniformly across municipalities and a beneficiary quota Q_m that varies by municipality; a continuum of municipalities indexed by m , each governed by an incumbent mayor with unobserved type $\theta_m \in \{0, 1\}$; and a continuum of voters in each municipality, indexed by i , who care about receiving the transfer and about whether the incumbent is competent.

The mayor of type θ_m has quadratic effort cost $\frac{1}{2}c(\theta_m)a_m^2$ with $c(\theta_m) = c_0 - c_1\theta_m$ and $c_0 > c_1 > 0$, faces enforcement pressure $\frac{1}{2}\tau(e_m - 1)^2$ around full quota utilization, and

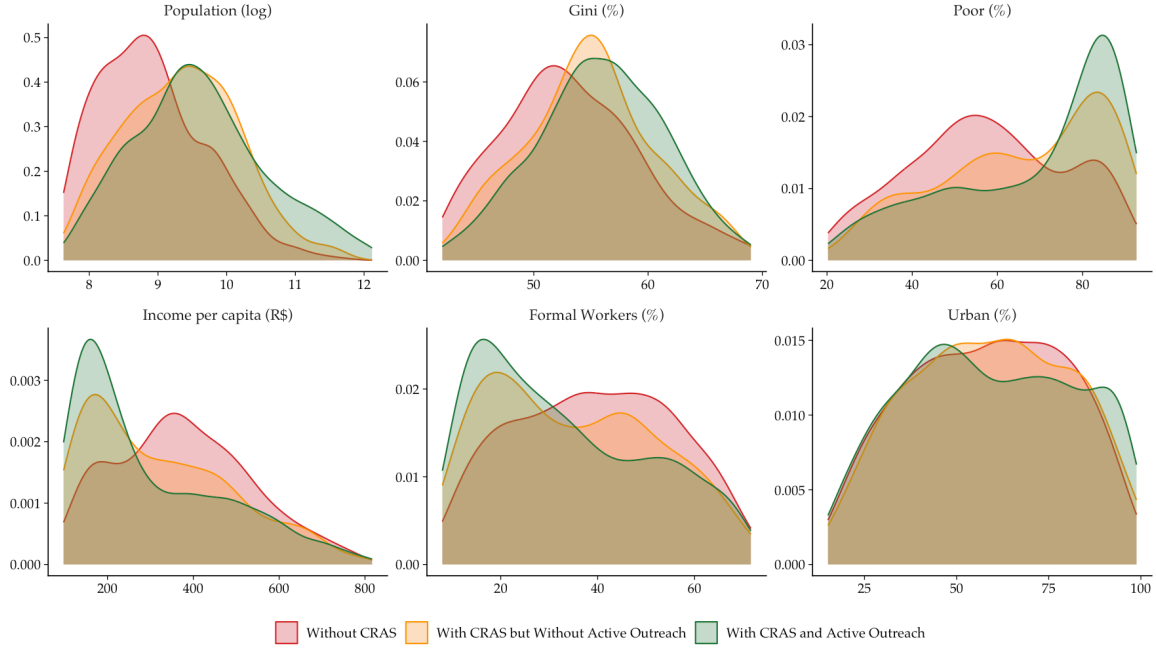


Figure E.6: Baseline covariates by active-outreach group

Notes: Kernel densities of the six Census 2000 baseline covariates entering the DiD-IV specification. Densities are truncated at the 2nd and 98th percentiles of each variable. Gini and urban share are displayed multiplied by 100 to align with the other percentage-scale variables. Sample: DiD-IV estimation sample of Section 5.

is rewarded electorally through the vote-share equation $V_m = V_0 + \lambda_1 B_m + \lambda_2 e_m$, with material weight $\lambda_1 > 0$ and inference weight $\lambda_2 \geq 0$. Voter i 's utility differential from supporting the incumbent rather than the challenger is $\Delta U_{im} = \alpha d_{im} + \delta \Pr(\theta_m = 1 | s_{im}) + \eta_{im}$, with material weight $\alpha > 0$, competence weight $\delta > 0$, and an idiosyncratic ideology shock η_{im} drawn from a continuous distribution.

Realized coverage takes the benchmark form $e_m = \psi_m + a_m$, with the scalar capacity index $\psi_m \in [\psi_0, \psi_0 + \bar{\psi}]$, or the augmented form

$$e_m = \psi_0 + g(\kappa_m^I, \kappa_m^A) \cdot (\bar{\psi} + a_m),$$

with multiplier $g(\kappa^I, \kappa^A) = \gamma + (1 - \gamma)\kappa^I \kappa^A$, productivity floor $\gamma \in [0, 1)$, infrastructure indicator $\kappa^I \in \{0, 1\}$, and alignment indicator $\kappa^A \in \{0, 1\}$. Realized beneficiaries are $B_m = e_m \cdot Q_m$ in both cases. The legacy floor ψ_0 is the passive utilization that operates through informal pre-existing channels, and the ceiling $\bar{\psi}$ is what an idealized CRAS at full operational intensity would add on top of the floor.

The game proceeds as in Section 3.1. Nature draws the mayor's type and the federal quota; inherited capacity is taken as given; the mayor chooses discretionary effort

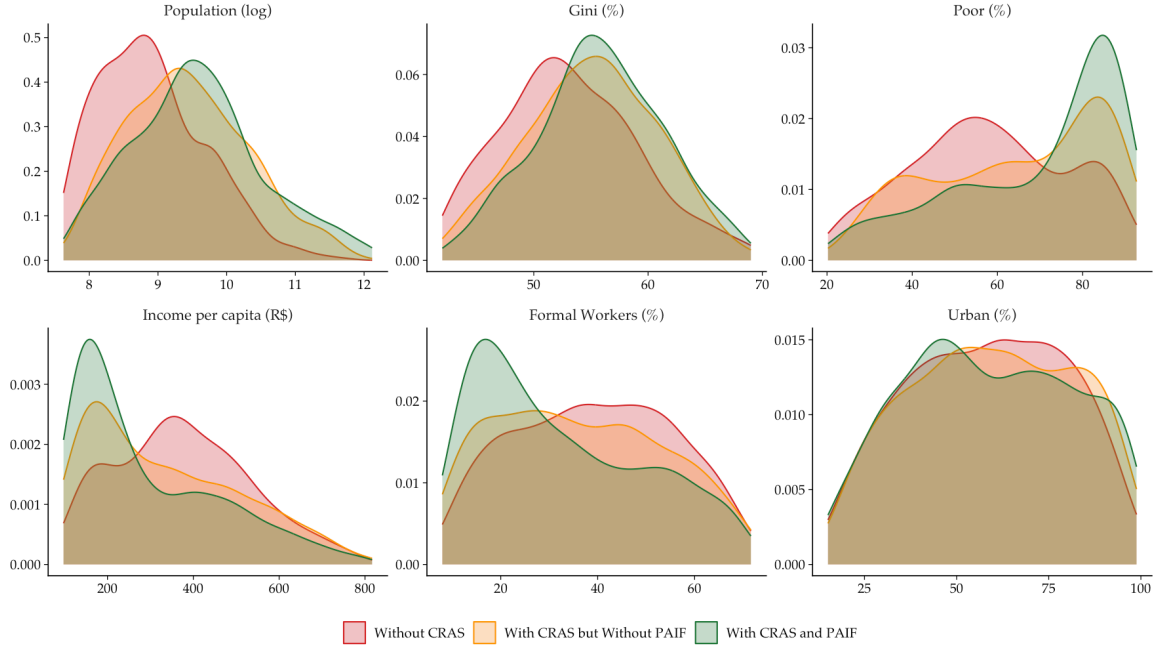


Figure E.7: Baseline covariates by PAIF-tracking group

Notes: Kernel densities of the six Census 2000 baseline covariates entering the DiD-IV specification. Densities are truncated at the 2nd and 98th percentiles of each variable. Gini and urban share are displayed multiplied by 100 to align with the other percentage-scale variables. Sample: DiD-IV estimation sample of Section 5.

$a_m \geq 0$ knowing his type; quota utilization e_m realizes; voters observe a noisy signal of program implementation and update beliefs about the mayor's type; the election takes place. The mayor maximizes

$$U_m(a_m; \theta_m) = V_m - \frac{1}{2}c(\theta_m)a_m^2 - \frac{1}{2}\tau(e_m - 1)^2,$$

taking Q_m , the capacity primitives, and the voter-inference function as given.

F.2 Voter Inference and the Vote-Share Equation

Voters do not observe the mayor's type θ_m directly. Each voter i in municipality m observes a noisy signal of program implementation,

$$s_{im} = \nu_0 + \nu_1 d_{im} + \nu_2 e_m + \varepsilon_{im}, \quad \nu_1, \nu_2 > 0,$$

where $d_{im} \in \{0, 1\}$ indicates whether household i in m is a beneficiary and ε_{im} is mean-zero noise independent across voters and conditional on e_m . The first informative component $\nu_1 d_{im}$ captures the voter's direct experience with the program; the second $\nu_2 e_m$

captures the aggregate visibility of program delivery, which is observable to beneficiaries and non-beneficiaries alike.

Given conditional signal densities $f_1(\cdot)$ and $f_0(\cdot)$ under the two types and a common prior $p_0 = \Pr(\theta_m = 1)$, the voter posterior is

$$\Pr(\theta_m = 1 \mid s_{im}) = \frac{f_1(s_{im})p_0}{f_1(s_{im})p_0 + f_0(s_{im})(1 - p_0)}.$$

Differentiating with respect to s_{im} and using the quotient rule, the posterior is strictly increasing in s_{im} if and only if the densities satisfy the monotone likelihood ratio property,

$$\frac{f_1(s)}{f_0(s)} \text{ strictly increasing in } s.$$

By Proposition 1, proved below, competent mayors choose strictly higher discretionary effort than incompetent mayors at every ψ_m, g, Q_m , and τ . Equilibrium realized coverage is therefore strictly larger under $\theta_m = 1$, $e_m^*(1) > e_m^*(0)$, which shifts the conditional mean of the signal upward by $\nu_2[e_m^*(1) - e_m^*(0)] > 0$. For any noise distribution with a log-concave density (the canonical case includes the Gaussian, the logistic, and a wide class of unimodal densities), a location shift in the conditional mean preserves the MLRP. The voter posterior $\Pr(\theta_m = 1 \mid s_{im})$ is therefore strictly increasing in s_{im} as an equilibrium property of the model rather than as a free parameter.

Aggregating $\Pr(\text{vote incumbent} \mid d_{im}, s_{im})$ across voters in the probabilistic-voting tradition of [Lindbeck and Weibull \(1987\)](#), with η_{im} inducing smoothing across the ideology distribution, municipal vote share takes the linear form

$$V_m = V_0 + \lambda_1 B_m + \lambda_2 e_m,$$

where $\lambda_1 > 0$ aggregates the material channel (αd_{im} across beneficiaries) and $\lambda_2 \geq 0$ aggregates the inference channel ($\delta \Pr(\theta_m = 1 \mid s_{im})$ across all voters). This is equation (4) of the body.

F.3 Proof of Proposition 1

The mayor's objective in the benchmark is $U_m = V_0 + (\psi_m + a_m)(\lambda_1 Q_m + \lambda_2) - \frac{1}{2}c(\theta_m)a_m^2 - \frac{1}{2}\tau(\psi_m + a_m - 1)^2$. Differentiating with respect to a_m and using $\partial e_m / \partial a_m = 1$,

$$\frac{\partial U_m}{\partial a_m} = (\lambda_1 Q_m + \lambda_2) - c(\theta_m)a_m - \tau(\psi_m + a_m - 1).$$

The first term is the marginal electoral benefit of effort: each additional unit of effort raises quota utilization by one, generating Q_m additional beneficiaries (each worth λ_1 in vote share) and one additional unit of visible implementation (worth λ_2). The second is the marginal effort cost. The third is the marginal enforcement cost, which is positive when $e_m > 1$ and negative when $e_m < 1$.

Setting the derivative to zero, expanding $\tau(\psi_m + a_m - 1) = \tau\psi_m + \tau a_m - \tau$, collecting terms in a_m , and dividing by $c(\theta_m) + \tau > 0$ yields

$$a_m^*(\theta_m) = \frac{(\lambda_1 Q_m + \lambda_2) + \tau(1 - \psi_m)}{c(\theta_m) + \tau}.$$

The second derivative, $-c(\theta_m) - \tau < 0$, confirms strict concavity, so a_m^* is the unique global maximum. Equilibrium realized coverage and beneficiaries follow from $e_m^*(\theta_m) = \psi_m + a_m^*(\theta_m)$ and $B_m^*(\theta_m) = e_m^*(\theta_m)Q_m$. The first stage is

$$\frac{\partial B_m^*}{\partial Q_m} = e_m^*(\theta_m) + Q_m \cdot \frac{\lambda_1}{c(\theta_m) + \tau},$$

strictly positive whenever $\psi_m \geq 0$, $Q_m > 0$, and the interior optimum holds.

The numerator of a_m^* does not depend on θ_m , and the denominator $c(\theta_m) + \tau$ is strictly smaller under $\theta_m = 1$ than under $\theta_m = 0$. Hence $a_m^*(1) > a_m^*(0)$, $e_m^*(1) > e_m^*(0)$, and $B_m^*(1) > B_m^*(0)$. This type-ordering establishes the equilibrium claim used in the MLRP closure of the previous subsection.

Remark 2 (Boundary Solutions). *If inherited capacity ψ_m is small and electoral returns are low, the interior solution may yield $a_m^*(\theta_m) < 0$, violating the non-negativity constraint $a_m \geq 0$. The constrained optimum is then $a_m^* = 0$, the municipality implements only its inherited capacity ($e_m^* = \psi_m$, $B_m^* = \psi_m Q_m$), and the quota shock has no effect on realized coverage. This corner solution corresponds to municipalities for which the instrument fails to generate a first stage; in the augmented model below, it is the limit of $g(\kappa^I, \kappa^A) \rightarrow \gamma$.*

F.4 Proof of Proposition 2

In the augmented model, $e_m = \psi_0 + g(\bar{\psi} + a_m)$, with $g = g(\kappa_m^I, \kappa_m^A)$. The mayor's objective becomes

$$U_m = V_0 + (\lambda_1 Q_m + \lambda_2) [\psi_0 + g(\bar{\psi} + a_m)] - \frac{1}{2} c(\theta_m) a_m^2 - \frac{1}{2} \tau [\psi_0 + g(\bar{\psi} + a_m) - 1]^2.$$

The marginal product of effort is now g rather than one: a unit of a_m raises e_m by g and raises B_m by $g \cdot Q_m$. Differentiating with respect to a_m and using the chain rule,

$$\frac{\partial U_m}{\partial a_m} = g(\lambda_1 Q_m + \lambda_2) - c(\theta_m)a_m - \tau g[\psi_0 + g(\bar{\psi} + a_m) - 1].$$

Setting the derivative to zero and solving,

$$a_m^*(\theta_m) = \frac{g(\lambda_1 Q_m + \lambda_2) + \tau g[1 - \psi_0 - g\bar{\psi}]}{c(\theta_m) + \tau g^2}.$$

The second derivative, $-c(\theta_m) - \tau g^2 < 0$ for any $g \geq 0$, confirms strict concavity and the uniqueness of the optimum. The numerator is strictly increasing in g at the relevant parameter range, and the denominator rises only quadratically with the second-order τg^2 term, so a_m^* is strictly increasing in g for any fixed θ_m .

The first stage of the empirical design follows from $\partial B_m^*/\partial Q_m = (\partial e_m^*/\partial Q_m) \cdot Q_m + e_m^*$ and the chain rule applied to the augmented production function:

$$\frac{\partial B_m^*}{\partial Q_m} = e_m^*(\theta_m) + Q_m \cdot \frac{\lambda_1 g^2}{c(\theta_m) + \tau g^2}.$$

As $\kappa_m^I \kappa_m^A \rightarrow 0$, $g \rightarrow \gamma$ and the marginal-effort term in the first stage attenuates by a factor of order γ^2 relative to the active subgroup. The attenuation operates on the marginal-productivity term and propagates through equilibrium effort. As $g \rightarrow \gamma$ with γ small, a_m^* shrinks proportionally, e_m^* approaches the legacy floor ψ_0 , and the first stage attenuates sharply relative to the active subgroup, exactly as Proposition 2 of the body states.

F.5 Frisch-Waugh-Lovell Decomposition of the 2SLS Coefficient

For the just-identified 2SLS estimator, the Frisch-Waugh-Lovell decomposition writes the headline coefficient as a weighted average of within-subgroup coefficients ([Angrist and Imbens, 1995](#), Proposition 1; [Goldsmith-Pinkham et al., 2022](#)). Let $\tilde{z}_m$ and $\tilde{d}_m$ denote the instrument and the endogenous variable residualized against all controls and fixed effects, and let w_m denote regression weights. Then

$$\hat{\beta}_{2SLS} = \sum_g \omega_g \beta_g^{\text{FWL}}, \quad \omega_g = \frac{\sum_{m \in g} w_m \tilde{z}_m \tilde{d}_m}{\sum_m w_m \tilde{z}_m \tilde{d}_m},$$

where the sum runs over a partition $\{g\}$ of the sample, β_g^{FWL} is the within-subgroup OLS coefficient of $\tilde{d}_m$ on the projection of $\tilde{z}_m$ onto $\tilde{d}_m$ within g , and $\sum_g \omega_g = 1$. Under

the regularity conditions of Goldsmith-Pinkham et al. (2022), the weights lie in $[0, 1]$ whenever the residualized full-sample covariance has the same sign as every within-group covariance, which holds whenever the headline first stage is positive.

The within-subgroup numerator of ω_g is approximately proportional to the population first-stage slope and the within-group instrument variance:

$$\sum_{m \in g} w_m \tilde{z}_m \tilde{d}_m \propto \mathbb{E} \left[\frac{\partial B_m^*}{\partial Q_m} \middle| m \in g \right] \cdot \text{Var}(\tilde{z}_m \mid m \in g).$$

The instrument-variance term is bounded below by the strict positivity of $\tilde{z}_m$ within the subgroup. The first-stage-slope term, by Proposition 2, attenuates as $g(\kappa^I, \kappa^A) \rightarrow \gamma$. Subgroups in which the structural multiplier collapses therefore contribute residualized covariance of order zero, $\omega_g \rightarrow 0$, and the remaining mass is absorbed by the active subgroup.

The proposition therefore states a structural mapping: $\omega_g \rightarrow 0$ as $\mathbb{E}[\partial B_m^* / \partial Q_m \mid m \in g] \rightarrow 0$. The empirical FWL weights computed in Appendix D.3 for the CRAS, turnover, and Frey partitions confirm this mapping numerically: roughly nine-tenths of the headline identification mass concentrates on the subgroup in which capacity and alignment are jointly favorable.

F.6 Comparative Statics

The augmented-model expression for a_m^* delivers four comparative statics that map directly onto the empirical design.

Quota. A larger federal quota raises optimal effort whenever the implementation chain operates: $\partial a_m^* / \partial Q_m = g \lambda_1 / [c(\theta_m) + \tau g^2] > 0$ for any $g > 0$. The response is stronger for competent mayors, who face a smaller denominator at the same numerator.

Structural margins. The composite multiplier $g(\kappa^I, \kappa^A)$ acts on the marginal product of effort. Increasing g from γ to one (the limit case where infrastructure and alignment are both active) raises both the unconditional level of a_m^* and the responsiveness of a_m^* to Q_m . Outside the active subgroup, g collapses to γ and the chain transmits federal authorization weakly.

Type. Competent mayors exert strictly more effort at every parameter combination: $\partial a_m^* / \partial \theta_m > 0$. This is the equilibrium foundation of the MLRP closure used by voter inference.

Enforcement. As $\tau \rightarrow \infty$, the enforcement penalty dominates: both types converge to $e_m^* = 1$ wherever the chain operates, the type differential vanishes, and voter inference becomes uninformative. The empirical relevance of the design relies on τ being finite, which is consistent with the cross-municipal heterogeneity of realized-versus-authorized utilization documented in Appendix B.6.

F.7 What the Empirical Design Identifies

The empirical strategy instruments realized expansion ΔB_m with the federal quota shock ΔQ_m . The model clarifies what each stage of the design measures and what the IV coefficient captures, while underscoring what it does not identify.

First stage. The first-stage coefficient measures the responsiveness of realized expansion to authorization. By Proposition 2, this responsiveness is governed by $g(\kappa^I, \kappa^A)$ and collapses where either of the two structural margins fails. The empirical first stage is therefore concentrated in the active subgroup, and the full-sample first-stage Kleibergen-Paap statistic summarizes responsiveness as a population average that is dominated by the active subgroup’s contribution. Section 6 documents this concentration directly through the dual-IV decomposition by CRAS presence and within-CRAS manager turnover.

Reduced form. The reduced form measures the intention-to-treat effect of the federal authorization on incumbent vote share. It is causal under instrument exogeneity alone, without invoking the exclusion restriction that the shock affects votes only through realized expansion. The reduced-form coefficient identifies the average electoral consequence of being authorized to expand, integrating over both compliers in the active subgroup and the no-effect mass concentrated in the inactive subgroups.

Second stage. The 2SLS coefficient identifies an average causal response in the sense of Angrist and Imbens (1995) and Callaway et al. (2024): a weighted average of per-beneficiary returns across the active subpopulation, with weights determined by the

residualized first-stage covariance through the Frisch-Waugh-Lovell decomposition derived in Appendix F.5. The headline estimate of +2.94 percentage points on vote share and +4.10 percentage points on reelection is the ACR for the subpopulation in which capacity and alignment are jointly favorable, in the sense made explicit by the FWL decomposition. The 2SLS does not separately identify the structural primitives λ_1 , λ_2 , c_0 , c_1 , τ , γ , ψ_0 , $\bar{\psi}$. The model's role is to discipline the interpretation of the reduced-form estimates, not to recover those primitives.

What the design cannot say. The design is silent on the per-beneficiary return in subgroups where the first stage collapses, because there is no instrument-induced variation in B_m to learn from. The empirical observation that FWL weights concentrate on the active subgroup is a statement about *where the implementation chain operates*, not about *where the welfare-to-vote elasticity is largest*. Likewise, the design identifies the extensive-margin effect of the 2009 reform conditional on the federally-set per-family transfer value remaining in place; Appendix C.10 documents that the headline is robust to the concurrent federal value adjustment of March 2011, but the design does not identify the causal effect of value on votes.